

ASNAT User Manual

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Welcome to ASNAT (Air Sensor Network Analysis Tool) -
a free application for retrieval, mapping, summarizing, comparing, plotting
and saving subsets of air sensor data read from web services and local files.

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Running ASNAT:

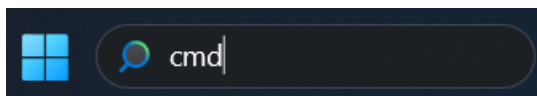
Running ASNAT on your local computer requires the free R programming environment. If you do not have R version 4.2 or later (including *Rscript* and *RTools*) installed on your computer, you will need to install it.

Optional but recommended: Verify R installation:

1. Open a shell window.

a. On Windows: this is called Command Prompt (CMD).

Enter **CMD** in the Search area at the bottom of the screen.

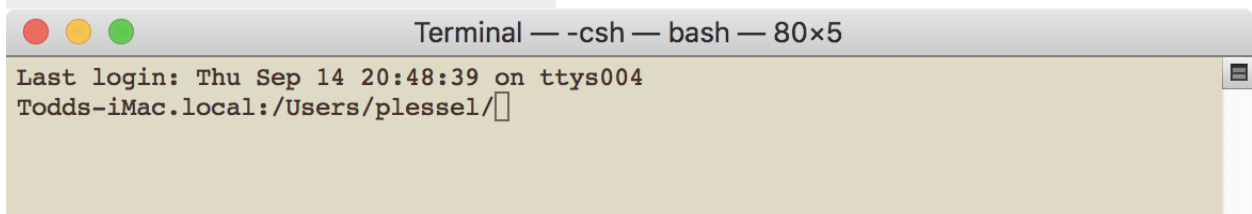
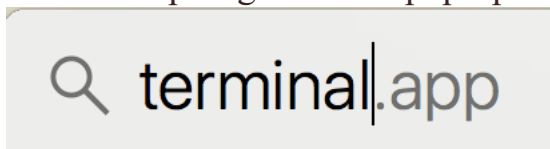


b. On Mac it is called Terminal.

Click the magnifying glass at the upper-right corner of the screen.



In the Spotlight Search pop-up enter **terminal**



c. On Linux it is called *xterm*. Click right-mouse button then select Shell.

2. Check that ***R 4.2 or later*** is installed and is in your **path**.

In the shell enter the following command:

Rscript --version

```
C:\Users\tplessel>Rscript --version
Rscript (R) version 4.2.2 (2022-10-31)
```

If the reported version is less than 4.2 then you need to update *R* to version 4.2 or later (including *Rscript* and *RTools*).

If the '*Rscript*' command is not recognized, but you think you have *R 4.2* or later

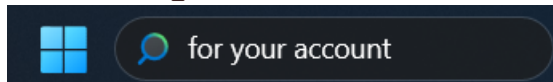
installed then edit your path to include *Rscript*.

- a. On Mac and Linux: edit your **path** environment variable in the appropriate file **~/.cshrc** or **~/.profile** or **~/.bashrc**, etc.

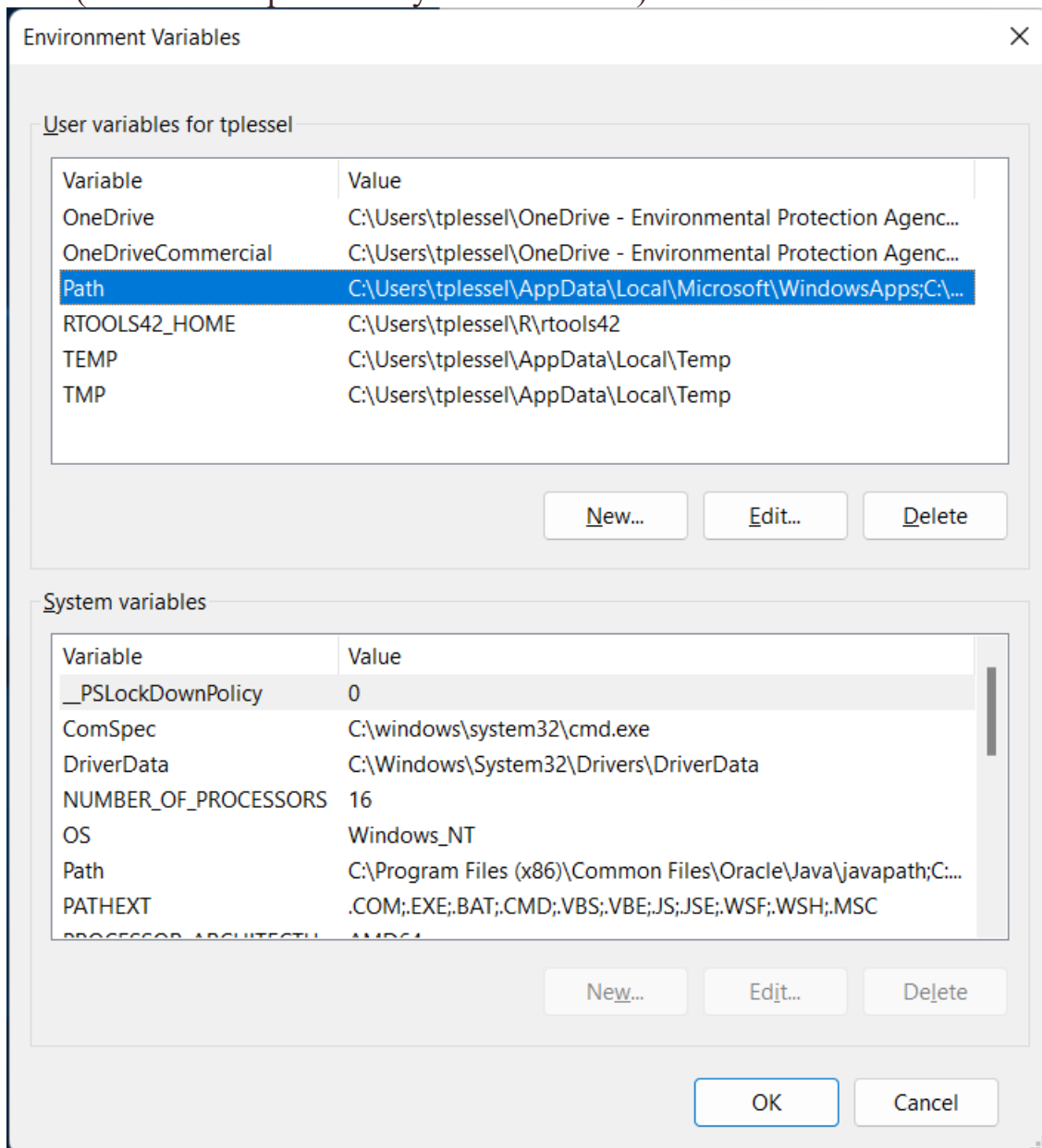
Note you do not need root/admin to install *R* in your home directory or edit files in your home directory.

- b. On Windows: edit your path as follows:

- b.1. In the Search area at the bottom of the screen type
for your account



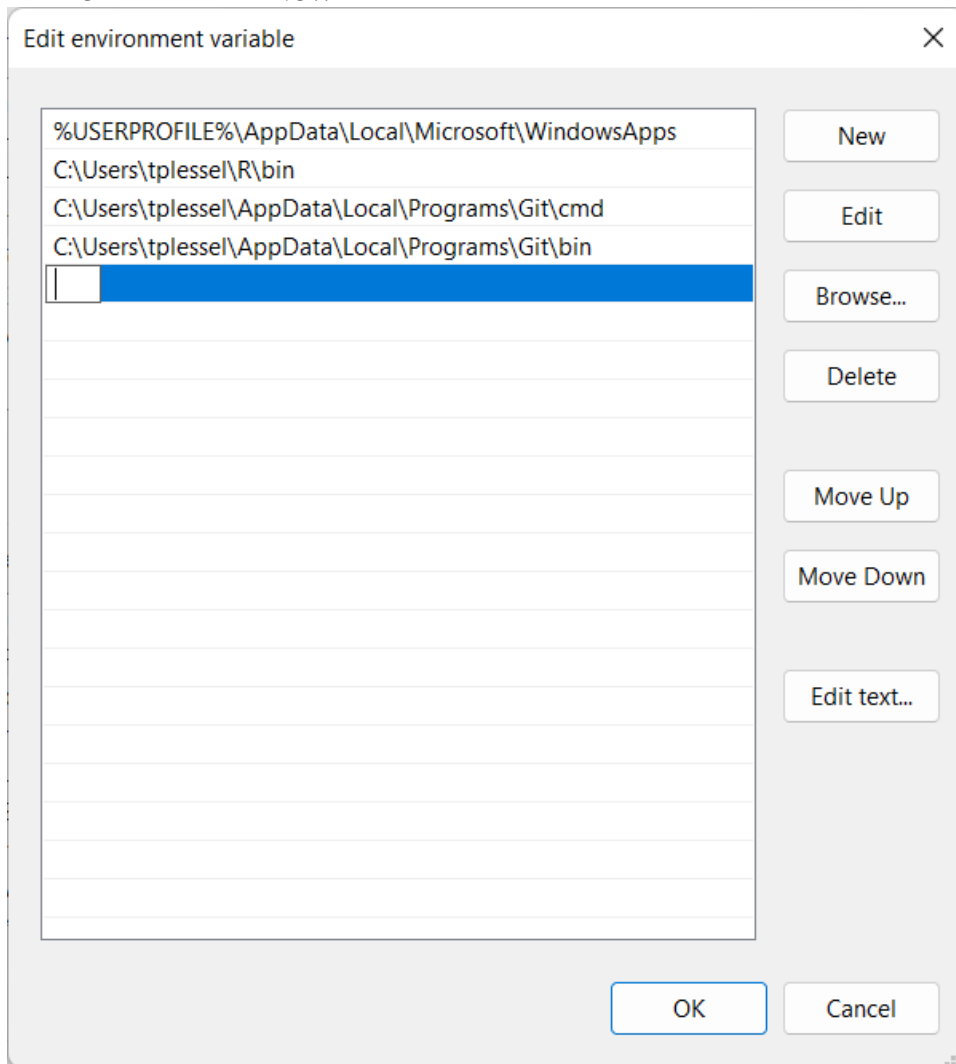
- b.2. Click on 'Edit environment variables for your account'
(Do not attempt to edit system variables.)



b.3. In the popup dialog, **click on Path**.

b.4. Click the **Edit** button.

b.5. Click the **New** button.



b.6. Carefully type the path to the 'R\bin' directory. E.g.,

C:\Users\logname\R\bin or wherever R is installed on your computer.

b.7. Click the **OK** button.

b.8. Click the **X** in the top-right corner of the dialog window to close it.

b.9. Close the command shell.

b.10. **On Windows you'll need to restart your computer.**

Then open a new shell and check again that

Rscript --version

now reports 4.2 or greater.

If not ask a system administrator for help installing and setting-up *R 4.2 or later* including Rscript and RTools. Rscript is a small shell-based *R interpreter*.

(*RTools* is a standard part of *R* that includes a free built-in *C++* compiler (g++/mingw) that *R* programs can use to call *C++* functions.

ASNAT is mostly written in *interpreted R* but contains some functions written in *native-compiled C++* which is 100 times faster than the equivalent *R* functions.)

Download ASNAT:

1. Please use the platform-specific ASNAT zip file. Ensure you have the correct file for your system (e.g., `Darwin.x86_64` for Mac with Intel processors, `Darwin.arm64` for Mac with M1 processors).

Next, move the `ASNAT.zip` file to your *home directory* if it was downloaded elsewhere such as a `Download` directory/folder.

On Mac and Linux: Enter the following command in the shell:

```
cd; mv ~/Downloads/ASNAT.zip .
```

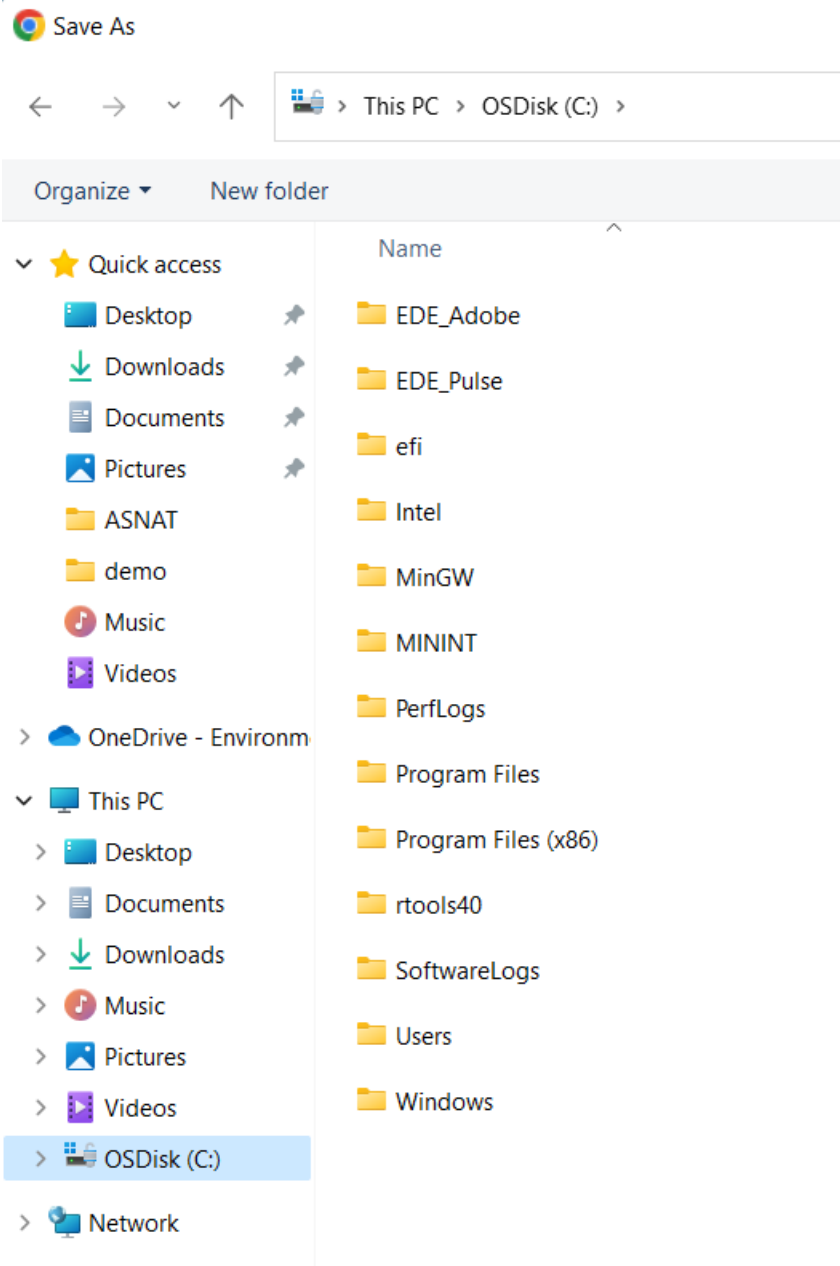
(Remove the old version of ASNAT directory, if it exists.)

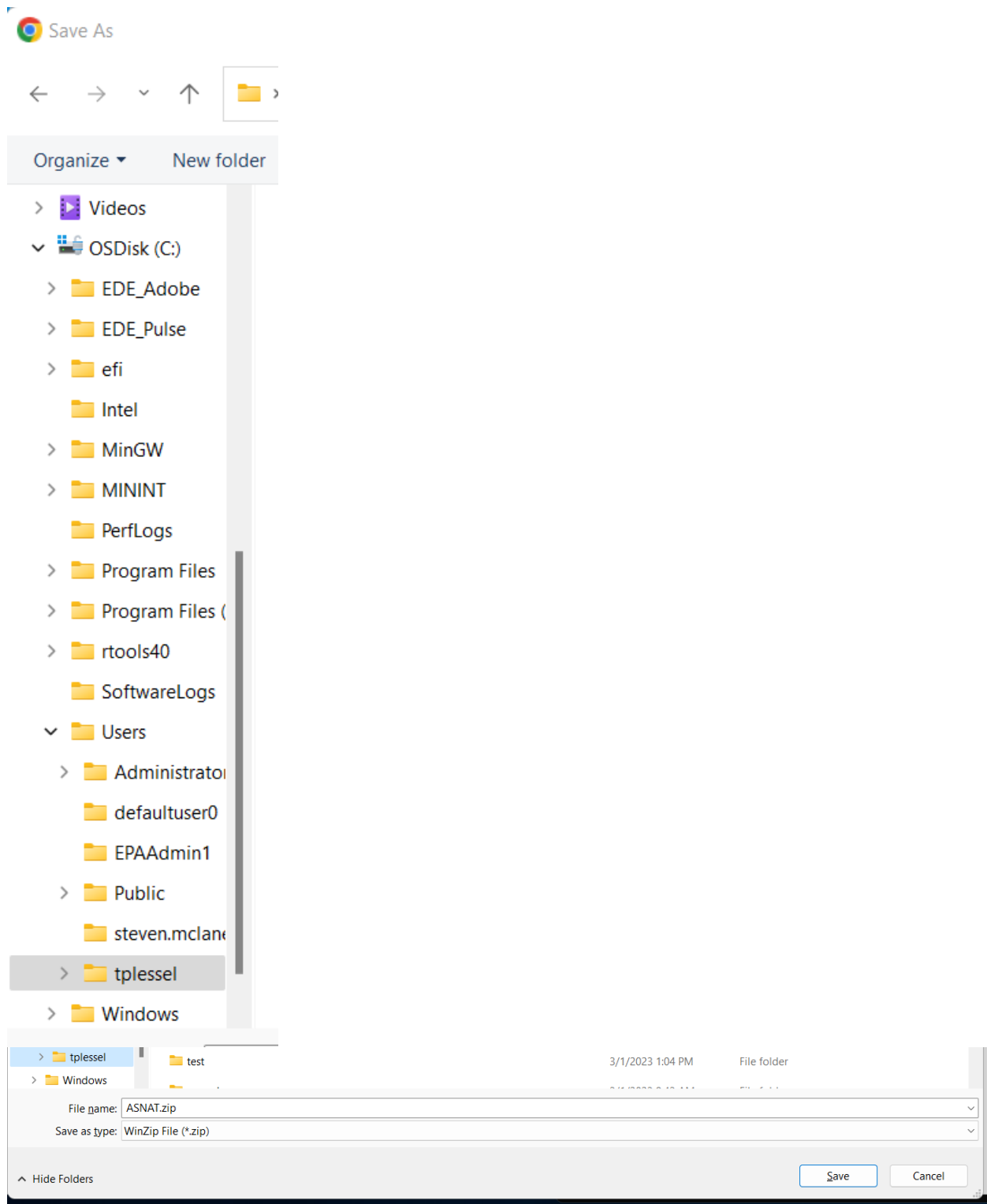
```
\rm -rf ASNAT
```

On Windows your home directory is `C:\Users\logname` (*Not OneDrive!*)

If the web browser prompts for a location to save the file, then

- a. Click **OSDisk (C:)**
- b. Double-click **Users** folder.
- c. Double-click your **logname** folder.
- d. Click **Save** button.





2. Unzip the ASNAT . zip file into ASNAT sub-directory/folder in your home directory.

a. On Mac in the shell enter (required for macOS users):

```
unzip ASNAT.zip  
xattr -cr .
```

b. On Linux:

```
unzip ASNAT.zip
```

c. On Windows:

Enter the following command in the shell

```
start ASNAT.zip
```

```
C:\Users\tplessel>start ASNAT.zip
```

(or use the File Explorer to find and double-click ASNAT.zip)

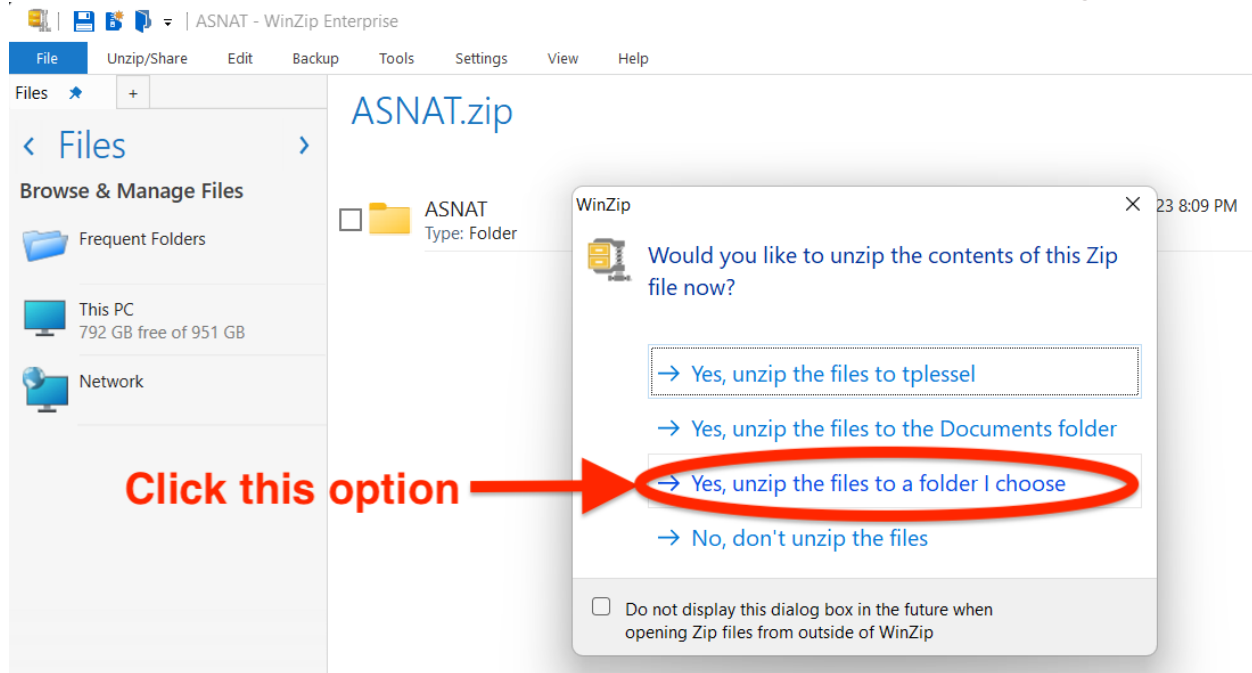
In the popup choose

Yes, unzip the files to a folder I choose.

Press return or Click the **Unzip** button.

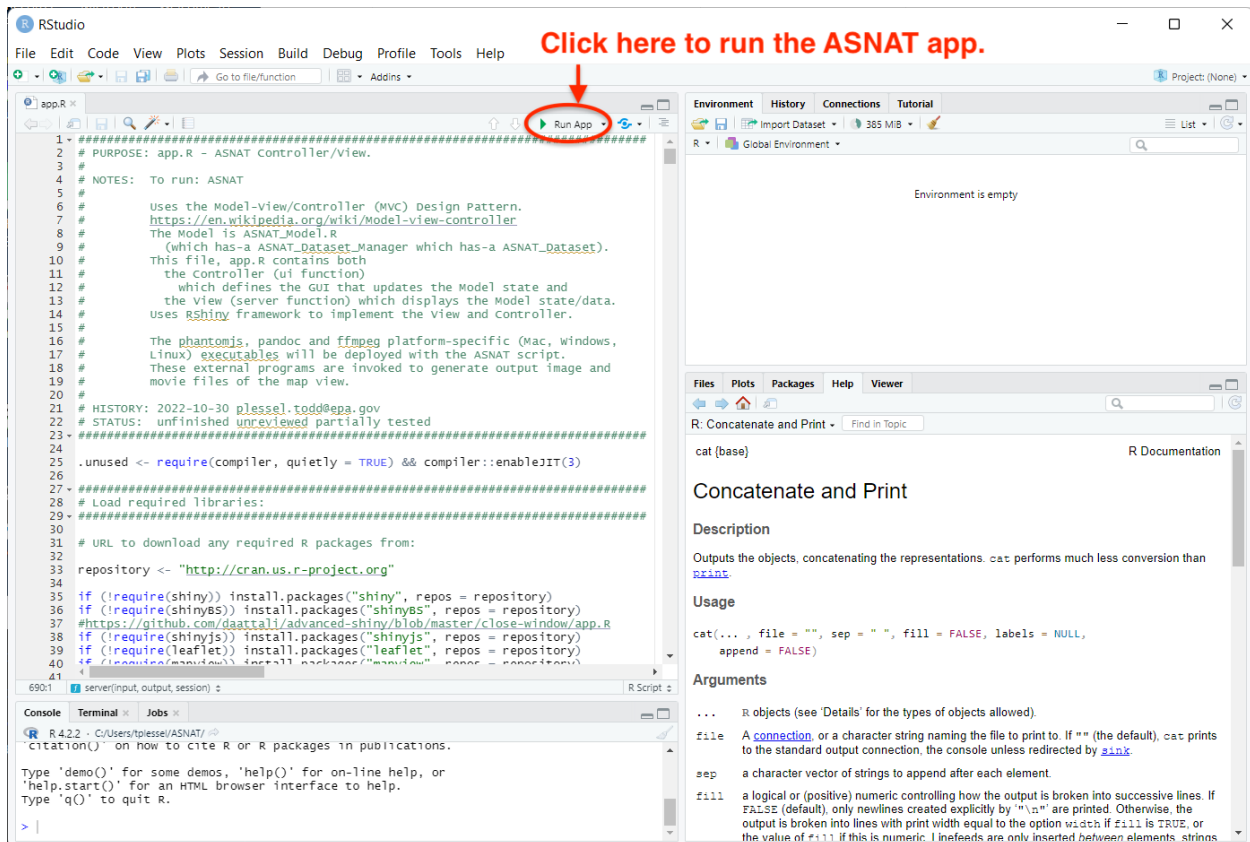
Click the **X** in the upper-right corner to close the File Explorer/WinZip windows.

This should create a folder **C:\Users\logname\ASNAT** containing R files.



3. Running ASNAT:

You can run ASNAT from within RStudio IDE. The first time you run the app it will download, and the installation required Packages then click **Run App**. A disadvantage of doing this is that the RStudio process uses more memory (and screen space) than the smaller Rscript interpreter used when running an app from a shell.



Alternatively, you can run ASNAT from the command prompts using the Rscript interpreter.

Use Rscript to install the required packages *once before you run ASNAT for the first time*.

Go to the ASNAT directory and use Rscript to install the various R packages that are required by ASNAT:

```
cd ASNAT
```

```
Rscript install_packages.R
```



```
C:\Users\tplessel>cd ASNAT
C:\Users\tplessel\ASNAT>Rscript install_packages.R
```

1. Enter ASNAT folder

2. Install required R packages

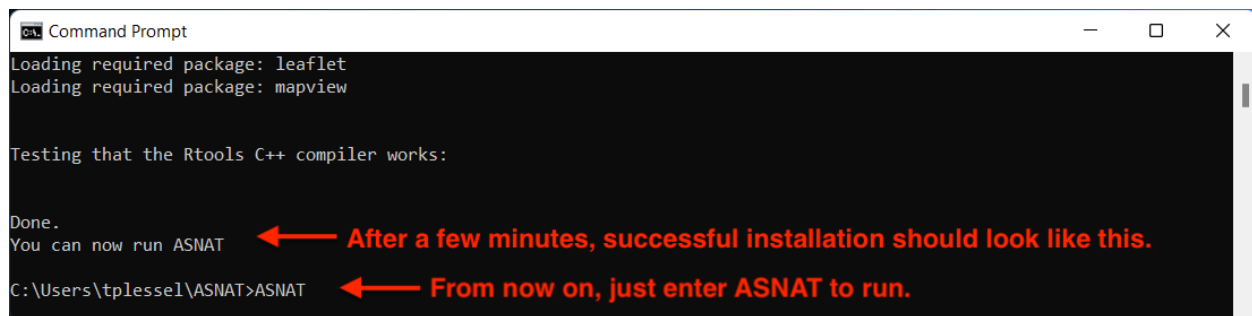
Installation need only be done once successfully.

It will take several minutes to install the many R packages needed by ASNAT. Check the output messages for any errors such as missing .h files or the message:

```
'RTools_is_installed_and_c_plus_plus_compiler_works()'
is not TRUE
```

The above error message means you must download and install *RTools*. Ask your system administrator for assistance resolving any problems before attempting to run ASNAT.

This step only needs to be done once successfully. Afterwards, just run ASNAT.



```
Loading required package: leaflet
Loading required package: mapview

Testing that the Rtools C++ compiler works:

Done.
You can now run ASNAT
C:\Users\tplessel\ASNAT>ASNAT
```

After a few minutes, successful installation should look like this.

From now on, just enter ASNAT to run.

Run ASNAT. This will display the buttons and menus on the left and a map of NYC on the right.

ASNAT

If you are unable to run ASNAT through Command Prompt, please try the following troubleshooting steps:

1. Ensure the folder name of the ASNAT application is 'ASNAT.'
2. Disable or pause any antivirus software.
3. Run Command Prompt as an Administrator.

Here is what ASNAT looks like when launched:

Note: The snippets below were taken from the first version of ASNAT. Since then, several new tabs and menus have been added. Users can view all the new developments and updates under the “Demos” options.

ASNAT

asnatt@epa.gov, 919-541-5500

Welcome, host: Todds-iMac.local wd: /Users/plesse/ASNAT pid: 2515 sessionToken: 60202cc3bb7bd98978dee1cb778d97bc

Reset map

1. Pan/zoom map to region of interest.
2. Start Date,Days to retrieve,Aggregate:

Start Date: 2023-09-01 Days: 1 Timestep: hours

Delete Loaded Data

Load Web (control/command-click two):
AirNow.pm25
AirNow.pm10
AirNow.rh
AirNow.temperature
AirNow.pressure
AirNow.ozone
AQS.pm25
AQS.pm10
AQS.rh
AQS.temperature
AQS.pressure
AQS.ozone
METAR.relativeHumidity
METAR.temperature
METAR.seaLevelPress
PurpleAir.pm25_corrected
PurpleAir.humidity
PurpleAir.temperature

PurpleAir Key:
your key

Purple Air Sites:
0 All sensors in view
7154 Quentin Rd Westport, CT USA
7634 University Dr
11308 NJ7347

Retrieve Data

Load data from a standard-format file:
Browse... No file selected

Load data from sensor instrument files

Select Comparison Dataset X:
▼

Select Variable Of Dataset X:
▼

Select Comparison Dataset Y:
▼

Select Variable Of Dataset Y:
▼

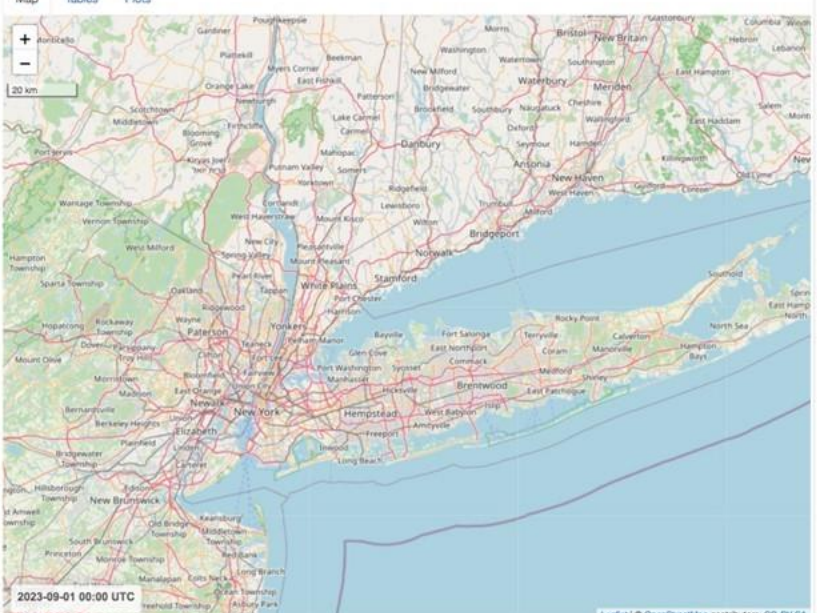
Neighbor distance(m): 1000

Save to Directory:
/Users/plesse/

Output File Format: tsv ▼

Save Data Quit

Map Tables Plots



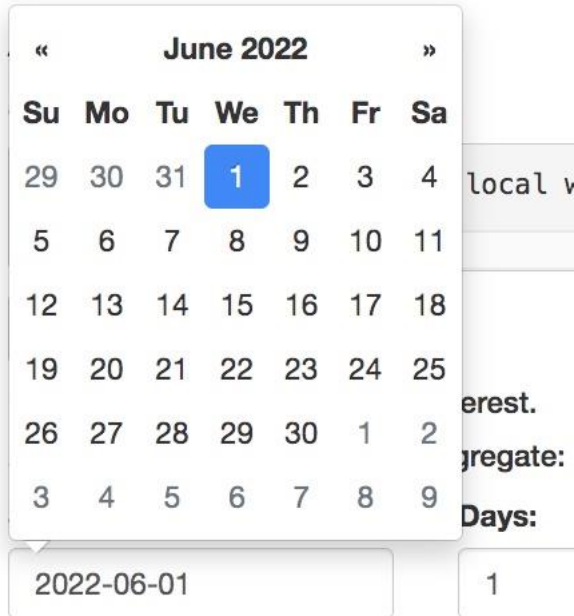
Timestep: 23
0 3 6 9 12 15 18 21 23

Zoom To Data Save Map Image Save Map Movie

Try the following demos:

Demo 1:

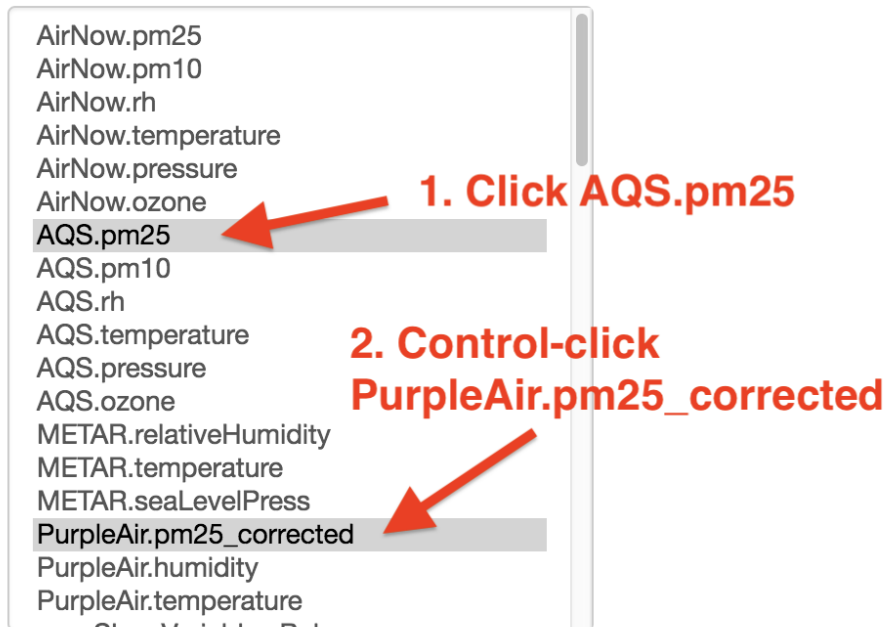
1.1. Use the mouse to select the **Start Date** then type **2022-06-01** and press enter (Make sure to select "hours" for the Timestep dropdown).



The screenshot shows a date selection interface. At the top, there is a calendar for June 2022. The days of the week are listed as Su, Mo, Tu, We, Th, Fr, Sa. The date 1 is highlighted in blue. Below the calendar, there is a text input field containing the date "2022-06-01". To the right of the date input, there is a dropdown menu with the value "1" selected. The dropdown menu is open, showing a list of options: "local v", "erest.", "ggregate:", and "Days:". The "Days:" option is highlighted.

1.2. In the list of variables, **click AQS.pm25** then hold the **control key** down (or **command key on Mac**) and **click PurpleAir.pm25_corrected**. (This allows you to retrieve multiple datasets. Alternatively, select one dataset at a time and click the Retrieve Data button then repeat for another dataset.)

Load Web (control/command-click two):



<https://www2.purpleair.com/pages/attribution>

PurpleAir Key: 3. Enter your PurpleAir key

Purple Air Sites:

 4. Click Retrieve Data

1.3. Enter your PurpleAir Read Key in the text area below the menu.

(Ask for a read key from contact@purpleair.com).

Note: The user will get an error if the user enters/uses an invalid Key.

1.4. Click Retrieve Data.

This will retrieve those datasets using the web service commands shown at the top of the window. These commands are also added to the file `ASNAT/log.txt` for possible use in your other scripts.

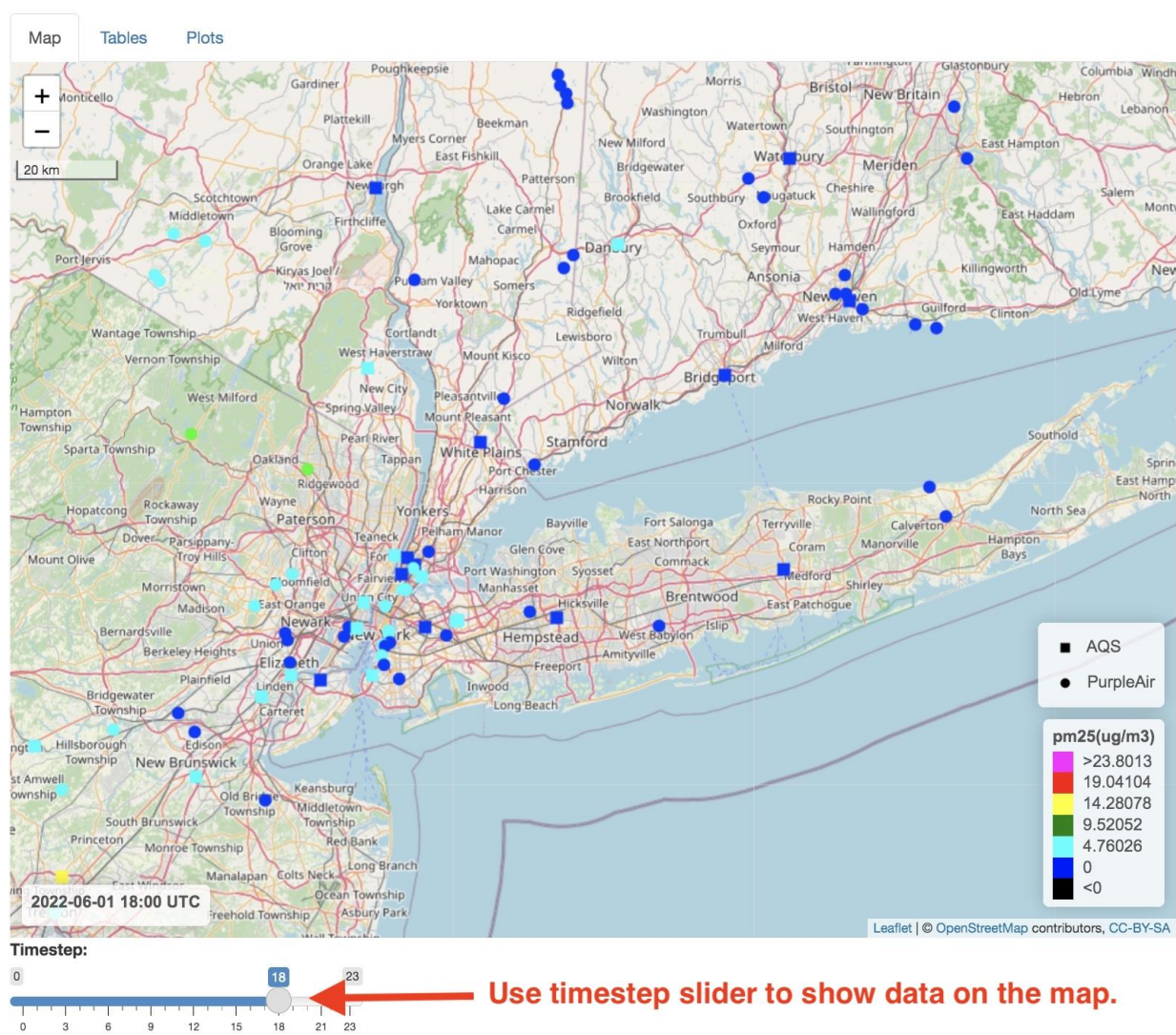
ASNAT

asn@epa.gov, 919-541-5500

Message area shows WCS requests that returned data.

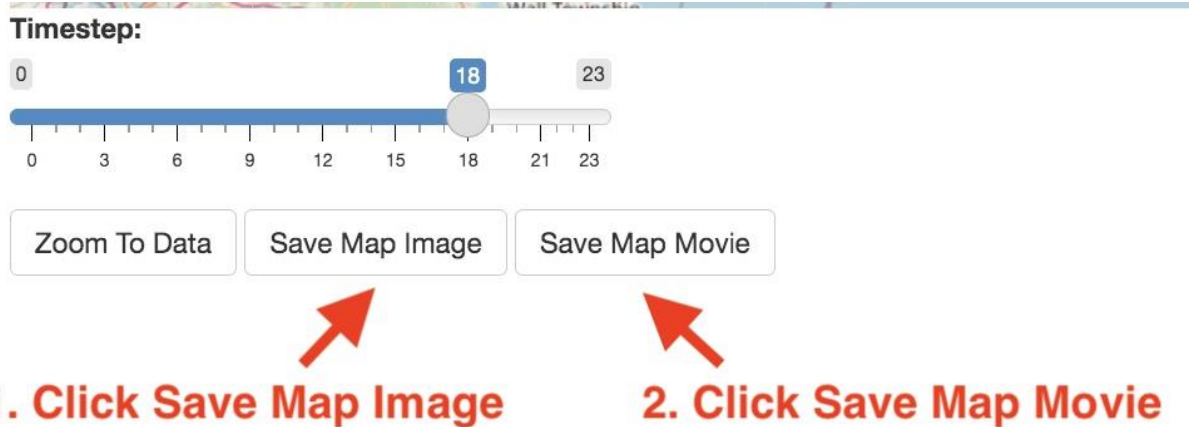
<https://ofmpub.epa.gov/rsig/rsigserver?SERVICE=wcs&VERSION=1.0.0&REQUEST=GetCoverage&FORMAT=asci11&COVERAGE=AQS.pm25&BOX=-74.8635864257813,40.1746762285634,-72.1389770507813,41.718030460>
https://ofmpub.epa.gov/rsig/rsigserver?SERVICE=wcs&VERSION=1.0.0&REQUEST=GetCoverage&FORMAT=asci11&COVERAGE=PurpleAir.pm25_corrected&BOX=-74.8635864257813,40.1746762285634,-72.1389770507813,41.718030460

1.5. Drag the Timestep slider to see mapped data for the given hour.



1.6. Click Save Map Image. This will save the displayed map image in the file `map_image_ttt.png` for timestep 'ttt' in your output directory/folder (your home directory by default).

Note: when running *remote-hosted* ASNAT instead of *locally* on your computer, There is no **Save To Directory** and **Save** buttons are **Download** buttons. This is because generated files (data output, tables, plots, images, movie) are created on the remote host, zipped into a zip file, and downloaded to the local computer. In this case, users will be prompted where to save these zip files. The zip files can then be unzipped to obtain the files.



1.7. Click Save Movie. This will create `map_movie.mp4` containing all timesteps. It will take a couple of minutes to complete. Later you can use a File Browser / Finder to click on these files to view them.

Saving map images (80% done) - please wait...

	map_movie.mp4	 Use File Browser and click to display files.	913 KB	MPEG-4 movie	Today at 8:01 PM
	map_image_018.png		1.1 MB	PNG image	Today at 8:00 PM

1.8. In the **Select Comparison Dataset Y** menu, select **PurpleAir.pm25_corrected_hourly**.

Select Comparison Dataset X:

AQS.pm25 ▼

Select Variable Of Dataset X:

pm25(ug/m3) ▼

Select Comparison Dataset Y:

PurpleAir.pm25_corrected_hourly

Select PurpleAir for Dataset Y



Select Variable Of Dataset Y:

pm25_corrected_hourly(ug/m3) ▼

(Defaults to last variable)

Neighbor distance(m):

1000

**Maximum distance between
time-matched X & Y neighbor points**

1.9. Click the **Tables** tab above the map.

1.10. Click the Summarize and Compare X and Y button to generate tables that summarize and compare AQS and PurpleAir data points.

Use the horizontal and vertical scrollbars to see these three tables.

1. Click Tables tab.

2. Click Summarize button to generate tables shown below.

Dataset Summary: AQS.pm25 vs PurpleAir.pm25_corrected_hourly
 AQS.pm25: 2022-06-01 to 2022-06-01 (-74.8067, -72.9029) (40.2224, 41.5506)

Summary of X vs Y

Station(-)	Timestamp_First_Measurement(UTC)	Timestamp_Last_Measurement(UTC)	Count(-)	Missing(%)	Mean_AQS.pm25(ug/m3)	Min_AQS.pm25(ug/m3)	P25_AQS.pm25(ug/m3)	Median_AQS.pm25(ug/m3)
90011123	2022-06-01T15:00:00-0000	2022-06-01T23:00:00-0000	9	62	4.10	3.00		3.60
340210008	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	12.30	5.10		9.90
360050080	2022-06-01T05:00:00-0000	2022-06-01T23:00:00-0000	19	21	2.40	0.00		1.30

Dataset Summary: PurpleAir.pm25_corrected_hourly vs AQS.pm25
 PurpleAir.pm25_corrected_hourly: 2022-06-01 to 2022-06-01 (-74.7602, -72.6266) (40.2183, 41.6962)

Summary of Y vs X

Station(-)	Timestamp_First_Measurement(UTC)	Timestamp_Last_Measurement(UTC)	Count(-)	Missing(%)	Mean_PurpleAir.pm25_corrected_hourly(ug/m3)	Min_PurpleAir.pm25_corrected_hourly(ug/m3)	P25_PurpleAir.pm25_corrected_hourly(ug/m3)	Median_PurpleAir.pm25_corrected_hourly(ug/m3)
91899	2022-06-01T00:01:08-0000	2022-06-01T23:00:16-0000	16	33		5.70		3.10
103376	2022-06-01T00:00:35-0000	2022-06-01T03:00:37-0000	4	83		6.30		4.00
131197	2022-06-01T00:00:59-0000	2022-06-01T22:07:09-0000	22	8		6.80		2.40

Neighboring Points (<= 1000m) Comparison:
 AQS.pm25 vs PurpleAir.pm25_corrected_hourly
 2022-06-01 (-74.8067, -72.9029) - (40.2224, 41.5506)

X vs Y time-matched neighboring point measures.

AQS.pm25(ug/m3)	PurpleAir.pm25_corrected_hourly(ug/m3)
7.60	8.08
11.60	10.05
12.20	11.30
17.50	12.85
15.30	13.72
18.60	14.22
3.80	5.23
19.50	12.41

Use scrollbars to see complete tables of statistics.

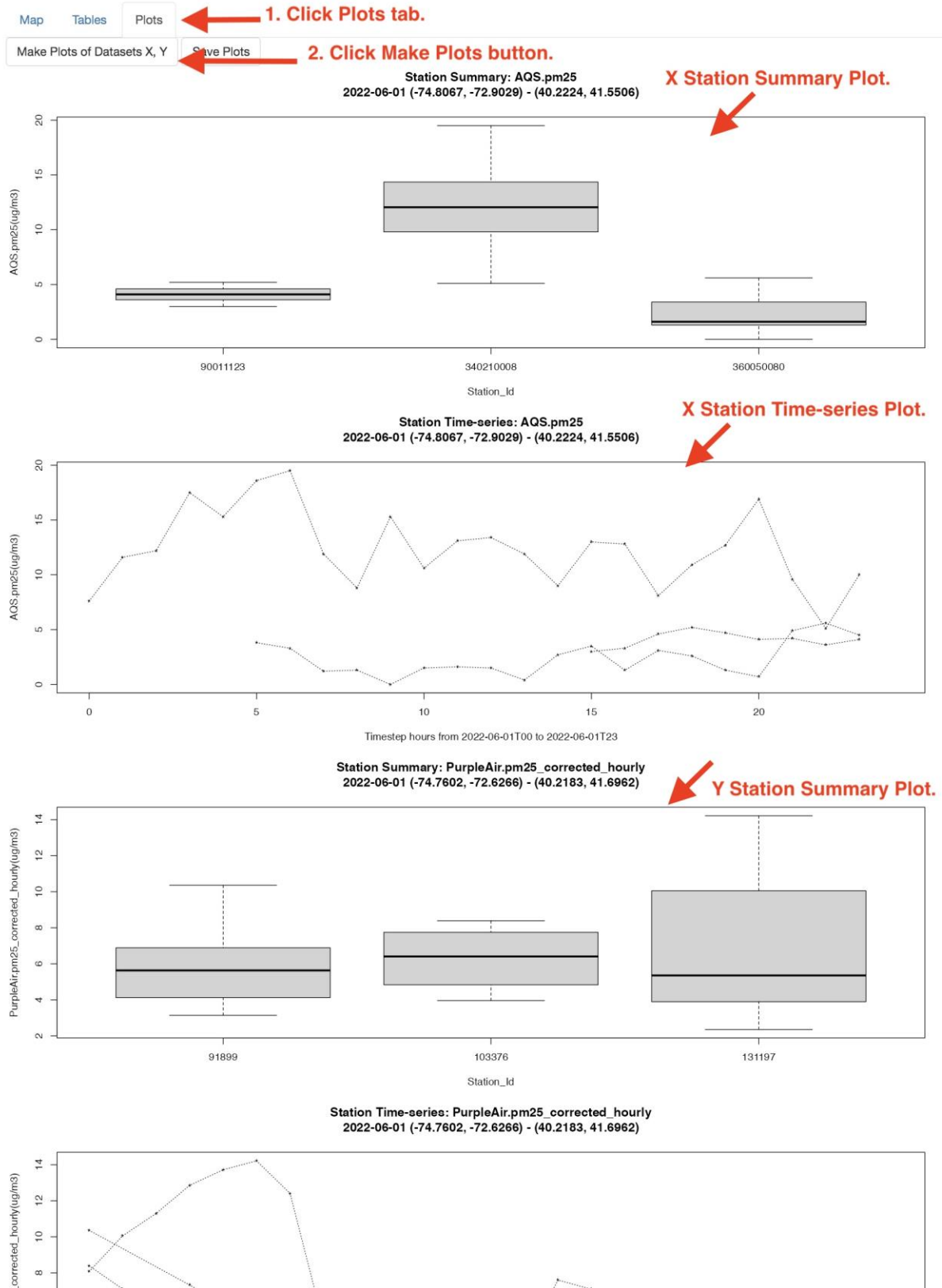
m3	P75_AQS.pm25(ug/m3)	Max_AQS.pm25(ug/m3)	Nearest_Y_Station(-)	Nearest_Y_Distance(m)
4.10	4.60	5.20	103376	17
2.10	13.90	19.50	131197	522
1.60	3.40	5.60	91899	672

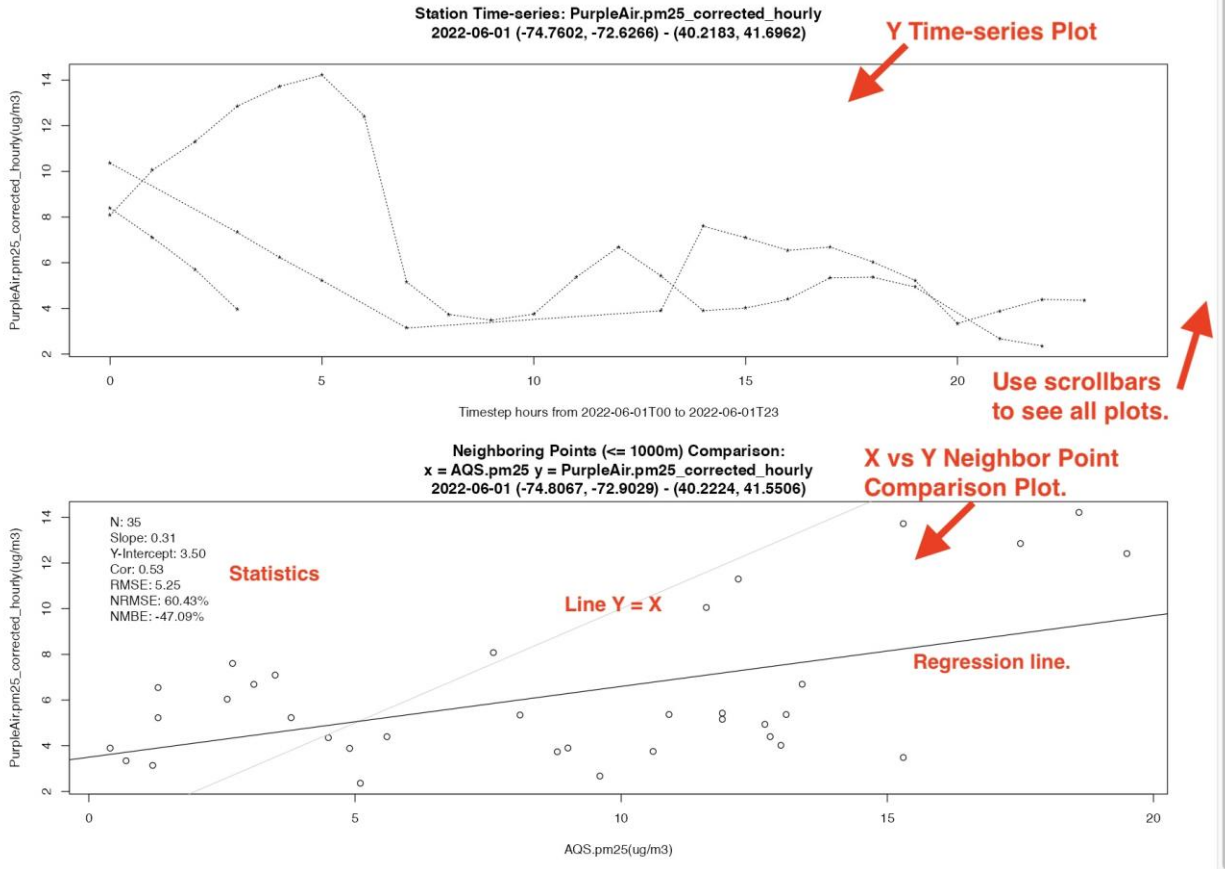
PurpleAir.pm25_corrected_hourly(ug/m3)	Median_PurpleAir.pm25_corrected_hourly(ug/m3)	P75_PurpleAir.pm25_corrected_hourly(ug/m3)	Max_PurpleAir.pm25_corrected_hourly(ug/m3)	Nearest_Y_Station(-)	Nearest_Y_Distance(m)
4.20		5.60	6.80	10.40	360050080
5.30		6.40	7.40	8.40	90011123
3.90		5.40	9.60	14.20	340210008

1.11. Click the Plots tab above the map.

1.12. Click the Make Plots of Datasets X, Y button to generate plots that summarize and compare AQS and PurpleAir data points.

Use the vertical scrollbars to see these five plots.





1.13. Click the **Save Plots** button to save pdf files of the plots.

You can use a File Browser / Finder to click on these files to view them.

/Users/plessel/AQS_pm25_boxplot.pdf

/Users/plessel/AQS_pm25_timeseries.pdf

2. Plot files saved.

1. Click Save Plots button.

Reset map

1. Pan/zoom map to region of interest.

2. Start Date,Days to retrieve,Aggregate:

Map Tables Plots

Make Plots of Datasets X, Y

Save Plots

1.14. Scroll to the **bottom** of the window and click the **Save Data** button (along the left edge of the window).

This will immediately save the datasets and the tables as plain text files in your home directory.

Scroll to the top of the window to see the list of files created.

File formats include tsv (tab-separated values), csv (comma-separated values), and json (JSON format). The tsv and csv files can be read into a spreadsheet program for further analysis.

Output File Format: tsv ▼

Default format is tab-delimited.

Save Data Quit

1. Click Save Data button.

ASNAT

asnat@epa.gov, 919-541-5500

```
/Users/plesse/AQS_pm25_2022-06-01_to_2022-06-01.tsv  
/Users/plesse/PurpleAir_pm25_corrected_hourly_2022-06-01_to_2022-06-01.tsv
```

Saved data files.

Later, they can also be re-loaded into ASNAT using the
'Load data from a standard-format file' Browse button.

1.15. Click the Quit button to quit the program and close the web browser tab.

Output File Format: tsv ▼

Save Data Quit

Click Quit button

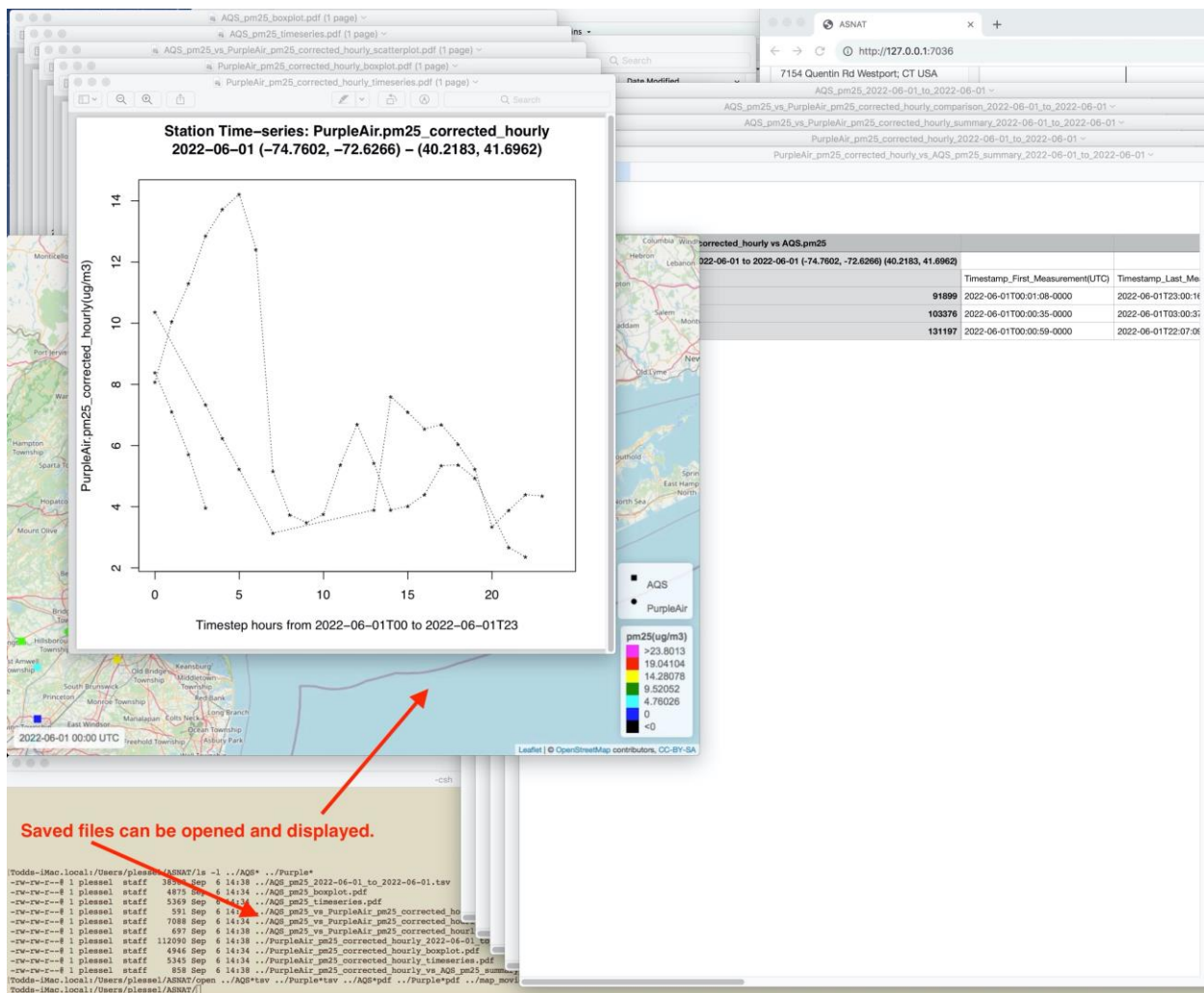
1.16 List the various output files generated:

Windows: `dir ..\*_to_*sv`

Mac/Linux: `ls -l ../*_to_*sv ../*.pdf`

On Mac these files can be displayed using the open command:

`open ../*_to_*sv ../*.pdf`

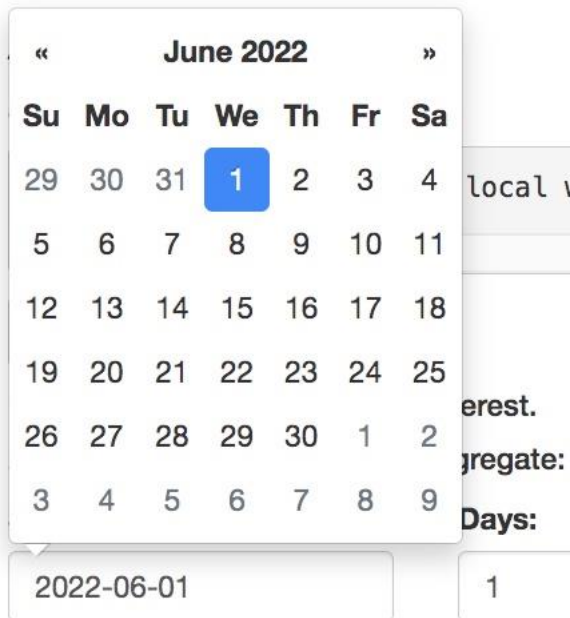


Demo 2: Load data from files:

2.1 Rerun ASNAT:

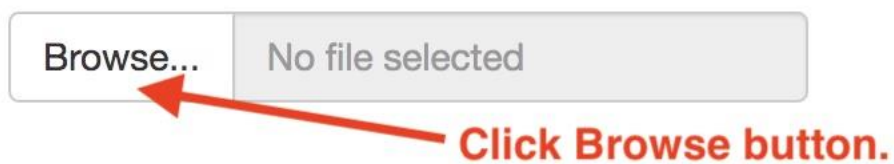
ASNAT

2.2. If the Start Date is not already 2022-06-01 then use the mouse to select the **Start Date** then type **2022-06-01** and press enter.



2.3 Scroll down to **Load data from a standard-format file:** and click **Browse**.

Load data from a standard-format file:



2.4 Navigate up one level to your home directory.



2.5 Select the previously saved (in Demo 1) file

AQS_pm25_2022-06-01_to_2022-06-01.tsv and click Open (or double-click or press return).

Load data from a standard-format file:

Browse...	AQS_pm25_2022-06-01_to_2022-06-01.tsv
Upload complete	

Load completed



2.6 Click Load data from sensor instrument files.

Load data from sensor instrument files
--

Click Load data...



2.7 Click Browse...

Load Data From Files

Load Files: **Click Browse...**

Browse... No file selected

Sensor longitude:

0

Sensor latitude:

0

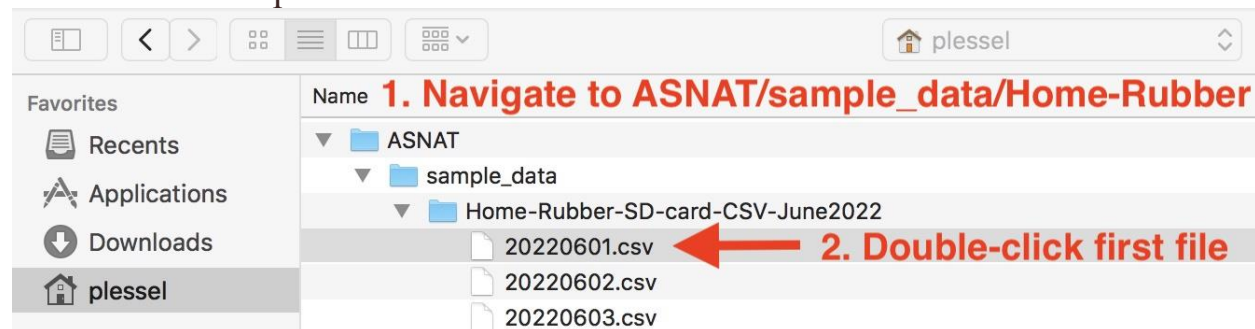
Dataset Name:

MySensor

Cancel Process and Load Selected Files

2.8 Navigate into `sample_data/Home-Rubber-SD-card-CSV-June2022`.

This contains PurpleAir sensor instrument files.



2.9 Double-click the file `20220601.csv`.

2.10 Since the file did not contain the location of the sensor, enter the
Sensor longitude **-74.05**
Sensor latitude **40.72**

Load Data From Files

Load Files:

Browse...	20220601.csv
Upload complete	

← **1. Load completed.**

**Sensor
longitude:**

← **2. Press tab key then enter longitude**

**Sensor
latitude:**

← **3. Press tab key then enter latitude**

Dataset Name:

← **4. Press tab key then enter HomeRubber**

5. Click Process button

Cancel	Process and Load Selected Files
--------	---------------------------------

2.11 Change the name of the dataset to **HomeRubber**.

2.12 Click the **Process and Load Selected Files** button (or **tab, tab enter**)

2.13 Change Select Comparison Dataset Y: from none to **HomeRubber.hourly_data**.

Note that the 'Select Variable Of Dataset Y' menu automatically selects the last variable **pm25_corrected(ug/m3)** which is *computed by ASNAT upon import*.

Select Comparison Dataset Y:

HomeRubber.hourly_data ▼

1. Select HomeRubber as Y

Select Variable Of Dataset Y:

pm25_corrected(ug/m3) ▼

Defaults to last variable

Neighbor distance(m):

1000

Default distance to neighbors

2.14 Click the Tables tab above the map.

2.15 Click the Summarize and Compare X and Y button.

1. Click Tables tab

2. Click Summarize button to generate tables below

Dataset Summary: AQS.pm25 vs HomeRubber.pm25_corrected **AQS vs HomeRubber**
 AQS.pm25: 2022-06-01 to 2022-06-01 (-74.8067, -72.9029) (40.2224, 41.5506)

Station(-)	Timestamp_First_Measurement(UTC)	Timestamp_Last_Measurement(UTC)	Count(-)	Missing(%)	Mean_AQ
340171003	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	

Dataset Summary: HomeRubber.pm25_corrected vs AQS.pm25 **HomeRubber vs AQS**
 HomeRubber.pm25_corrected: 2022-06-01 to 2022-06-01 (-74.05, -74.05) (40.72, 40.72)

Station(-)	Timestamp_First_Measurement(UTC)	Timestamp_Last_Measurement(UTC)	Count(-)	Missing(%)	Mean_Hc
202438586	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	

Neighboring Points (<= 1000m) Comparison: AQS.pm25 vs HomeRubber.pm25_corrected **AQS vs HomeRubber neighboring points**
 2022-06-01 (-74.8067, -72.9029) - (40.2224, 41.5506)

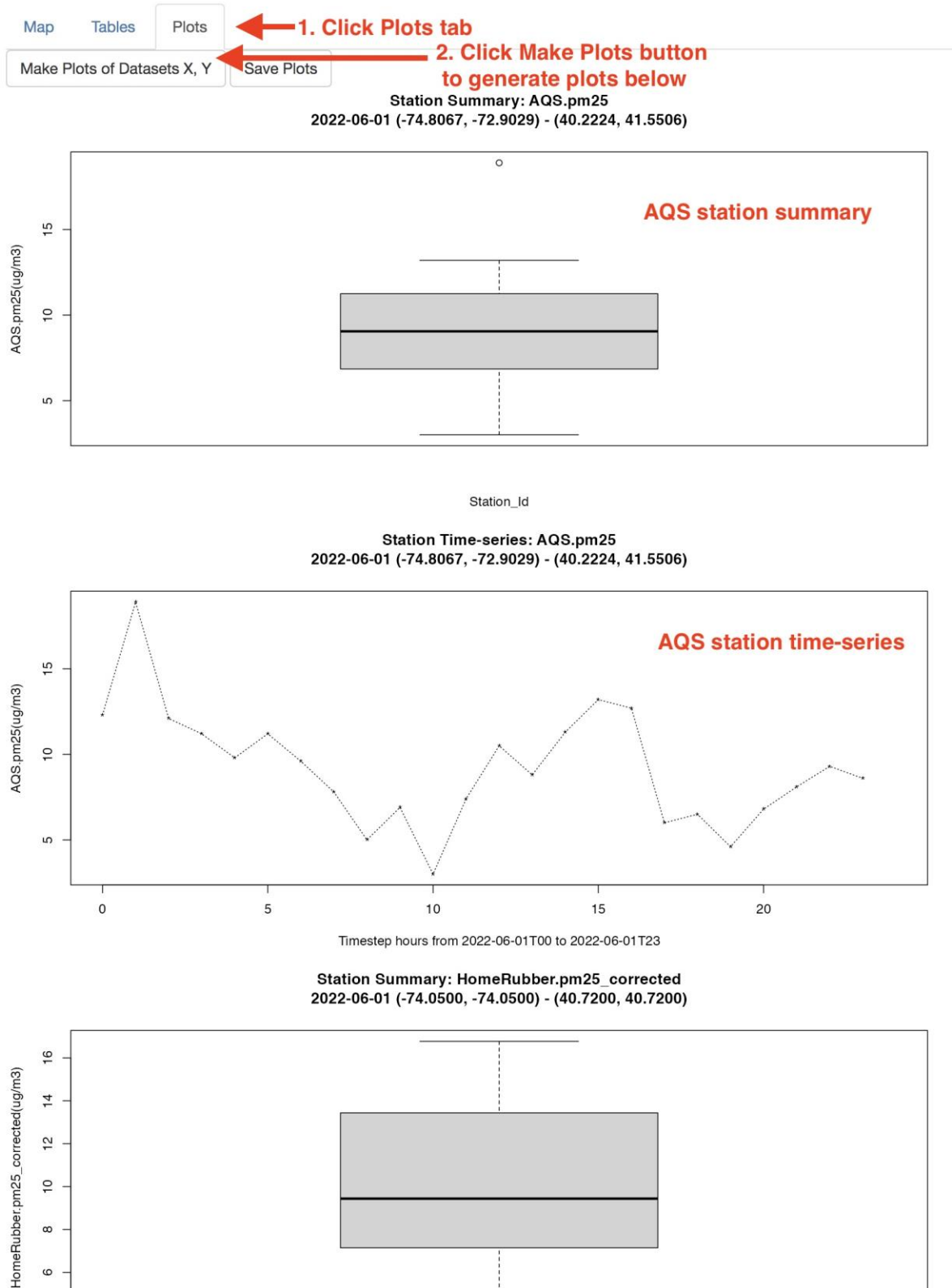
AQS.pm25(ug/m3)	HomeRubber.pm25_corrected(ug/m3)
12.30	11.31
18.90	13.94
12.10	14.76
11.20	15.76
0.00	16.52

Use scrollbars to see full table

2.16 See that there is one pair of sites within 630 m apart and their paired pm25 measures.

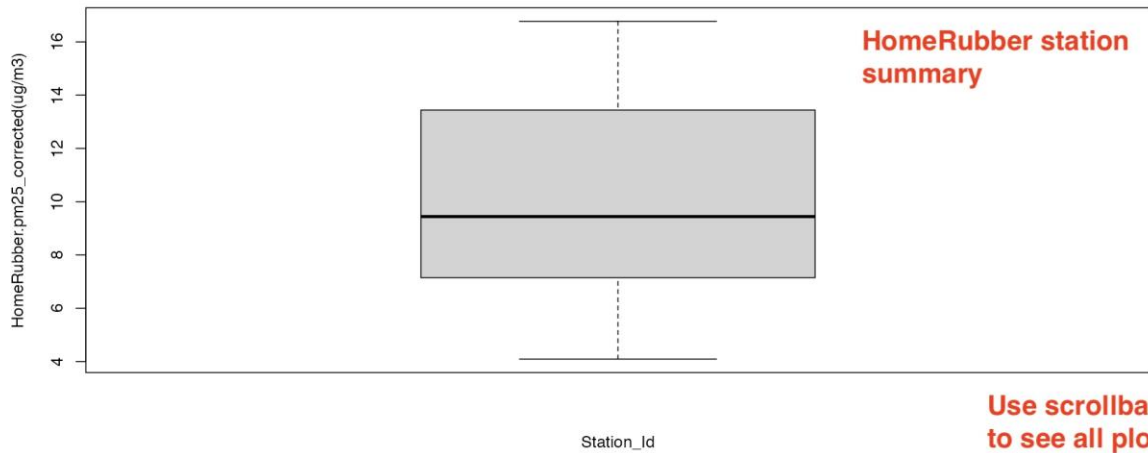
2.17 Click the **Plots** tab above the map.

2.18 Click the **Make Plots of Datasets X, Y** button.

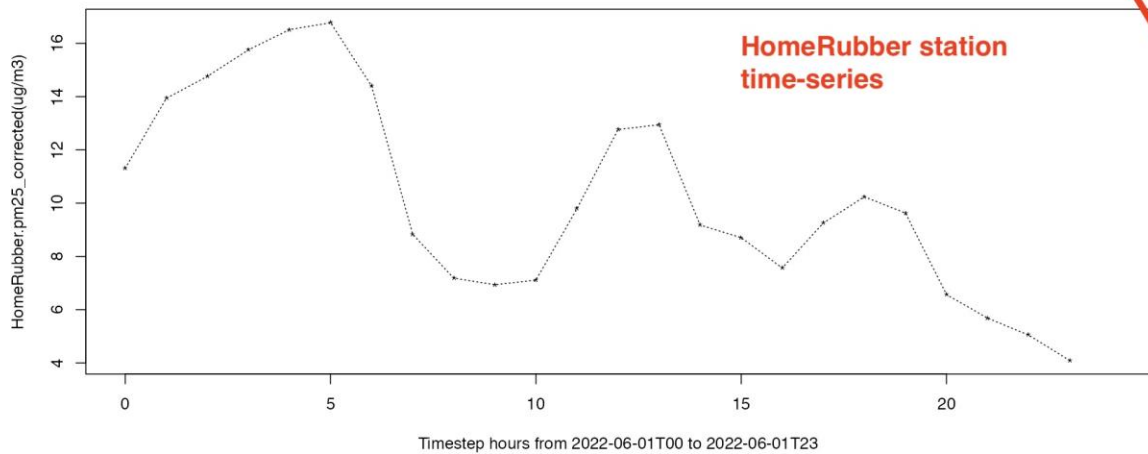


2.19 Scroll to see the five plots.

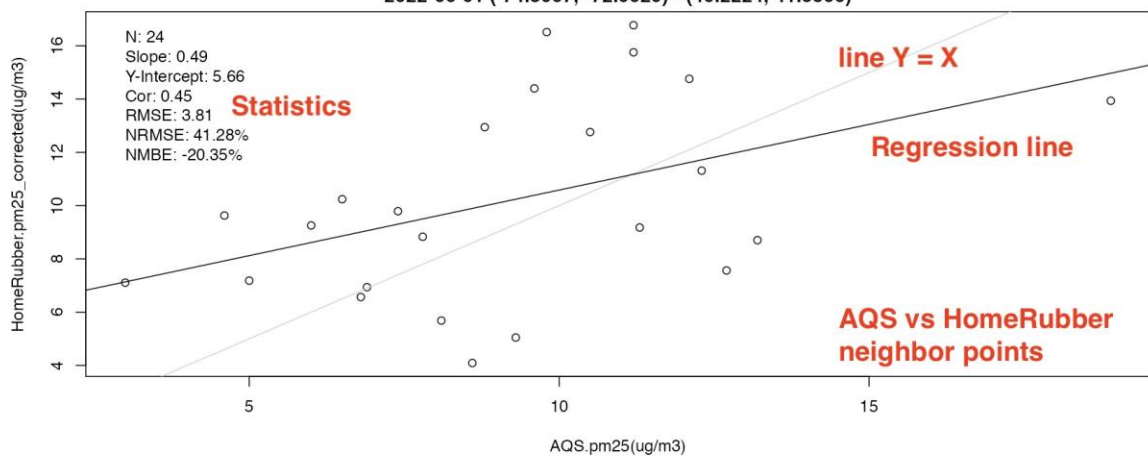
Station Summary: HomeRubber.pm25_corrected
2022-06-01 (-74.0500, -74.0500) - (40.7200, 40.7200)



Station Time-series: HomeRubber.pm25_corrected
2022-06-01 (-74.0500, -74.0500) - (40.7200, 40.7200)



Neighboring Points (<= 1000m) Comparison:
x = AQS.pm25 y = HomeRubber.pm25_corrected
2022-06-01 (-74.8067, -72.9029) - (40.2224, 41.5506)



2.20 Click the Save Plots button.

The screenshot shows a file list at the top with two entries: `/Users/plessel/AQS_pm25_boxplot.pdf` and `/Users/plessel/AQS_pm25_timeseries.pdf`. A red arrow points from the text "Saved plot files" to these entries. Below the file list is a "Reset map" button. To the right are three tabs: "Map", "Tables", and "Plots". The "Plots" tab is active. Below the tabs is a button labeled "Make Plots of Datasets X," and to its right is a "Save Plots" button. A red arrow points from the text "1. Click Save Plots button" to the "Save Plots" button.

/Users/plessel/AQS_pm25_boxplot.pdf
/Users/plessel/AQS_pm25_timeseries.pdf ← **Saved plot files**

Reset map

1. Click Save Plots button

Map Tables Plots

Make Plots of Datasets X, Save Plots

1. Pan/zoom map to region of interest.
2. Start Date,Days to retrieve,Aggregate:

2.21 Click the Save Data button.

2.22 Click the Quit button.

The screenshot shows an "Output File Format:" label followed by a dropdown menu with "tsv" selected. Below this are two buttons: "Save Data" and "Quit". A red arrow points from the text "1. Click Save Data button" to the "Save Data" button. Another red arrow points from the text "2. Click Quit button" to the "Quit" button.

Output File Format: tsv ▼

Save Data Quit

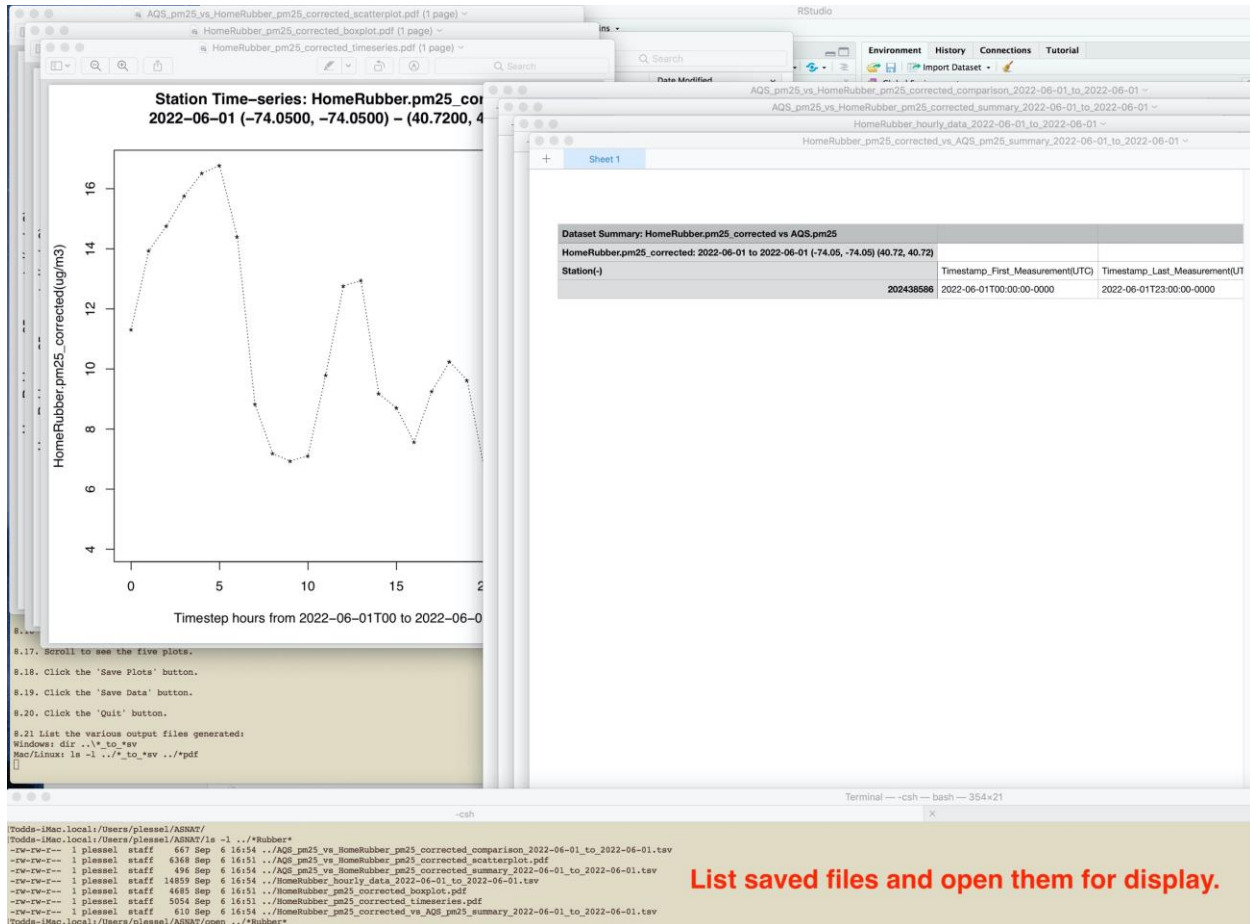
1. Click Save Data button

2. Click Quit button

2.23 List the various output files generated:

Windows: `dir ..\*_to_*sv`

Mac/Linux: `ls -l ../*_to_*sv ../_*.pdf`



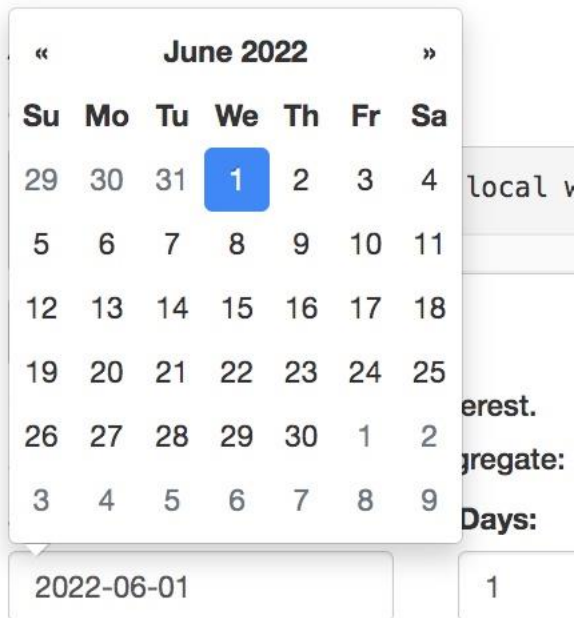
List saved files and open them for display.

Demo 3: Select single PurpleAir sensor:

3.1 Rerun ASNAT:

ASNAT

3.2 If the Start Date is not already 2022-06-01 then use the mouse to select the **Start Date** then type **2022-06-01** and press enter.



June 2022						
Su	Mo	Tu	We	Th	Fr	Sa
29	30	31	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	1	2
3	4	5	6	7	8	9

2022-06-01

1

3.3 Click **+** in the top-left corner of the map or use the **mouse middle button or wheel to zoom in to Hoboken** and **drag the mouse left button to pan the map view as shown.**

Note the list of Purple Air Sites updates to show only those sites within the view area.

map 57367m x 57367m [-74.418423, -73.737271] [40.532469, 40.919619] [-73.990868, 40.762948] ← The message area shows the viewing area dimensions.

Reset map

1. Pan/zoom map to region of interest.
2. Start Date, Days to retrieve, Aggregate:

Start Date: 2022-06-01 Days: 1 Timestep: hours

Delete Loaded Data

Load Web (control/command-click two):

- AirNow.pm25
- AirNow.pm10
- AirNow.rh
- AirNow.temperature
- AirNow.pressure
- AirNow.ozone
- AQS.pm25
- AQS.pm10
- AQS.rh
- AQS.temperature
- AQS.pressure
- AQS.ozone
- METAR.relativeHumidity
- METAR.temperature
- METAR.seaLevelPress
- PurpleAir.pm25_corrected
- PurpleAir.humidity
- PurpleAir.temperature

PurpleAir Key: your_purpleair_read_key

Purple Air Sites:

- 0 All sensors in view
- 22749 Hoboken, NJ
- 28285 Southwest Hoboken
- 29183 Hell's Kitchen

1. Use mouse middle button/wheel or map +/- controls to zoom in to Hoboken as shown.

2. Drag mouse left button to pan view area.

Note the list of PurpleAir sensors updates to show only those in the view area.

3.4 Click the Delete Loaded Data button to delete previously cached data/files before retrieving data for a new viewing area.

Delete Loaded Data ← Click Delete button. This deletes previously cached data/files before retrieving data from a new viewing area.

Load Web (control/command-click two):

Note: temporary data files retrieved from web services are written to ASNAT/data/tmp

and are named by data source, variable and time range. For example:

```
/Users/plessel/ASNAT/ls -l data/tmp
```

```
plessel staff 3303 Sep 6 17:22 AQS_pm25_2022-06-01_to_2022-06-01.txt
```

```
plessel staff 878 Sep 6 17:22 PurpleAir_pm25_corrected_hourly_2022-06-01_to_2022-06-01.txt
```

Subsequent web retrievals are avoided if the file is already cached.

However, since the file name does not encode the viewing bounds, these *cached files must be deleted before retrievals for a different viewing area.*

3.5 Select AQS.pm25 and control-click (command-click on Mac) on **PurpleAir.pm25_corrected**.

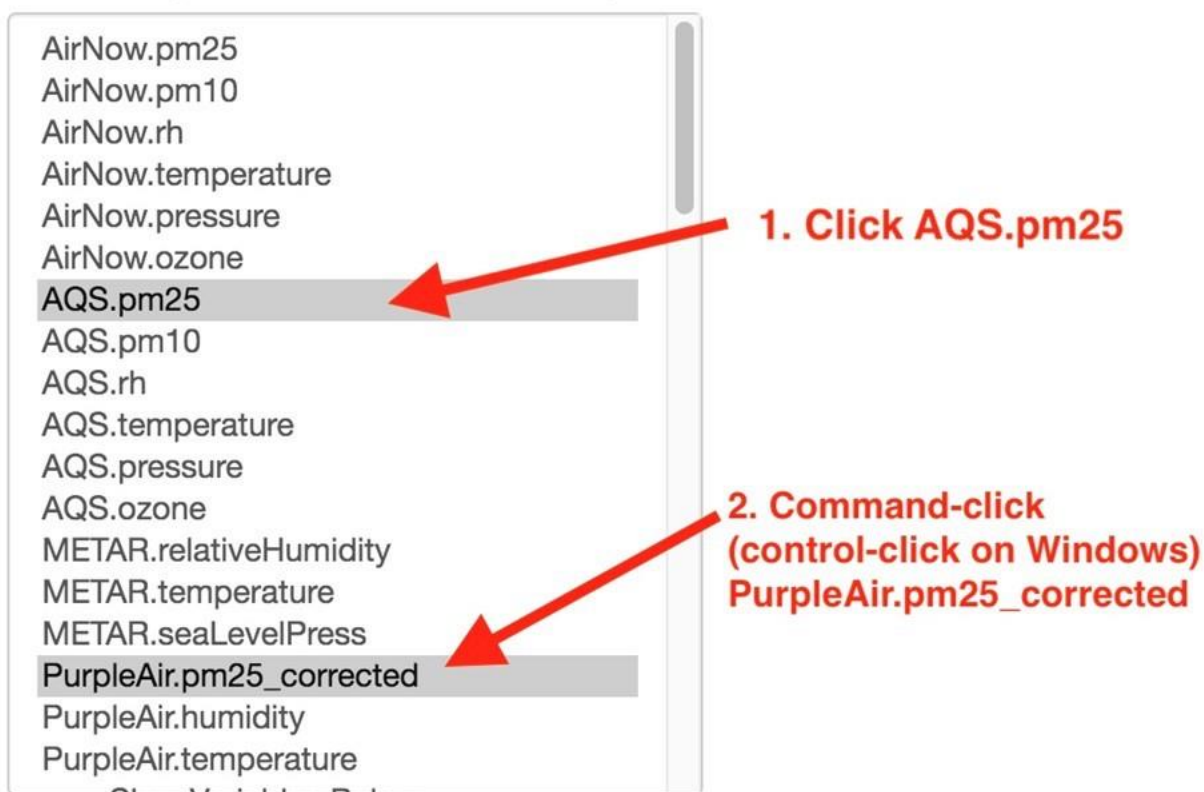
3.6 Enter your PurpleAir read key (obtained from contact@purpleair.com).

3.7 Click on Purple Air site 22749 Hoboken; NJ.

The default is *0 All sensors in view*, but in this demo we want to compare AQS to a single PurpleAir sensor.

3.8 Click Retrieve Data button to retrieve AQS and PurpleAir data. Note the message are shows the retrieved PurpleAir data is only for SENSOR=22749.

Load Web (control/command-click two):



PurpleAir Key:



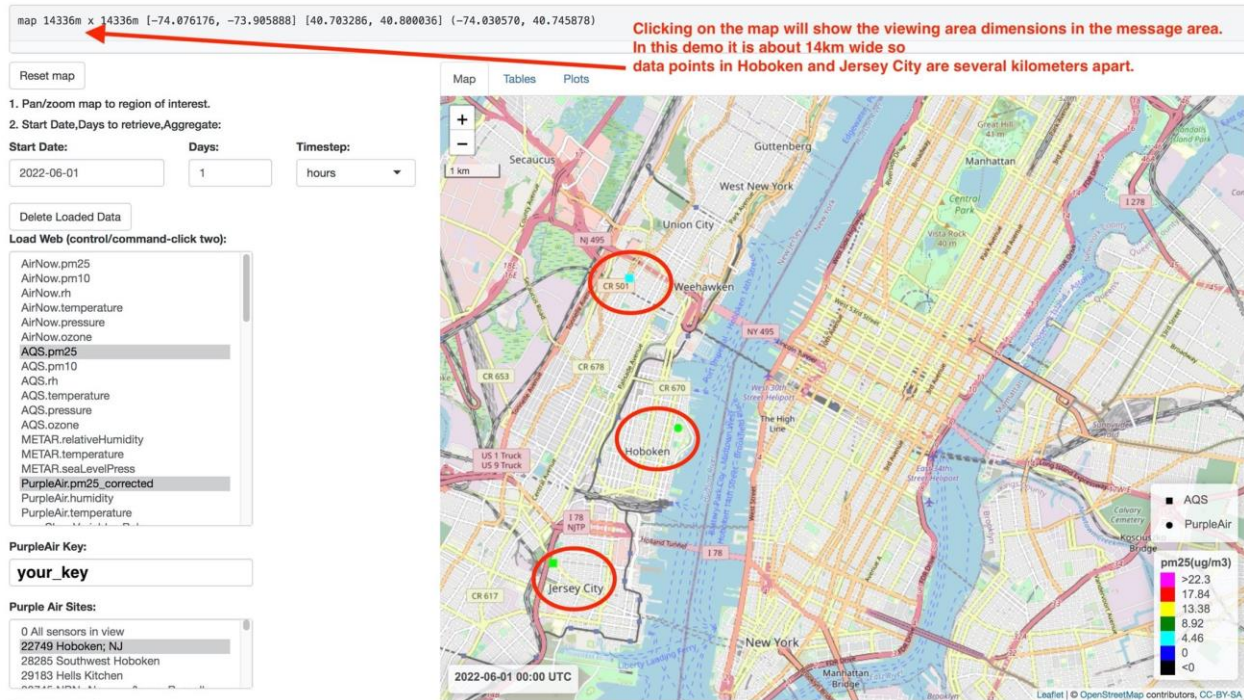
Purple Air Sites:



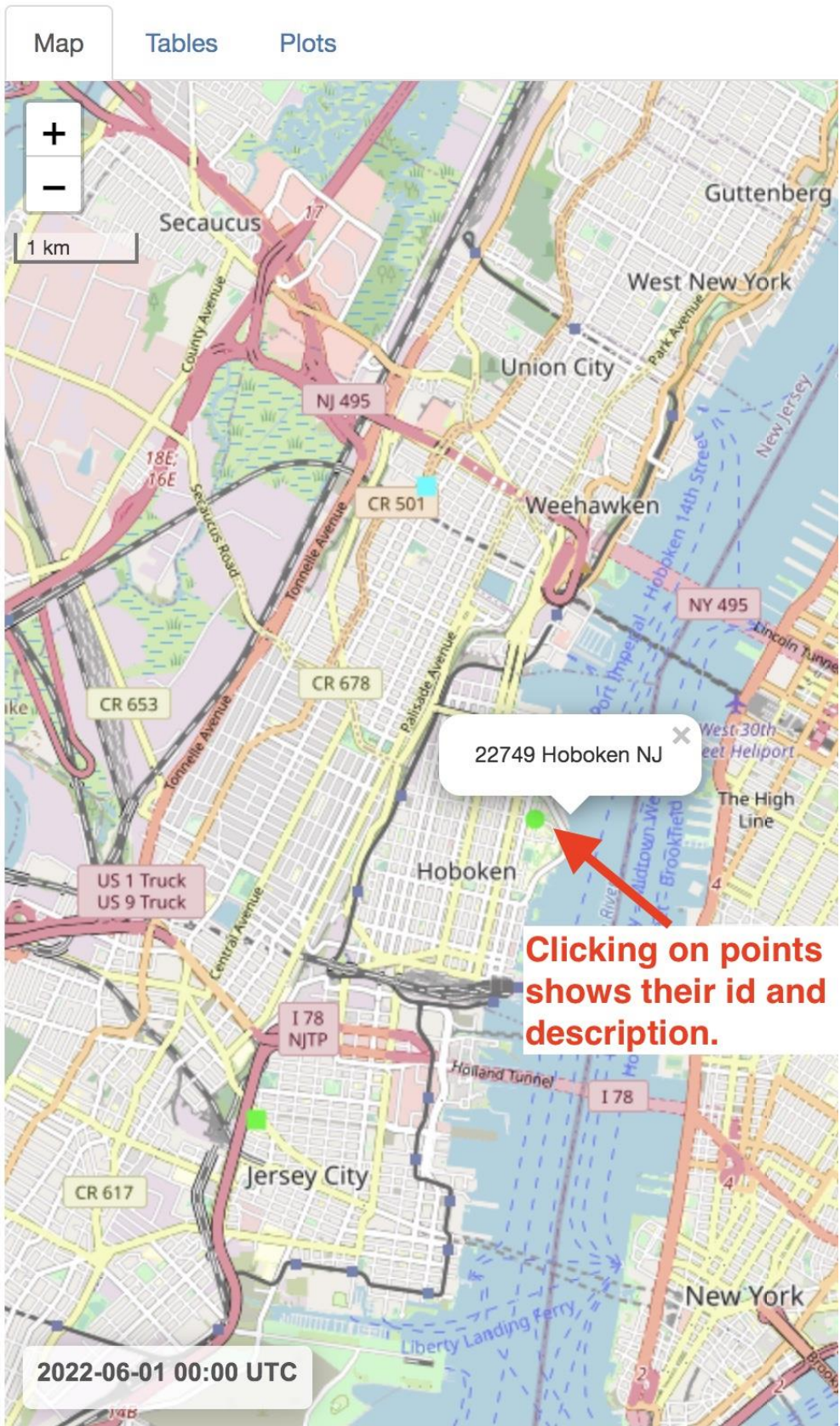
Note PurpleAir data retrieval is for sensor 22749 only.

```
i1&COVERAGE=AQS.pm25&BBOX=-74.0842437744141,40.6880578838636,-73.9139556884766,40.7848307950502,&TIME=2022-06-01T00:00:00Z/2022-06-01T23:59:59Z&FILTER_MISSING=1
```

3.9 Clicking on the map shows the viewing are dimensions in the message area. From this we can see that the data points are several kilometers apart.



3.10 Clicking on a data point in the map will show its *ID* and *description*.
(Caution: AQS data are vetted, but not all PurpleAir descriptions are reasonable.)



Timestep:

3.11 Select PurpleAir.pm25_corrected_hourly for Select Comparison Dataset Y.

The variable defaults to the single or last variable available.

3.12 Enter 5000 for the Neighbor distance since it is clear from the map that the PurpleAir site is several kilometers from the two AQS sites.

Select Comparison Dataset Y:

PurpleAir.pm25_corrected_hourly  **1. Select PurpleAir for Y**

Select Variable Of Dataset Y:

pm25_corrected_hourly(ug/m3)  **Defaults to single variable retrieved.**

Neighbor distance(m): 5000  **2. Enter 5000 for Neighbor distance**

3.13 Click the Tables tab.

3.14 Click the Summarize and Compare X and Y button to generate tables.

Note that there are two AQS sites that are neighbors of the single PurpleAir site. This results in two AQS measures for the single hour-matched PurpleAir measure as shown in the Neighboring Points Comparison Table.

Map Tables ← **1. Click Tables tab**

Summarize and Compare X and Y ← **2. Click Summarize button to generate tables below**

Dataset Summary: AQS.pm25 vs PurpleAir.pm25_corrected_hourly
 AQS.pm25: 2022-06-01 to 2022-06-01 (-74.0523, -74.0362) (40.7254, 40.7709)

Station(-)	Timestamp_First_Measurement(UTC)	Timestamp_Last_Measurement(UTC)	Count(-)	Missing(%)	Mean_AQS.pm25(ug/m3)	Min_
340170008	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	7.10	
340171003	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	9.20	

Dataset Summary: PurpleAir.pm25_corrected_hourly vs AQS.pm25
 PurpleAir.pm25_corrected_hourly: 2022-06-01 to 2022-06-01 (-74.0259, -74.0259) (40.747, 40.747)

Station(-)	Timestamp_First_Measurement(UTC)	Timestamp_Last_Measurement(UTC)	Count(-)	Missing(%)	Mean_PurpleAir.pm25_corrected_h
22749	2022-06-01T00:01:11-0000	2022-06-01T23:00:49-0000	10	58	

Neighboring Points (<= 5000m) Comparison:
 AQS.pm25 vs PurpleAir.pm25_corrected_hourly
 2022-06-01 (-74.0523, -74.0362) - (40.7254, 40.7709)

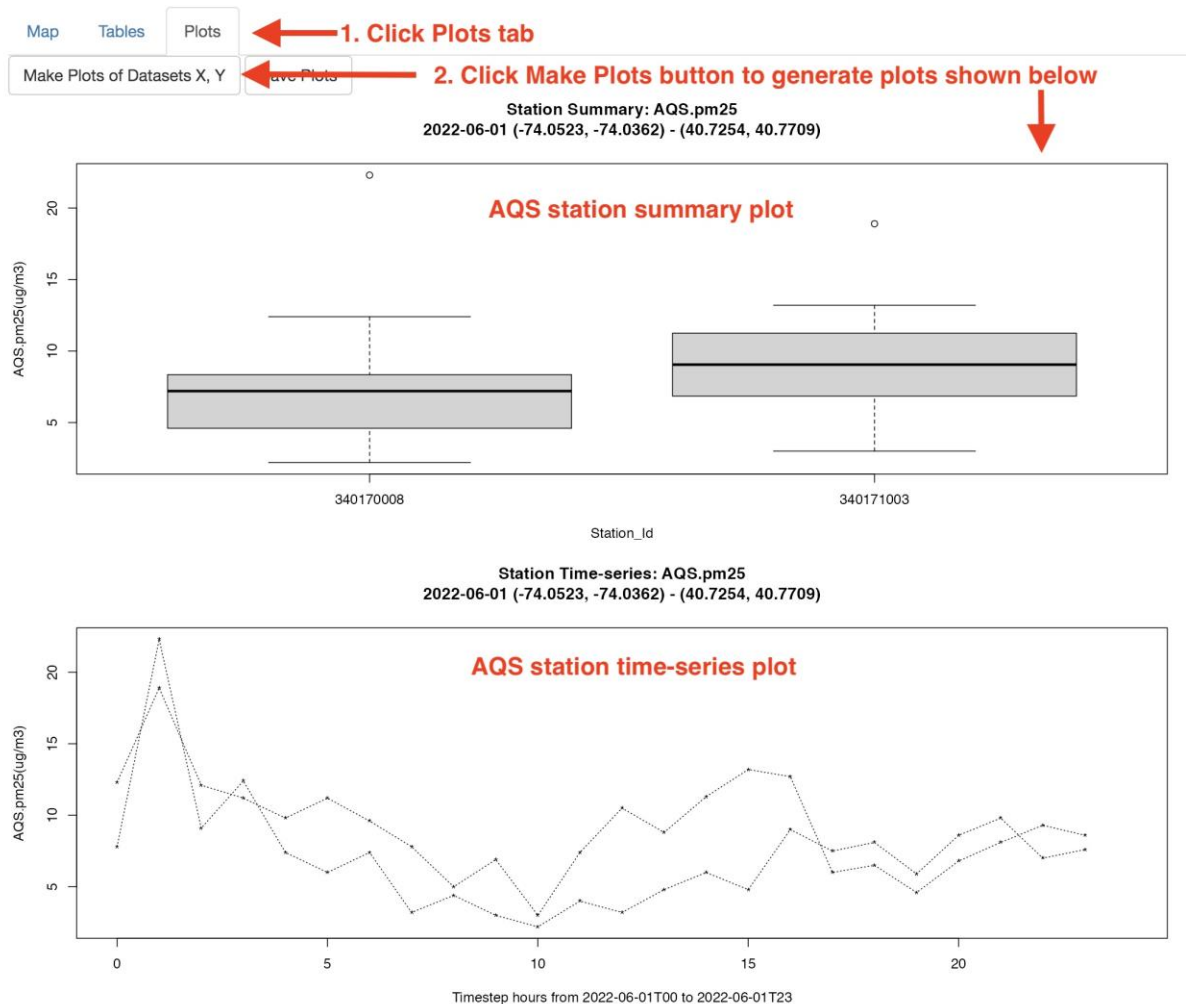
AQS.pm25(ug/m3)	PurpleAir.pm25_corrected_hourly(ug/m3)
7.80	10.81
12.30	10.81
12.40	8.12
11.20	8.12
7.40	7.27
9.80	7.27
6.00	6.51
11.20	6.51
3.20	4.42
7.80	4.42

Note there are two AQS stations within 5,000m of the single PurpleAir sensor 22749.

This yields two AQS measures per hour-matched PurpleAir measure as seen in the Neighboring Points Comparison Table.

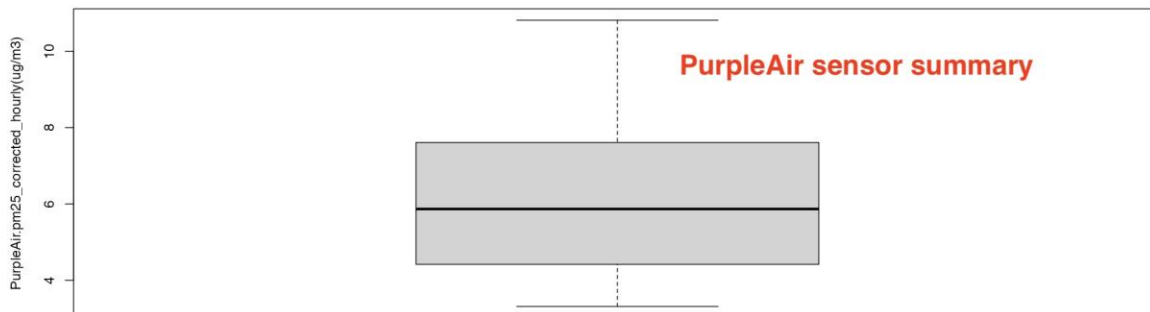
3.15 Click the Plots tab.

3.16 Click the Make Plots of Datasets X, Y button to generate the plots.



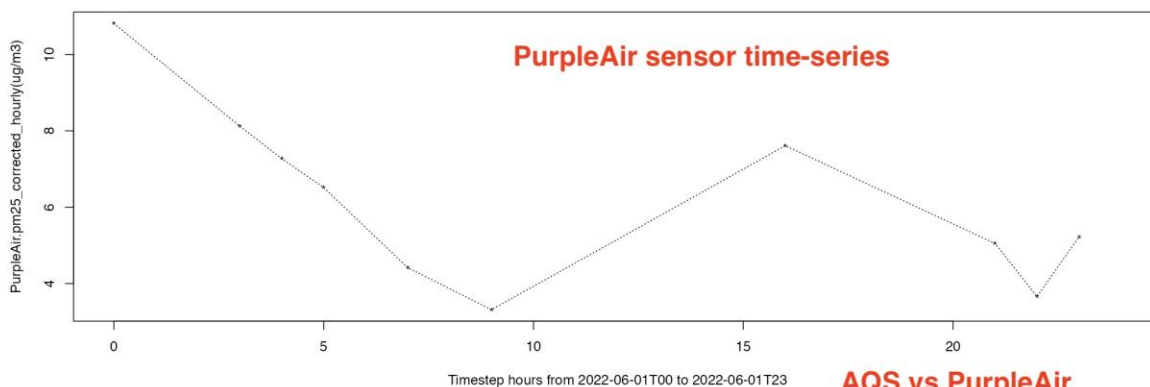
3.17 Use the **scrollbar** to see all five plots.
(Extending to 30 days takes a few minutes to retrieve,
yields many more comparison points and slightly better correlation.)

Station Summary: PurpleAir.pm25_corrected_hourly
2022-06-01 (-74.0259, -74.0259) - (40.7470, 40.7470)



Station_Id

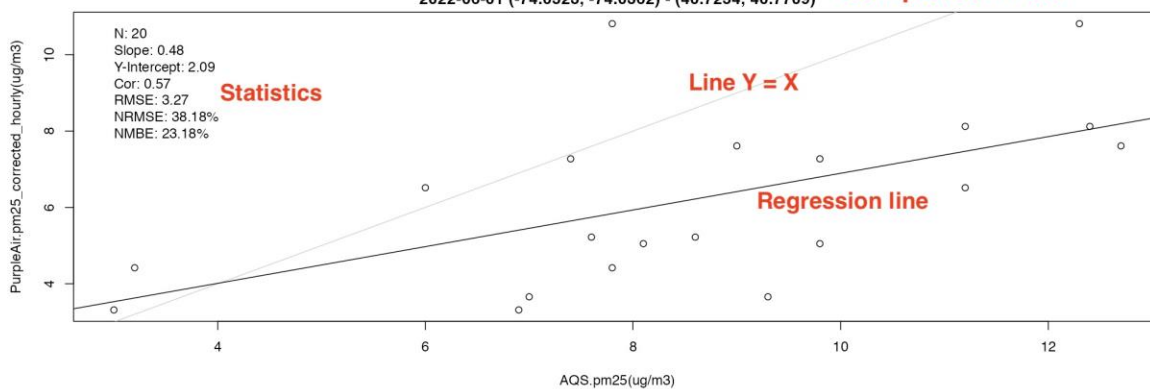
Station Time-series: PurpleAir.pm25_corrected_hourly
2022-06-01 (-74.0259, -74.0259) - (40.7470, 40.7470)



1. Use scrollbar
to see all plots.

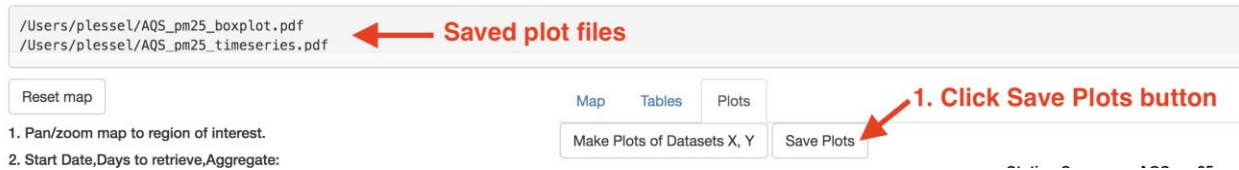
Neighboring Points (<= 5000m) Comparison:
x = AQS.pm25 y = PurpleAir.pm25_corrected_hourly
2022-06-01 (-74.0523, -74.0362) - (40.7254, 40.7709)

AQS vs PurpleAir
Neighboring Points
Comparison Plot



3.18 Click Save Plots button.

Note the names of the saved files in the message area.
(Scroll for complete list.)



3.19 Click Save Data button to save the retrieved data files and tables as plain text (tab-delimited) files.

Output File Format: tsv ▼

Save Data Quit


Click Save Data button

ASNAT

asn@epa.gov, 919-541-5500

Saved data files

`/Users/plessel/AQS_pm25_2022-06-01_to_2022-06-01.tsv`
`/Users/plessel/PurpleAir_pm25_corrected_hourly_2022-06-01_to_2022-06-01.tsv`

3.20 Click Quit button to quit ASNAT.

Output File Format: tsv ▼

Save Data Quit

 **Click Quit button**

Demo 4: Flagging:

ASNAT allows for a dataset's data lines to be marked with one or more integer codes (separated with semi-colons without spaces) indicating the line is suspect and plotted differently (as gray points) and statistics computed after omitting these flagged lines.

There are three ways to flag data lines:

4.1 Manually selecting lines either in ASNAT or editing/computing the file's flagged column outside ASNAT and loading it into ASNAT.

4.2 Specifying flag conditions on the available columns of the dataset.

4.3 Selecting Dataset Y Neighbor Flagging which compares time-matched neighboring measures from Dataset X and Dataset Y and flags measures in Dataset Y based on comparison to the measures in Dataset X.

To demo flagging:

Quit/restart ASNAT and load the same datasets as in **Demo 1**:

Reset map

Start Date: 2022-06-01

Days: 1

Timestep: hours

Delete Loaded Data

Load Web:

AQS.pm25

PurpleAir.pm25_corrected

Purple Air Key:

your_key

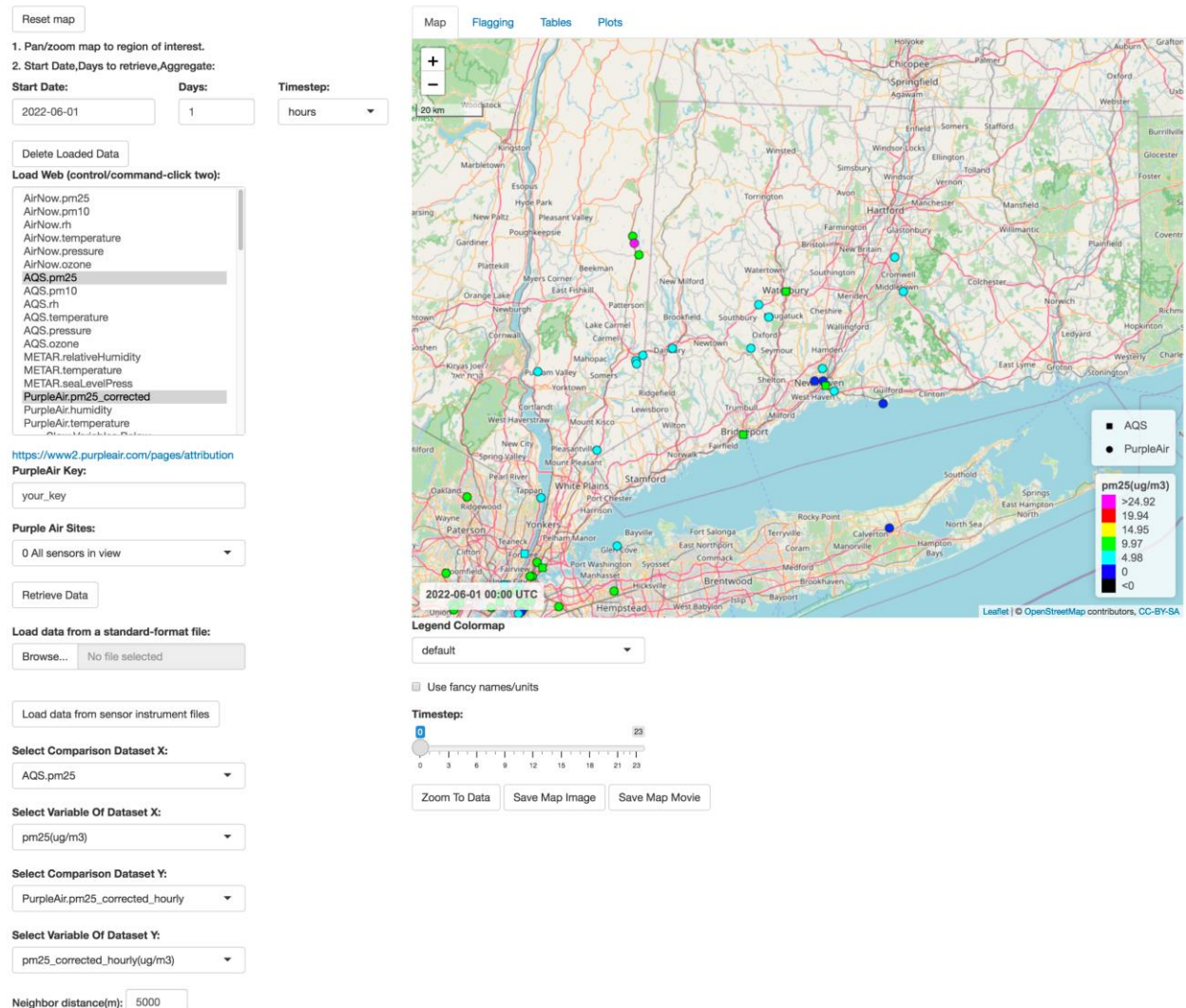
Retrieve Data

Select Comparison Dataset Y:

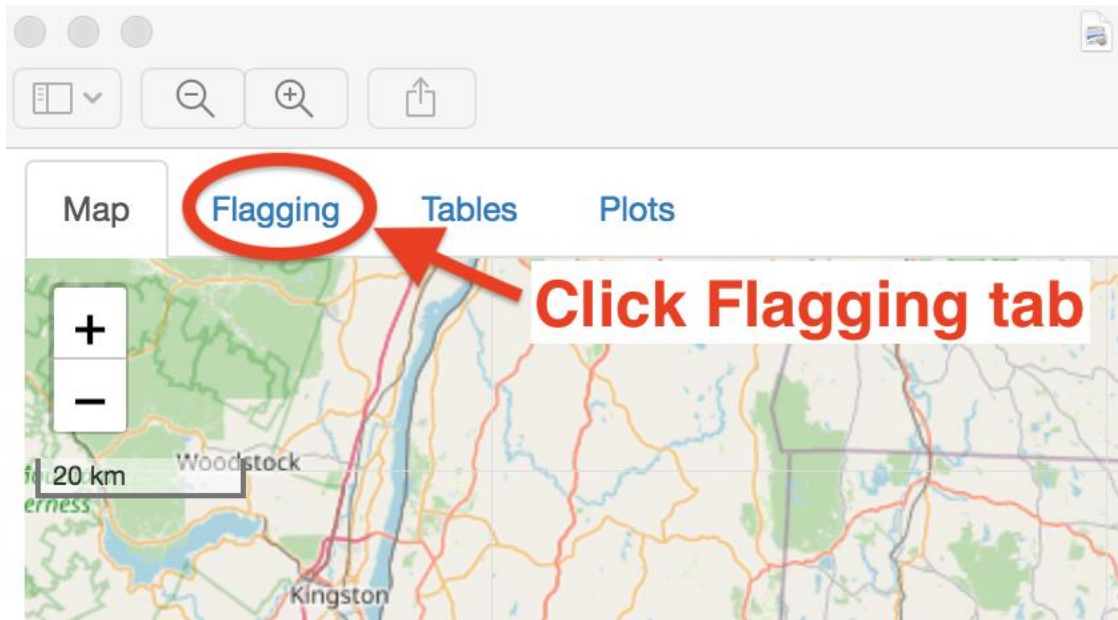
PurpleAir.pm25_corrected_hourly

Neighbor Distance(m): 5000

ASNAT should appear as the image below.



Once the data is loaded as shown above,
Click the Flagging tab.



This will display the data lines of Dataset Y – in this example, PurpleAir.pm25_corrected_hourly as shown in the image below.

Map
Flagging
Tables
Plots

Dataset to Flag:
Dataset Y

Show 10 entries
Search:

	timestamp(UTC)	longitude(deg)	latitude(deg)	elevation(m)	id(-)	count(-)	pm25_corrected_hourly(ug/m3)	flagged(-)	note(-)
1	2022-06-01T00:00:00-0000	-73.84475	40.7117	21.6408	24311	30	13.73281	0	Forest Hills
2	2022-06-01T00:00:04-0000	-72.74895	42.12907	43.2816	133515	30	5.38771	0	Public Works Sackett St
3	2022-06-01T00:00:05-0000	-74.2219	40.71593	53.0352	99355	30	10.93961	0	S6F21
4	2022-06-01T00:00:06-0000	-72.67964	41.77156	17.3736	102474	30	5.81694	0	Hartford CTDEEP
5	2022-06-01T00:00:09-0000	-72.52486	42.05951	67.9704	131553	30	4.1586	0	East Longmeadow High School
6	2022-06-01T00:00:11-0000	-73.08233	42.00181	272.4912	3439	29	8.73617	0	Sandy Brook Colebrook CT
7	2022-06-01T00:00:14-0000	-73.63896	40.87506	37.4904	44077	29	7.86249	0	Pine Low Glen Cove NY
8	2022-06-01T00:00:15-0000	-73.9389	40.79346	5.4864	91443	30	12.9475	2	FA-O7
9	2022-06-01T00:00:20-0000	-71.72055	41.61822	66.1416	16181	30	3.63288	2	URI Alton Jones Camp
10	2022-06-01T00:00:20-0000	-73.97508	40.699	5.7912	131845	30	2.29338	99	New Lab - Voltaic Systems #2

Showing 1 to 10 of 3,016 entries
Previous
1
2
3
4
5
...
302
Next

Exclude 99 from flagged column
Include 99 in flagged column

Load flag conditions from a file:
Browse...
No file selected

Flag Conditions:
pm25_corrected_hourly < 0 or pm25_corrected_hourly > 1000
(id = 16181 or id = 91443) and timestamp <= 2022-06-01T12

Apply Flagging

Save Flag Conditions

Dataset Y Neighbor Flagging:
☒ Apply flag 80 if difference > 5
☒ Apply flag 81 if percent difference > 70
☒ Apply flag 82 if R-squared < 0.5

Since the dataset can have many data lines, they are paged, by default 10 lines at a time.

This can be increased with the Show entries menu.

Pages can be navigated by clicking on the page numbers or the Next button.

Note the 'flagged (-)' column added before the last 'note(-)' column.

ASNAT adds this column when loading a dataset if it does not already exist.

0 indicates the data line is unflagged (the default).

Map Flagging Tables Plots

Dataset to Flag:
Dataset Y

Show 10 entries

If comparing two datasets - X and Y - then by default, Dataset Y is selected for flagging.

In this demo, Dataset X is AQS.pm25 and Dataset Y is PurpleAir.pm25_corrected_hourly

Note the added 'flagged(-)' column (default 0 means unflagged).

	timestamp(UTC)	longitude(deg)	latitude(deg)	elevation(m)	id(-)	count(-)	pm25_corrected_hourly(ug/m3)	flagged(-)	note(-)
1	2022-06-01T00:00:00-0000	-73.84475	40.7117	21.6408	24311	30	13.73281	0	Forest Hills
2	2022-06-01T00:00:01-0000	-73.9541	40.6356	9.144	136692	29	8.08192	0	Gotham Professional Arts Academy
3	2022-06-01T00:00:05-0000	-74.2219	40.71593	53.0352	99355	30	10.93961	0	S6F21
4	2022-06-01T00:00:09-0000	-74.52499	41.34552	188.6712	117237	30	10.64445	0	MCS west
5	2022-06-01T00:00:13-0000	-74.43156	40.5406	21.9456	97935	29	10.13941	0	Brookside Rd
6	2022-06-01T00:00:13-0000	-74.47177	40.57411	19.5072	44275	23	9.90582	0	Piscataway
7	2022-06-01T00:00:14-0000	-73.63896	40.87506	37.4904	44077	29	7.86249	0	Pine Low Glen Cove NY
8	2022-06-01T00:00:15-0000	-73.9389	40.79346	5.4864	91443	30	12.9475	0	FA-O7
9	2022-06-01T00:00:16-0000	-74.48189	41.41832	225.2472	31187	30	11.4438	0	RM
10	2022-06-01T00:00:20-0000	-73.97508	40.699	5.7912	131845	30	2.29338	0	New Lab - Voltaic Systems #2

Showing 1 to 10 of 1,392 entries

Exclude 99 from flagged column Include 99 in flagged column

Load flag conditions from a file:
Browse... No file selected

Flag Conditions:

Apply Flagging

Save Flag Conditions

Dataset Y Neighbor Flagging:

☐ Apply flag 80 if difference > 5

☐ Apply flag 81 if percent difference > 70

☐ Apply flag 82 if R-squared < 0.5

Previous 1 2 3 4 5 ... 140 Next

The Dataset Y columns are displayed - by default 10 rows at a time - and can be paged with the page or Next button.

4.1 Manual Flagging:

Click on line 7 and then click on line 10.

(Re-clicking toggles line selection.

Shift-clicking selects/toggles a range of lines.)

Click the 'Include 99 in flagged column' button.

This will change the flagged column of these lines from 0 to 99.

99 is the integer code used to indicate manually flagged lines.

	timestamp(UTC)	longitude(deg)	latitude(deg)	elevation(m)	id(-)	count(-)	pm25_corrected_hourly(ug/m3)	flagged(-)	note(-)
1	2022-06-01T00:00:00-0000	-73.84475	40.7117	21.6408	24311	30	13.73281	0	Forest Hills
2	2022-06-01T00:00:01-0000	-73.9541	40.6356	9.144	136692	29	8.08192	0	Gotham Professional Arts Academy
3	2022-06-01T00:00:05-0000	-74.2219	40.71593	53.0352	99355	30	10.93961	0	S6F21
4	2022-06-01T00:00:09-0000	-74.52499	41.34552	188.6712	117237	30	10.64445	0	MCS west
5	2022-06-01T00:00:13-0000	-74.43156	40.5406	21.9456	97935	29	10.13941	0	Brookside Rd
6	2022-06-01T00:00:13-0000	-74.47177	40.57411	19.5072	44275	23	9.90582	0	Piscataway
7	2022-06-01T00:00:14-0000	-73.63896	40.87506	37.4904	44077	29	7.86249	99	Pine Low Glen Cove NY
8	2022-06-01T00:00:15-0000	-73.9389	40.79346	5.4864	91443	30	12.9475	0	FA-O7
9	2022-06-01T00:00:16-0000	-74.48189	41.41832					0	RM
10	2022-06-01T00:00:20-0000	-73.97508	40.699	5.7912	131845	30	2.29338	99	New Lab - Voltaic Systems #2

Showing 1 to 10 of 1,392 entries

Previous 1 2 3 4 5 ... 140 Next

Exclude 99 from flagged column Include 99 in flagged column

3. This changes flagged value 0 to 99

1. Click on data lines 7 and 10.

2. Click 'Include 99 in flagged column' button.

4.2 Flag Conditions:

Flag conditions are Boolean expressions of the available columns of a dataset. These expressions must be evaluated as either true or false. They are evaluated on each data line and, if true, they include an integer code for each true flag condition. The numbers are 1 for the first condition, 2 for the second condition, and so on.

Let's specify a few flag conditions on Dataset Y using the named columns shown in the table.

These are timestamp, longitude, latitude, elevation, id, count, pm25_corrected_hourly and note.

The column names are operands.

Available relational operators are: <, <=, >, >=, =, !=, or, and, (,).

One may also use R operators such as: ||, &&, -, +, *, /, ^, etc.

Note that * and 'and' have higher precedence than + and 'or' and ^ is right-associative.

(E.g., $2 + 3 * 4 = 14$ and $2^{3^2} = 512$.)

Use parentheses only when necessary.

can be used for comment lines.

Click in the text editing area labeled 'Flag Conditions:'

Enter the following lines:

(id = 44275 or id = 99449) and count < 25

Filter-out negative ug/m3:

pm25_corrected_hourly < 0

2022-06-01T00 <= timestamp and timestamp <= 2022-06-01T05

Click the 'Apply Flagging' button.

ASNAT should appear as shown in the next image.

The screenshot shows the ASNAT interface with a table of 10 data entries. The 'flagged' column contains integer values: 3, 3, 3, 3, 3, 1,3, 3,99, 3, 3,99. A red oval highlights the 'flagged' column, and a red arrow points to the 'Apply Flagging' button. Below the table, the 'Flag Conditions' section is visible, containing the following text:

```
(id = 44275 or id = 99449) and count < 25
# Filter-out negative ug/m3:
pm25_corrected_hourly < 0
2022-06-01T00 <= timestamp and timestamp <= 2022-06-01T05
```

Red annotations include:

- 1. Enter these flag conditions (pointing to the flag conditions text area)
- 2. Click 'Apply Flagging' button (pointing to the 'Apply Flagging' button)
- 3. Note the added flagged codes (pointing to the 'flagged' column values)

Note the integer codes included in the flagged column.

Regarding timestamps, the literal comparison values can be any starting sub-length of the timestamp.

In the example here, all data lines between hours 00 and 05 are flagged.

For multi-day datasets, one may omit the hour to omit entire days.

E.g., for a 30-day dataset, omit the first 3 and last 6 days with:

timestamp <= 2022-06-03 or timestamp >= 2022-06-25

ASNAT's unique ability to evaluate arbitrary expressions of possibly multiple columns is more powerful than simple range specifications on single columns.

These flag conditions can be saved to plain text files and reloaded into ASNAT later.

Click the 'Save Flag Conditions' button to save them to the file 'flags.txt' in the 'Save to Directory'. Rename this file if desired.

Click the 'Browse...' button to reload them later.

Flag conditions can only be applied to a dataset containing the columns named in the flag conditions.

Load flag conditions from a file:

The Flag Conditions can be reloaded from a file.

Flag Conditions:

```
(id = 44275 or id = 99449) and count < 25
# Filter-out negative ug/m3:
pm25_corrected_hourly < 0
2022-06-01T00 <= timestamp and timestamp <= 2022-06-01T05
```

The Flag Conditions are saved as 'flags.txt' in the 'Save to Directory'.

4.3 Neighbor Flagging:

Manual flagging and condition-based flagging operate only on single lines of the 'Dataset to Flag'.

ASNAT also includes flagging based on differences between matched Dataset X and Dataset Y measures.

In this example, '**Select Variable of Dataset Y**' is pm25_corrected_hourly.

This value is compared to Dataset X (AQS.pm25).

For each Dataset Y time-matched point within the specified '**Neighbor Distance**' of a Dataset X point, one can flag based on the absolute difference, percent difference, or R-squared value of the matched time series of X and Y measures. (R-squared is called the coefficient of determination and equals the square of r, the Pearson sample correlation coefficient.

https://en.wikipedia.org/wiki/Pearson_correlation_coefficient)

These are the buttons shown at the bottom and the default values are shown but may be changed.

One can select/deselect these additional flagging options which are applied after the Summary tables are computed.

Select all three buttons and use the default values.

As shown in the tooltip for these values, the flagged integer codes are 80, 81, 82 respectively.

Save Flag Conditions

Dataset Y Neighbor Flagging:

☒ Apply flag 80 if difference >

☒ Apply flag 81 if percent difference >

☒ Apply flag 82 if R-squared <

Mark Dataset Y flagged column with 80 if the absolute difference of the time-matched comparison variable with Dataset X is greater than value.

5

70

0.5

To apply Neighbor Flagging, compute the Summary and Comparison tables.
Click the Tables tab.

Click the 'Summarize and Compare X and Y' button.

This generates four tables:

Dataset Summary of X vs Y

Dataset Summary of Y vs X

Neighboring Points of Y for each point in X

Neighboring Points of Y for each point in X R-squared values

As shown in the following images.

Map Flagging Tables

Summarize and Compare X and Y

1. Click Tables tab

2. Click 'Summarize and Compare X and Y' button

This generates summary tables X vs Y and Y vs X

Dataset Summary: AQS.pm25 vs PurpleAir.pm25_corrected_hourly
AQS.pm25: 2022-06-01 to 2022-06-01 (-74.8067, -72.9029) (40.2224, 41.5506)

Site_Id(-)	Timestamp_First_Measurement(UTC)	Timestamp_Last_Measurement(UTC)	Count(-)	Missing(%)	Mean_AQS.pm25(ug/m3)	Min_AQS.pm25(ug/m3)	P25_AQS.pm25(ug/m3)
90011123	2022-06-01T15:00:00-0000	2022-06-01T23:00:00-0000	9	62	4.10	3.00	3.60
90090027	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	5.10	3.00	3.50
340030010	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	7.60	1.90	5.90
340170008	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	7.10	2.20	4.70
340171003	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	9.20	3.00	6.90
340210008	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	12.30	5.10	9.90
340390004	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	11.70	5.30	8.10
360050080	2022-06-01T05:00:00-0000	2022-06-01T23:00:00-0000	19	21	2.40	0.00	1.30
360050110	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000	24	0	8.80	5.60	6.90
360470052	2022-06-01T05:00:00-0000	2022-06-01T23:00:00-0000	19	21	11.70	8.30	10.60
360610115	2022-06-01T05:00:00-0000	2022-06-01T23:00:00-0000	19	21	3.10	0.10	1.80
360610135	2022-06-01T05:00:00-0000	2022-06-01T23:00:00-0000	19	21	2.90	0.20	2.00
360810120	2022-06-01T05:00:00-0000	2022-06-01T23:00:00-0000	19	21	2.70	0.10	1.60
360810124	2022-06-01T05:00:00-0000	2022-06-01T23:00:00-0000	19	21	9.40	7.00	8.40
360810125	2022-06-01T05:00:00-0000	2022-06-01T23:00:00-0000	19	21	4.80	1.50	2.90

Dataset Summary: PurpleAir.pm25_corrected_hourly vs AQS.pm25
PurpleAir.pm25_corrected_hourly: 2022-06-01 to 2022-06-01 (-74.76022, -72.62656) (40.2183, 41.69623)

Site_Id(-)	Timestamp_First_Measurement(UTC)	Timestamp_Last_Measurement(UTC)	Count(-)	Missing(%)	Mean_PurpleAir.pm25_corrected_hourly(ug/m3)	Min_PurpleAir.pm25_corrected
22749	2022-06-01T00:01:11-0000	2022-06-01T23:00:49-0000	10	58	6.50	
24063	2022-06-01T00:00:56-0000	2022-06-01T23:01:04-0000	24	0	2.50	

Neighboring Points (<= 5000m) Comparison: AQS.pm25 vs PurpleAir.pm25_corrected_hourly

2022-06-01 (-74.8067, -72.9029) - (40.2224, 41.5506)

And Neighbor Points table

timestamp(UTC)	AQS.id(-)	AQS.pm25(ug/m3)	PurpleAir.id(-)	PurpleAir.pm25_corrected_hourly(ug/m3)	PurpleAir.flagged(-)
2022-06-01T00:00:00-0000	90090027	10.30	39689	7.12	3
2022-06-01T00:00:00-0000	90090027	10.30	24063	4.06	3;80;81
2022-06-01T00:00:00-0000	90090027	10.30	106862	4.54	3;80;81
2022-06-01T00:00:00-0000	340030010	6.90	91899	10.79	3
2022-06-01T00:00:00-0000	340170008	7.80	98723	11.23	3
2022-06-01T00:00:00-0000	340170008	7.80	22749	11.24	3
2022-06-01T00:00:00-0000	340170008	7.80	101655	8.60	3

Neighboring Points (<= 5000m) Comparison R2: AQS.pm25 vs PurpleAir.pm25_corrected_hourly

2022-06-01 (-74.8067, -72.9029) - (40.2224, 41.5506)

And R-squared table

AQS.id(-)	AQS.longitude(deg)	AQS.latitude(deg)	PurpleAir.id(-)	PurpleAir.longitude(deg)	PurpleAir.latitude(deg)	distance(m)	R2(-)
340170008	-74.0362	40.7709	22749	-74.0259	40.7470	2793.1682	0.3943
340171003	-74.0523	40.7254	22749	-74.0259	40.7470	3269.6087	0.9564
340170008	-74.0362	40.7709	58069	-73.9949	40.7544	3932.7059	0.7242
360050080	-73.9201	40.8361	91423	-73.8916	40.8611	3671.8532	0.0000
360610115	-73.9356	40.8496	91423	-73.8916	40.8611	3917.0500	0.9051
340390004	-74.2084	40.6414	96741	-74.2089	40.6635	2455.6155	0.7617
360470052	-74.0187	40.6418	97641	-73.9894	40.6596	3159.8462	0.9179
340170008	-74.0362	40.7709	101655	-73.9836	40.7643	4484.1806	0.9104
340171003	-74.0523	40.7254	110490	-74.0845	40.7104	3178.8728	0.5163
340171003	-74.0523	40.7254	114593	-74.0727	40.7269	1730.5630	0.8508
360470052	-74.0187	40.6418	129163	-73.9962	40.6760	4248.7797	0.6746
340210008	-74.7632	40.2224	131197	-74.7602	40.2183	520.7753	1.0000

Finally, plot the results from these tables.

Click the 'Plots' tab.

Click the 'Make Points of Datasets X, Y' button.

Note in the time-series plot of Dataset Y the flagged points are gray.

The scatter plot flagged points are also gray.

There is an additional scatter plot of just the non-flagged points (N = 81).

[Map](#)
[Flagging](#)
[Tables](#)
[Plots](#)

1. Click the 'Plots' tab.

Neighbors Map - Shows Neighbors of (Single) Selected Dataset X Site

Select Dataset X Site With Neighbors:

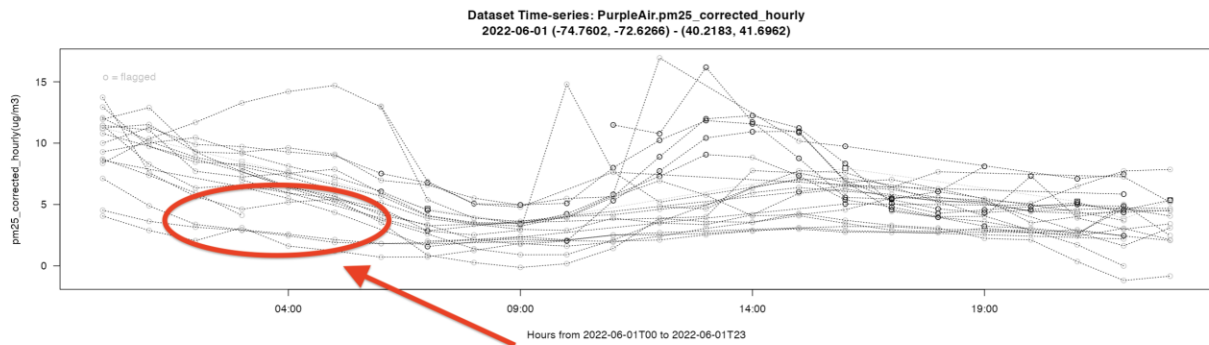
0 All Sites

☐ Use interactive plots

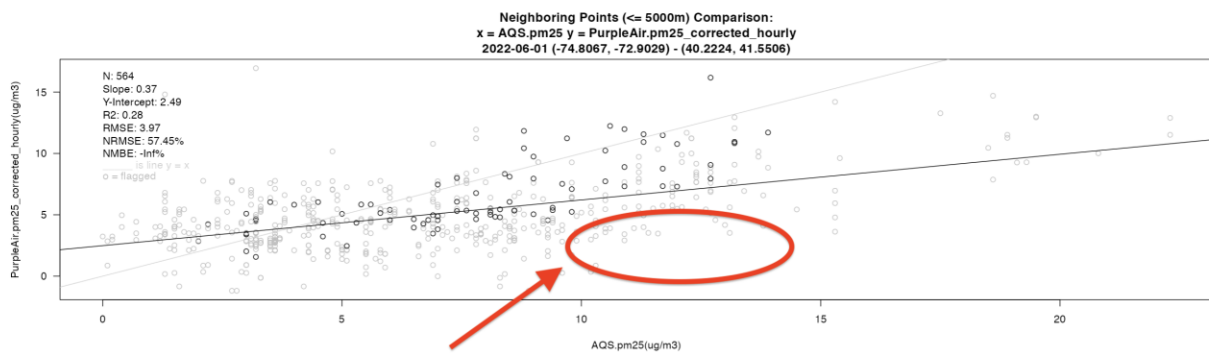
2. Click the 'Make Plots of Datasets X, Y' button.

Make Plots of Datasets X, Y

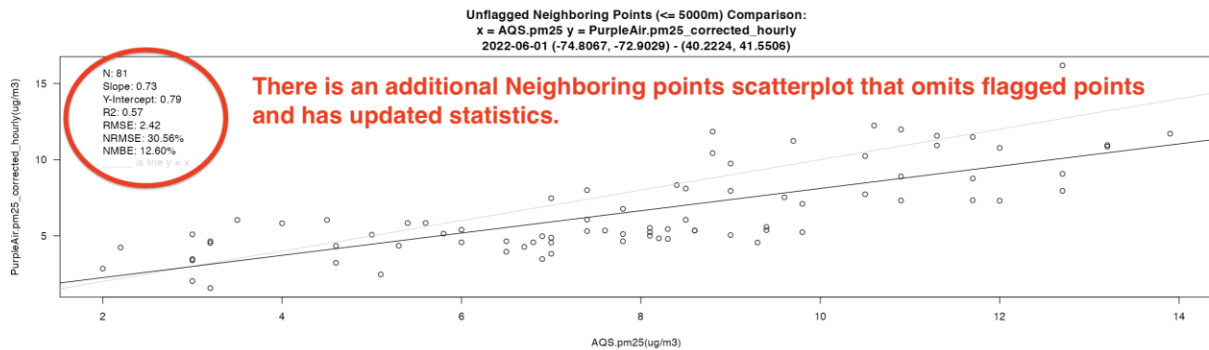
Save Plots



The Time-series plot of Dataset Y shows flagged points in gray.



The Neighboring points scatterplot shows flagged points in gray.



There is an additional Neighboring points scatterplot that omits flagged points and has updated statistics.

Demo 5: Interactive Plots:

By default, ASNAT uses the basic plots which are grayscale, fast, clear, and best for use in publication-quality PDF documents. Lacking color, they are also understandable by those with color vision impairment. PDF files are printed at the highest resolution of the printer rather than the screen resolution they were captured at (as is the case with pixel image formats).

However, ASNAT also provides the option of colored 'interactive plots'.

These allow the user to zoom, probe, select, and hide parts of plots and save the modified plot as a PNG image file.

Continuing from the previous demo, you can explore interactive plots.

On the **Plots tab**,

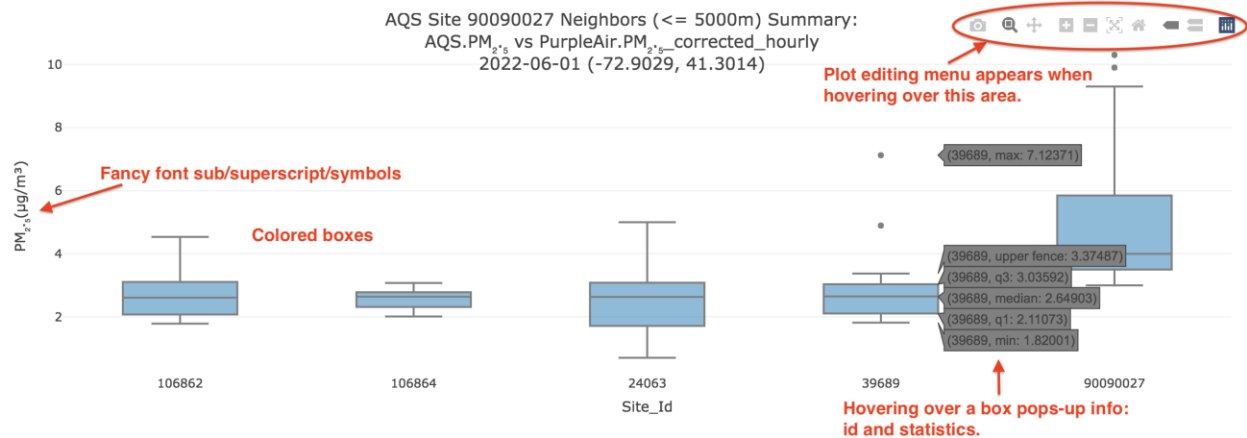
Click the 'Use interactive plots' button.

Click the 'Make Plots of Datasets X, Y' button.

(You may have to click twice for all plots to appear/update.)



The summary boxplot now has blue boxes, fancy fonts (with sub/superscripts and symbols), and hovering over a box pops-up ID and statistics for that plot element. Hovering over the top right corner of an interactive plot reveals a plot editing menu.

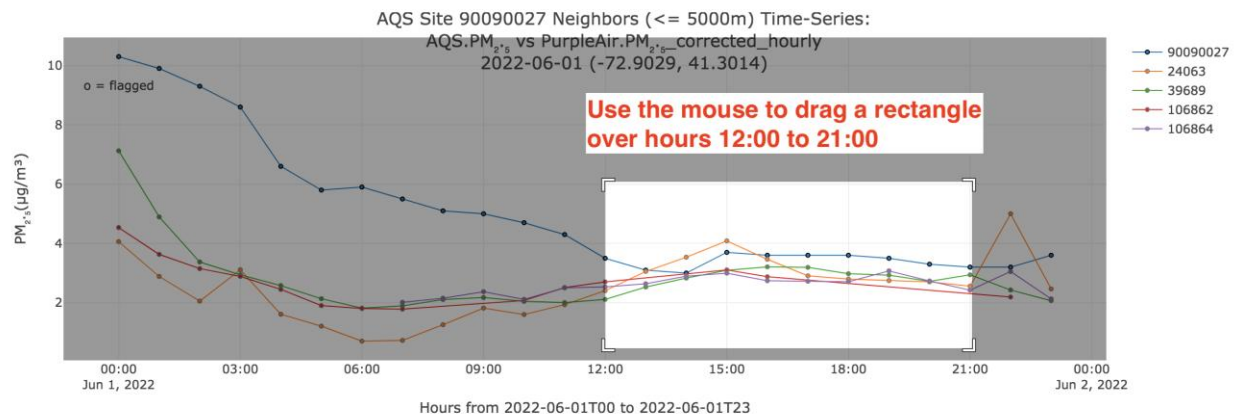


Explore these plot editing features with the Time-series plot.

5.1 Zoom:

The first feature is zooming.

Use the mouse to drag a rectangular area on the plot from hours 12:00 to 21:00.



Releasing the mouse button causes the plot to be zoomed in to show just these hours.

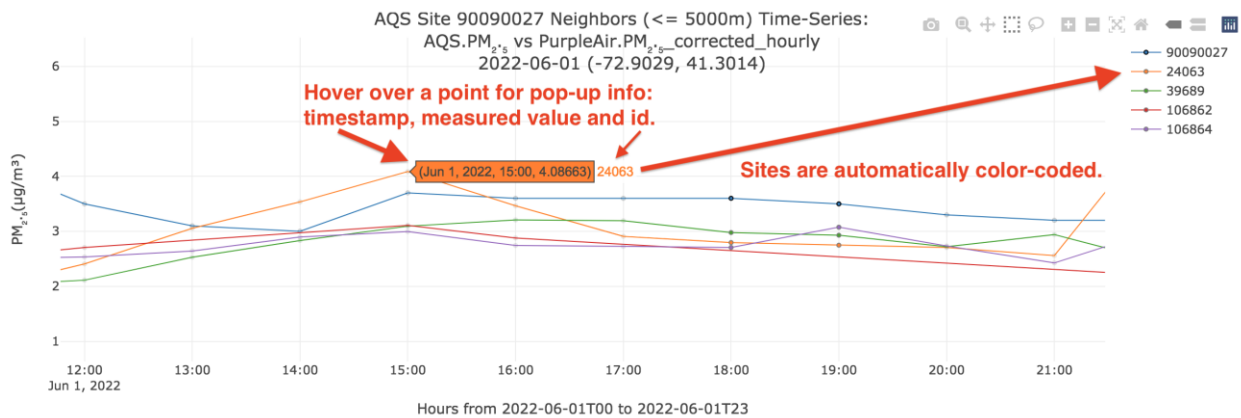
In a dense plot, zooming in can help you visualize the data for small areas at a time.

If you mess up, just click the 'home' symbol to reset everything back to the default.

Click 'home' symbol to reset to default view.

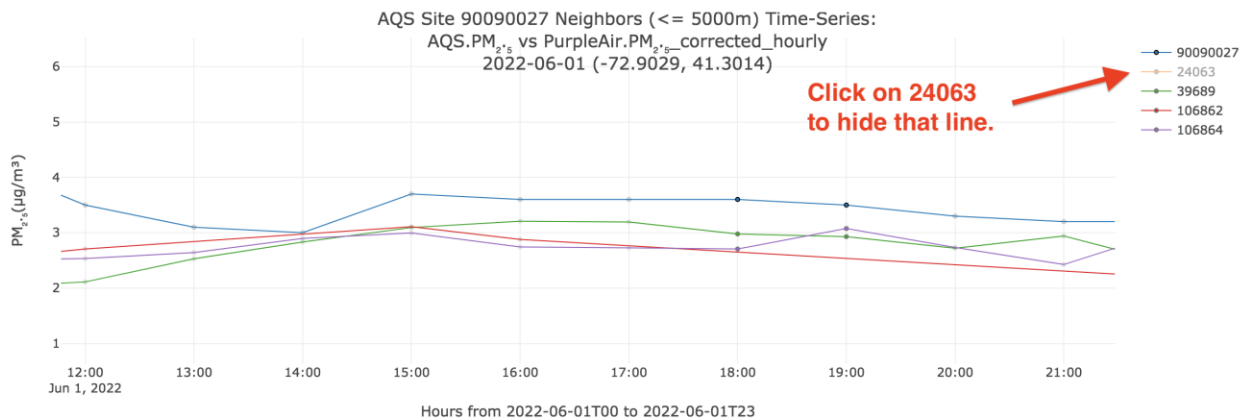


Hovering pops up info about a data point: timestamp, measured value, and ID. Note that each site is automatically assigned a unique color as shown in the legend on the right.



Plot elements, in this case, a time-series line for each site, can be toggled visible or invisible.

Click on the legend for site 24063 to hide the time-series line for that site.



You may click it again to make it re-appear.

Several time-series lines may be toggled at a time.

Click the '+' symbol to allow panning the plot.

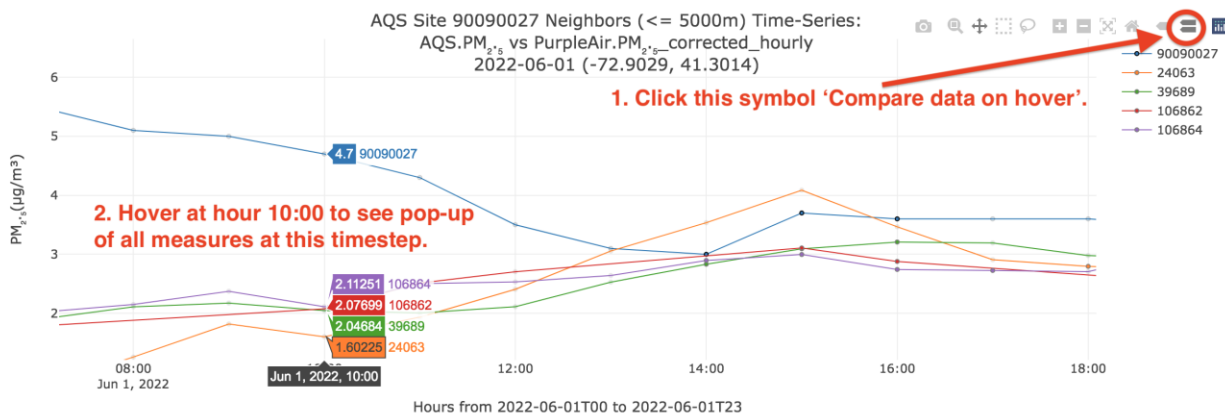
Then **drag the mouse to the right** to show earlier timesteps while still zooming in.



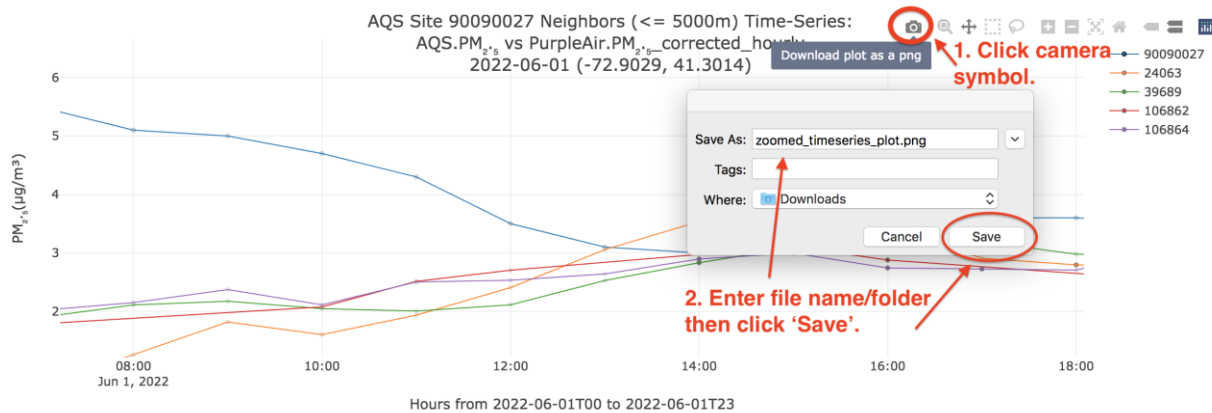
5.2 Compare data on hover:

Click the second to last symbol in the plot edit menu.

Then **hover over hour 10:00** to see a pop-up showing the values of all measurements for that timestep.



Finally, **click the camera symbol** to save this plot as a PNG image file.



The last plot - a scatterplot - is similarly interactive but the time-series plot is most appropriate for interaction.

Demo 6: Data Correction:

ASNAT allows for data correction between two datasets. The user can select the device ID in the dataset and perform corrections based on the selected models (currently, only linear regression is supported).

The correction results will be displayed in both static and interactive plots, as well as in a table.

6.1 Relaunch **RStudio** and **ASNAT**.

6.2 Use the mouse to select the **Start Date**, then type **2022-06-01** and press Enter.

6.3 Use the mouse to select **Days**, then enter **2**.

ASNAT

« June 2022 »

Su	Mo	Tu	We	Th	Fr	Sa
29	30	31	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	1	2
3	4	5	6	7	8	9

2022-06-01|

Days: 2

Timestep: hours

Delete Loaded Data

6.4 Click **Browse** under **Load data from a standard format file**.

Purple Air Sites:

0 All sensors in view

Retrieve Data

Load data from a standard-format file:

Browse... No file selected

Load data from sensor instrument files

6.5 In the newly opened window, navigate to the `\ASNAT\sample_data\standard_format_file`, and select **AQS_pm25_2022-06-01_to_2022-06-02.tsv**.

Name	Date modified	Type	Size
Home-Rubber-SD-card-CSV-June2022	10/18/2024 2:04 PM	File folder	
standard_format_file	10/18/2024 2:05 PM	File folder	

Name	Date modified	Type	Size
AQS_pm25_2022-06-01_to_2022-06-02.tsv	10/18/2024 2:03 PM	TSV File	504 KB
PurpleAir_pm25_corrected_hourly_2022-06-01_t...	10/18/2024 2:03 PM	TSV File	2,479 KB

6.6 Repeat the process and select **PurpleAir_pm25_corrected_hourly_2022-06-01_to_2022-06-02.tsv**.

Name	Date modified	Type	Size
AQS_pm25_2022-06-01_to_2022-06-02.tsv	10/18/2024 2:03 PM	TSV File	504 KB
PurpleAir_pm25_corrected_hourly_2022-06-01_t...	10/18/2024 2:03 PM	TSV File	2,479 KB

6.7 In the **Select Comparison Dataset Y** menu, select **PurpleAir.hourly_data**, and change the **Neighbor distance(m)** to 1000.

The screenshot shows a web interface for selecting comparison datasets. It contains four dropdown menus and one text input field. Red arrows and text annotations highlight specific selections:

- Select Comparison Dataset X:** The dropdown menu is set to "AQS.hourly_data". A red arrow points to it with the text "Select AQS for DatasetX".
- Select Variable Of Dataset X:** The dropdown menu is set to "pm25(ug/m3)".
- Select Comparison Dataset Y:** The dropdown menu is set to "PurpleAir.hourly_data". A red arrow points to it with the text "Select PurpleAir for DatasetY".
- Select Variable Of Dataset Y:** The dropdown menu is set to "pm25_corrected_hourly(ug/m3)".
- Neighbor distance(m):** The text input field contains "1000", which is circled in red.

6.8 Click the **Corrections** tab.

6.9 Click the **Regression Correction** tab.

The screenshot shows a row of navigation tabs: "Map", "Flagging", "Tables", "Plots", "Corrections", and "Network Summary". Below "Corrections" are two sub-tabs: "Regression Correction" and "Moving Averages". Red circles highlight the "Corrections" tab and the "Regression Correction" sub-tab.

6.10 Use the mouse to click the **Select All** and **Unselect All** buttons to select or unselect all devices.

6.11 Use the mouse to select rows 2 to 5.

Map Flagging Tables Plots

Summarize and Compare X and Y Data Correction

Data Correction Panel

(Please ensure that dataset X is the reference data and dataset Y is the target data.)

Neighboring Points

Show entries Search:

Neighboring Points (<= 1000 m)

	PurpleAir.id(-)	PurpleAir.longitude(deg)	PurpleAir.latitude(deg)	R2(-)	distance(m)	AQS.id(-)	AQS.longitude(deg)	AQS.latitude(deg)
1	5498	-75.14166	39.96417	0.3750108594751062	469.8836453067628	421010057	-75.1426	39.96417
2	9652	-79.88554	40.29301	0.9726384263695701	144.8794009592215	420033007	-79.8853	40.29301
3	10264	-77.39969	37.55658	0.9813492986655461	54.47009594887508	510870014	-77.4003	37.55658
4	12168	-72.27245	42.93051	0.360531259037658	12.25585752394077	330050007	-72.2723	42.93051
5	12192	-72.26555	42.92347	0.8687635453518151	954.7878325351485	330050007	-72.2723	42.93051
6	12204	-72.26815	42.93333	0.1325453649764496	461.3726875498235	330050007	-72.2723	42.93051
7	12210	-72.27411	42.92475	0	655.5354188705153	330050007	-72.2723	42.93051
8	12212	-72.27235	42.93053	0.1382644151988632	5.259239097004517	330050007	-72.2723	42.93051
9	16181	-71.72055	41.61822	0.06215979632614628	338.517259241299	440030002	-71.72	41.61822
10	21109	-77.40051	37.5566	0.9599086500865532	21.57492706095181	510870014	-77.4003	37.55658

Showing 1 to 10 of 48 entries

Previous 2 3 4 5 Next

 **Select and Unselect All**

Scroll down to review the selected devices.

Showing 1 to 10 of 48 entries

Previous 2 3 4 5 Next

Selected IDs:

Number of points selected: 4

PurpleAir ID: 10264 | AQS ID: 510870014 | R-squared: 0.981

PurpleAir ID: 9652 | AQS ID: 420033007 | R-squared: 0.973

PurpleAir ID: 12192 | AQS ID: 330050007 | R-squared: 0.869

PurpleAir ID: 12168 | AQS ID: 330050007 | R-squared: 0.361

6.12 Select **Linear** from the **Select Regression Type** dropdown menu, then click **Generate Coefficients**.

Correction Models

Select Regression Type

Linear

Equations:

Corrected Data

6.13 Click the **Select All** button to select all newly generated equations.

6.14 Click **Export to CSV** to export the details of the selected equations to a CSV file under `\ASNAT\data\equation`

6.15 Click **Apply Regression** to generate correction plots and a table.

Correction Models

Apply Correction

Select Regression Type

Linear

Generate Coefficients

Equations:

Show 10 entries

Search:

Regression Equations and Statistics

Device_ID	Equation	R_Squared
10264	$y = -3.1301 + 1.2550x$	0.9813
9652	$y = -0.8688 + 0.9538x$	0.9726
12192	$y = 1.4108 + 0.3236x$	0.8688
12168	$y = 1.4441 + 0.3189x$	0.3605

Showing 1 to 4 of 4 entries

Select All Unselect All

Apply Regression Export to CSV

Corrected Data

Previous 1 Next

Click Select All to select all equations

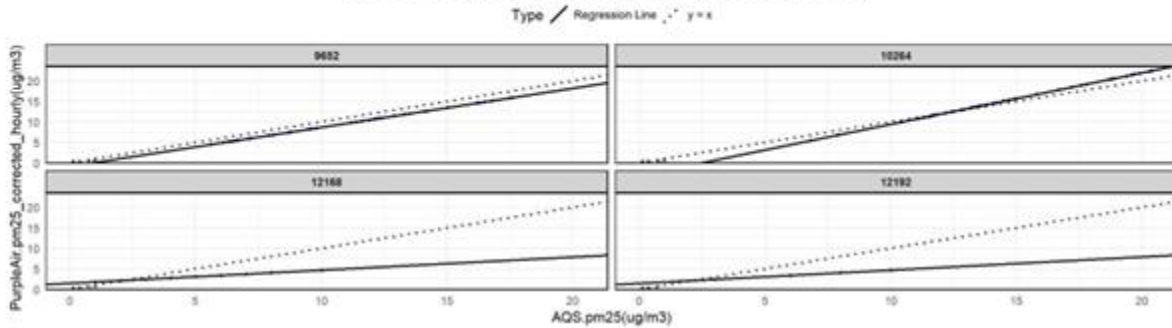
Click Export to CSV to export equation details to a csv file

Click Apply Regression to generate the correction data

Apply Regression Export to CSV

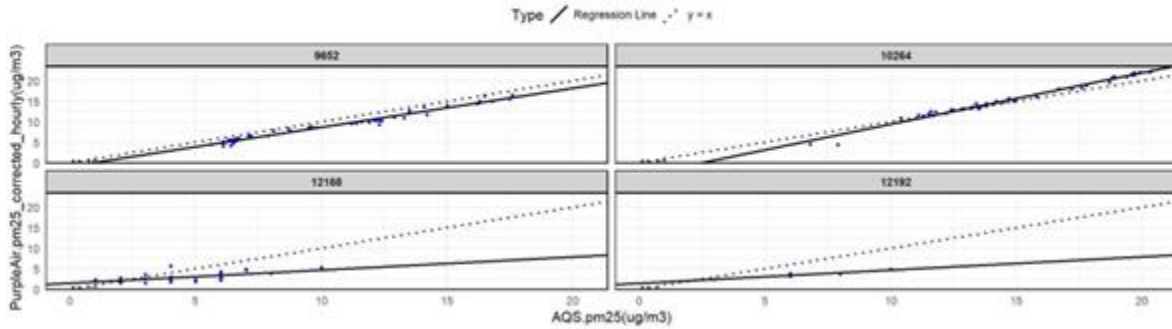
Corrected Data

Neighboring Points (<= 1000m) Comparison R2: AQS.pm25 vs PurpleAir.pm25_corrected_hourly
2022-06-01 - 2022-06-02 (-80.8012, -67.0613) - (37.5565, 47.3553)



Original Data

Neighboring Points (<= 1000m) Comparison R2: AQS.pm25 vs PurpleAir.pm25_corrected_hourly
2022-06-01 - 2022-06-02 (-80.8012, -67.0613) - (37.5565, 47.3553)



6.16 Scroll down to **Device Details**, and click the **Select Device ID** dropdown menu to choose a device ID to view detailed interactive plots and the data table.

Static plots (before and after correction) will be generated for each device, while interactive plots and the details table will only be displayed for the selected device in the **Select Device ID** menu.

Device Details

Select Device ID:

9652

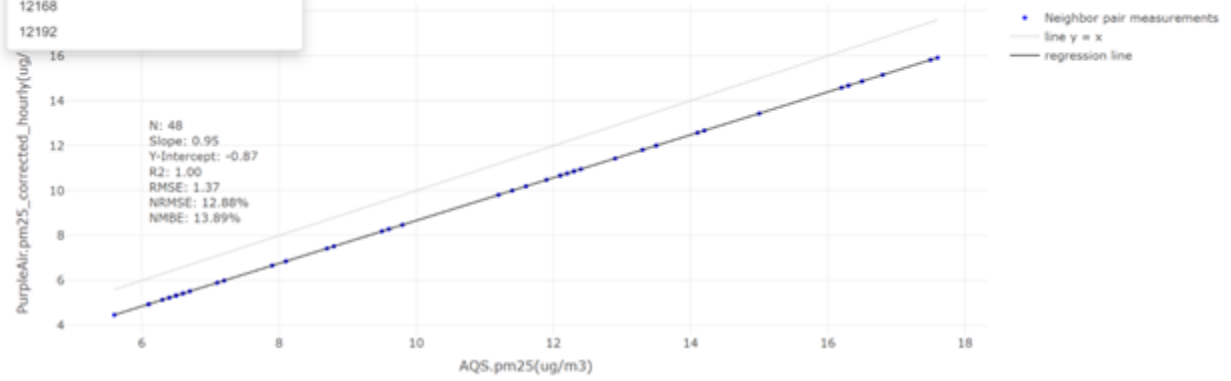
9652

10264

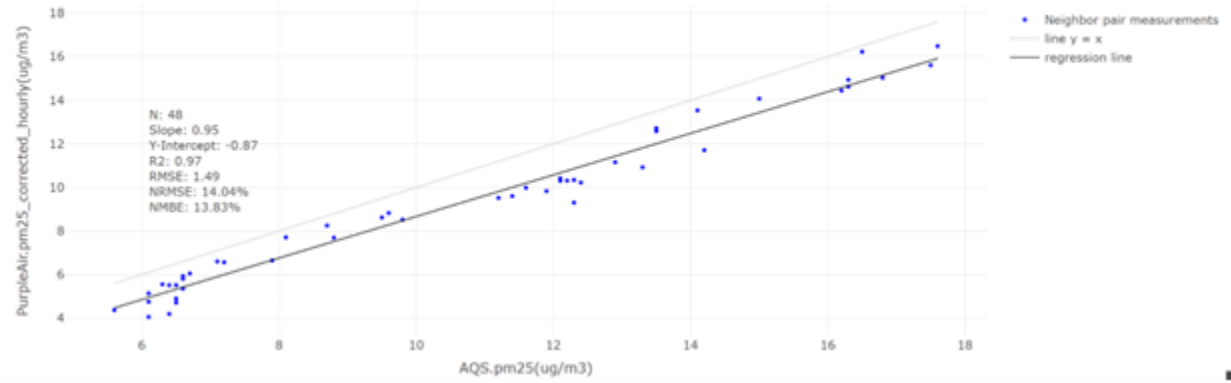
12168

12192

Corrected Data for Device 9652



Original Data for Device 9652

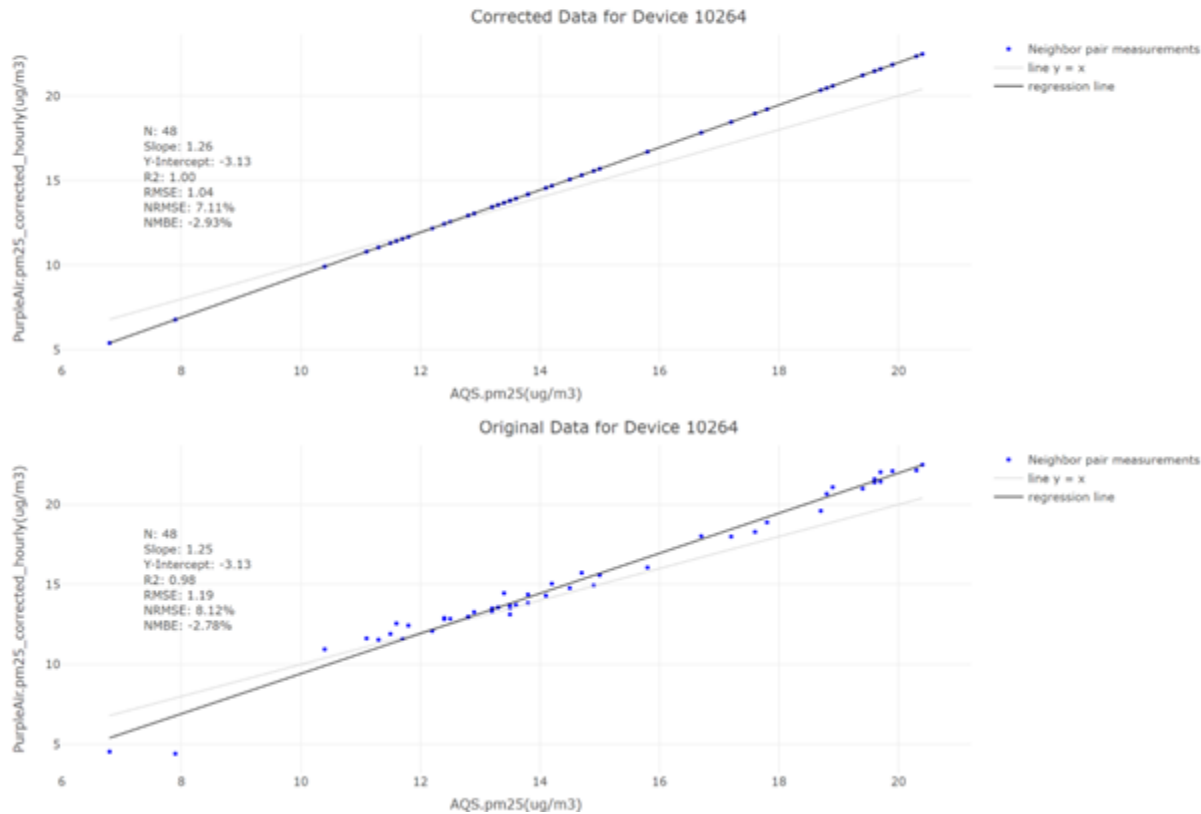


Device Details

Select Device ID:

10264

Interactive Plot



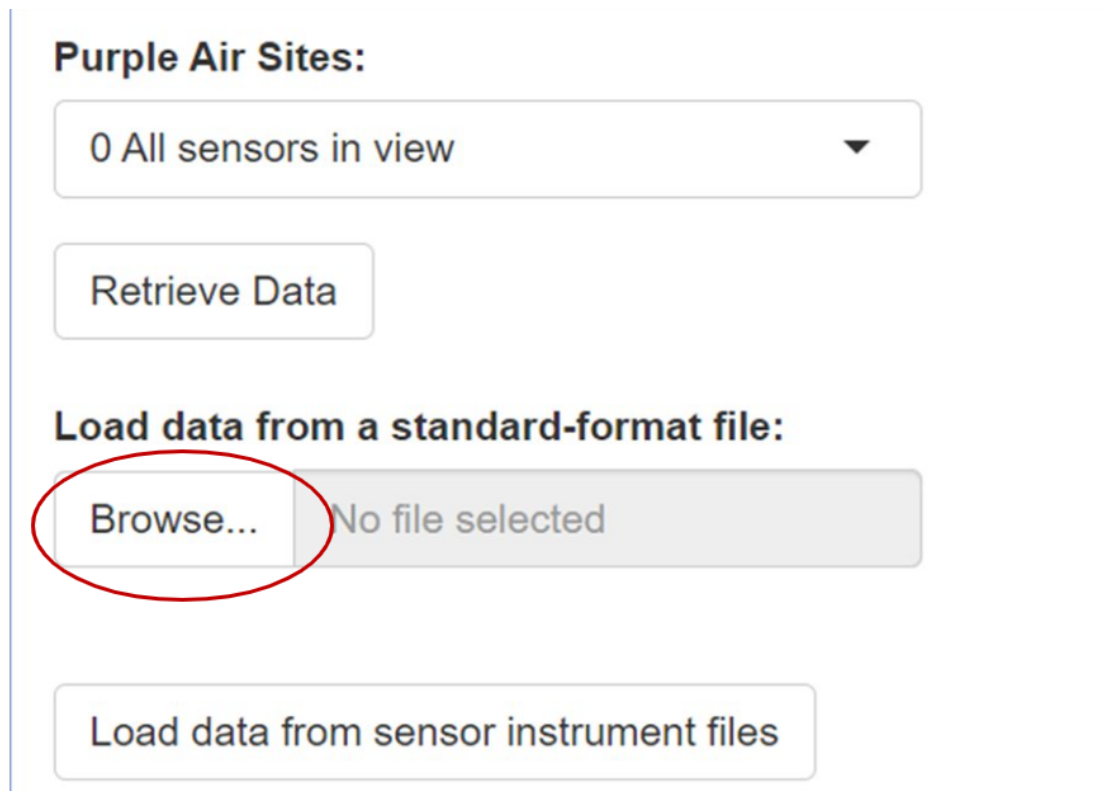
timestamp(UTC)	AQS.id(-)	AQS.pm25(ug/m3)	PurpleAir.id(-)	PurpleAir.pm25_corrected_hourly(ug/m3)	PurpleAir.flagged(-)	PurpleAir.pm25_corrected_hourly(ug/m3)_corrected
2022-06-01T00:00:00.0000	420033007	11.40	9652	9.60	0	10.00
2022-06-01T01:00:00.0000	420033007	12.30	9652	10.35	0	10.86
2022-06-01T02:00:00.0000	420033007	12.90	9652	11.16	0	11.44
2022-06-01T03:00:00.0000	420033007	13.50	9652	12.58	0	12.01
2022-06-01T04:00:00.0000	420033007	13.50	9652	12.73	0	12.01
2022-06-01T05:00:00.0000	420033007	14.10	9652	13.54	0	12.58
2022-06-01T06:00:00.0000	420033007	15.00	9652	14.08	0	13.44
2022-06-01T07:00:00.0000	420033007	16.30	9652	14.63	0	14.68
2022-06-01T08:00:00.0000	420033007	16.20	9652	14.44	0	14.58
2022-06-01T09:00:00.0000	420033007	16.30	9652	14.94	0	14.68
2022-06-01T10:00:00.0000	420033007	16.80	9652	15.04	0	15.16
2022-06-01T11:00:00.0000	420033007	17.60	9652	16.49	0	15.92
timestamp(UTC)	AQS.id(-)	AQS.pm25(ug/m3)	PurpleAir.id(-)	PurpleAir.pm25_corrected_hourly(ug/m3)	PurpleAir.flagged(-)	PurpleAir.pm25_corrected_hourly(ug/m3)_corrected
2022-06-01T00:00:00.0000	510870014	17.60	10264	18.29	0	18.96
2022-06-01T01:00:00.0000	510870014	18.70	10264	19.60	0	20.34
2022-06-01T02:00:00.0000	510870014	19.60	10264	21.38	0	21.47
2022-06-01T03:00:00.0000	510870014	19.60	10264	21.59	0	21.47
2022-06-01T04:00:00.0000	510870014	19.70	10264	21.43	0	21.59
2022-06-01T05:00:00.0000	510870014	20.30	10264	22.14	0	22.35
2022-06-01T06:00:00.0000	510870014	20.40	10264	22.49	0	22.47
2022-06-01T07:00:00.0000	510870014	19.90	10264	22.10	0	21.84
2022-06-01T08:00:00.0000	510870014	19.70	10264	22.03	0	21.59
2022-06-01T09:00:00.0000	510870014	19.40	10264	21.00	0	21.22
2022-06-01T10:00:00.0000	510870014	18.90	10264	21.09	0	20.59
2022-06-01T11:00:00.0000	510870014	18.80	10264	20.67	0	20.46

Demo 7: Additional Flagging:

7.1 Rerun ASNAT:

ASNAT

7.2 Load the sample data PurpleAir_pm25_corrected_hourly_2022-06-01_to_2022-06-02.tsv through "**Load data from a standard-format file**" in \ASNAT\ sample_data\ standard_format_file.



The screenshot shows a web interface for 'Purple Air Sites'. At the top, there is a dropdown menu showing '0 All sensors in view' with a downward arrow. Below this is a 'Retrieve Data' button. The next section is titled 'Load data from a standard-format file:'. In this section, there is a 'Browse...' button circled in red, followed by a greyed-out area that says 'No file selected'. At the bottom of this section is a button labeled 'Load data from sensor instrument files'.

Scroll down to the **Dataset Y Neighbor Flagging** section and change the **Neighbor distance(m)** to 5000.

Select Dataset X:

PurpleAir.hourly_data ▼

Select Variable Of Dataset X:

pm25_corrected_hourly(ug/m3) ▼

Select Comparison Dataset Y: (optional)

none ▼

Select Variable Of Dataset Y:

▼

Neighbor distance(m): ⓘ

5000

Save to Directory:

7.3 Use the mouse to click the **Flagging** tab.

Map **Flagging** Tables Plots Corrections Network Summary

Dataset to Flag: Dataset X

Search:

	timestamp(UTC)	longitude(deg)	latitude(deg)	elevation(m)	id(-)	count(-)	pm25_corrected_hourly(ug/m3)	flagged(-)
1	2022-06-01T00:00:00-0000	-66.5041	45.75441	56.6928	91257	30	2.1157	0
2	2022-06-01T00:00:00-0000	-70.52792	44.55543	193.8528	133491	30	1.05933	0
3	2022-06-01T00:00:00-0000	-71.65979	42.32031	93.2688	132589	30	1.90827	0
4	2022-06-01T00:00:00-0000	-73.84475	40.7117	21.6408	24311	30	13.73281	0
5	2022-06-01T00:00:00-0000	-76.60104	39.19797	11.5824	4602	30	17.86321	0
6	2022-06-01T00:00:00-0000	-77.20436	38.88765	140.208	91157	29	16.95945	0
7	2022-06-01T00:00:00-0000	-78.4348	38.5231	1069.2384	98407	29	12.23101	0
8	2022-06-01T00:00:00-0000	-79.8745	40.34514	291.6936	5528	30	8.98656	0

Scroll down to the bottom, and then click the **Additional Flags** tab.

Dataset Y Neighbor Flagging **Additional Flags** Invalid Values Flags Cross-field validation flags Suspicious changes flags Redundancy check

Invalid formatting and units

☐ Apply flag 83 if a constant value persists for three or more consecutive measurements.

☐ Apply flag 84 if missing data for more than x consecutive timesteps

☐ Apply flag 85 if data point is a statistical outlier.

☐ Apply outlier detection within a specified time window

☐ Apply flag 86 if data point is an outlier by hampel filter. Window Threshold

7.4 Click **Apply Flag 83 to enable the flag, which checks if a constant value persists for three or more consecutive measurements.**

Dataset Y Neighbor Flagging

Additional Flags

Invalid Values Flags

Cross-field validation flags

Suspicious changes flags

Redundancy check

Invalid formatting and units

☒ Apply flag 83 if a constant value persists for three or more consecutive measurements.

3

☐ Apply flag 84 if missing data for more than x consecutive timesteps

3

☐ Apply flag 85 if data point is a statistical outlier.

3

☐ Apply outlier detection within a specified time window

7.5 Scroll up and click the **Refresh Table Using Settings Below button to apply the preset flag conditions listed below to the flag table.**

Exclude 99 from flagged column

Include 99 in flagged column

Load flag conditions from a file:

Browse...

No file selected

Flag Conditions:

Apply Flag Conditions Above

Clear Flagging

Save Flag Conditions

Refresh Table Using Settings Below

Dataset Y Neighbor Flagging

Additional Flags

Invalid Values Flags

Cross-field validation flags

Suspicious changes flags

Redundancy check

Invalid formatting and units

☒ Apply flag 83 if a constant value persists for three or more consecutive measurements.

3

☐ Apply flag 84 if missing data for more than x consecutive timesteps

3

7.6 Scroll up and review the new flags in the table.

Show entries

Search:

	timestamp(UTC)	longitude(deg)	latitude(deg)	elevation(m)	id(-)	count(-)	pm25_corrected_hourly(ug/m3)	flagged(-)	note(-)
1	2022-06-01T00:00:00-0000	-66.5041	45.75441	56.6928	91257	30	2.1157	83	AQSU-ASD6
2	2022-06-01T00:00:00-0000	-70.52792	44.55543	193.8528	133491	30	1.05933	0	Mountain Valley Middle School
3	2022-06-01T00:00:00-0000	-71.65979	42.32031	93.2688	132589	30	1.90827	0	Melican Middle School
4	2022-06-01T00:00:00-0000	-73.84475	40.7117	21.6408	24311	30	13.73281	0	Forest Hills
5	2022-06-01T00:00:00-0000	-76.60104	39.19797	11.5824	4602	30	17.86321	0	ex
6	2022-06-01T00:00:00-0000	-77.20436	38.88765	140.208	91157	29	16.95945	0	Falls Church - Westwood Park
7	2022-06-01T00:00:00-0000	-78.4348	38.5231	1069.2384	98407	29	12.23101	0	SHEN- Big Meadows
8	2022-06-01T00:00:00-0000	-79.8745	40.34514	291.6936	5528	30	8.98656	0	Port Vue PA
9	2022-06-01T00:00:00-0000	-80.0476	40.30428	358.7496	9036	30	9.02467	0	ROCIS_h5w7_O
10	2022-06-01T00:00:00-0000	-80.30667	40.69105	240.1824	94491	30	8.58554	0	CMP11 - Beaver

Showing 1 to 10 of 28,674 entries

Previous 2 3 4 5 ... 2,868 Next

Load flag conditions from a file:

7.7 Scroll down to Flag Conditions and click the **Clear Flagging** button to reset the flags in the table.

Exclude 99 from flagged column

Include 99 in flagged column

Load flag conditions from a file:

Browse...

No file selected

Flag Conditions:

Apply Flag Conditions Above

Clear Flagging

Save Flag Conditions

Refresh Table Using Settings Below

7.8 Scroll down to the Dataset Y Neighbor Flagging section.

7.9 Click Apply Flag 83 again to disable the flag.

Dataset Y Neighbor Flagging

Additional Flags

Invalid Values Flags

Cross-field validation flags

Suspicious changes flags

Redundancy check

Invalid formatting and units

☐ Apply flag 83 if a constant value persists for three or more consecutive measurements.

3

☐ Apply flag 84 if missing data for more than x consecutive timesteps

3

☐ Apply flag 85 if data point is a statistical outlier.

3

☐ Apply outlier detection within a specified time window

☐ Apply flag 86 if data point is an outlier by hamper filter.

Window

3

Threshold

3

7.10 Click Apply Flag 84 to enable the, which checks if a missing value persists for x or more consecutive measurements. The value for x can be adjusted by entering a new number in the text input field.

Save Flag Conditions

Refresh Table Using Settings Below

Dataset Y Neighbor Flagging

Additional Flags

Invalid Values Flags

Cross-field validation flags

Suspicious changes flags

Redundancy check

Invalid formatting and units

☐ Apply flag 83 if a constant value persists for three or more consecutive measurements.

3

☒ Apply flag 84 if missing data for more than x consecutive timesteps

3

☐ Apply flag 85 if data point is a statistical outlier.

3

☐ Apply outlier detection within a specified time window

☐ Apply flag 86 if data point is an outlier by hampel filter.

Window

3

Threshold

3

7.11 Scroll up and click the **Refresh Table Using Settings Below** button. Review the new flags in the table. (Since ASNAT currently does not load missing data into the dataset, no changes will be displayed.)

Map

Flagging

Tables

Plots

Dataset to Flag:

Dataset Y

Show

10

entries

Search:

	timestamp(UTC)	longitude(deg)	latitude(deg)	elevation(m)	id(-)	count(-)	pm25_corrected_hourly(ug/m3)	flagged(-)	note(-)
1	2022-06-01T00:00:0000	-66.5041	45.75441	56.6928	91257	30	2.1157	0	AQSU-A8D6
2	2022-06-01T00:00:0000	-70.52792	44.55543	193.8528	133491	30	1.05933	0	Mountain Valley Middle School
3	2022-06-01T00:00:0000	-71.65979	42.32031	93.2688	132589	30	1.90827	0	Melican Middle School
4	2022-06-01T00:00:0000	-73.84475	40.7117	21.6408	24311	30	13.73281	0	Forest Hills
5	2022-06-01T00:00:0000	-76.60104	39.19797	11.5824	4802	30	17.86321	0	ex
6	2022-06-01T00:00:0000	-77.20436	38.88765	140.208	91157	29	16.95945	0	Falls Church - Westwood Park
7	2022-06-01T00:00:0000	-78.4348	38.5231	1069.2384	98407	29	12.23101	0	SHEN- Big Meadows
8	2022-06-01T00:00:0000	-79.8745	40.34514	291.6936	5528	30	8.98656	0	Port Vue PA
9	2022-06-01T00:00:0000	-80.0476	40.30428	358.7496	9036	30	9.02467	0	ROCIS_h5w7_O
10	2022-06-01T00:00:0000	-80.30667	40.69105	240.1824	94491	30	8.58554	0	CMP11 - Beaver

Showing 1 to 10 of 28,674 entries

Previous

1

2

3

4

5

...

2,868

Next

Exclude 99 from flagged column

Include 99 in flagged column

7.12 Scroll down to the **Dataset Y Neighbor Flagging** section.

7.13 Click Apply Flag 84 again to disable the flag.

Dataset Y Neighbor Flagging Additional Flags Invalid Values Flags Cross-field validation flags Suspicious changes flags Redundancy check

Invalid formatting and units

☐ Apply flag 83 if a constant value persists for three or more consecutive measurements.

☐ Apply flag 84 if missing data for more than x consecutive timesteps

☐ Apply flag 85 if data point is a statistical outlier.

☐ Apply outlier detection within a specified time window

☐ Apply flag 86 if data point is an outlier by hampel filter. Window Threshold

7.14 Click Apply Flag 85 to enable the flag, which identifies data as a statistical outlier if the standard deviation is a factor of **X** greater than the mean.

7.15 Change the corresponding value X to 1.

7.16 Click Apply Outlier Detection within a Specified Time Window to enable the time window filter. In the **Start Time** field (YYYY-MM-DDTHH:MM), type **2022-06-01T00:00:00-0000**, and in the **End Time** field (YYYY-MM-DDTHH:MM), type **2022-06-01T08:00:00-0000**.

☐ Apply flag 83 if a constant value persists for three or more consecutive measurements. **Click to enable the flag**

☐ Apply flag 84 if missing data for more than x consecutive timesteps **Change the value to adjust the X**

☒ Apply flag 85 if data point is a statistical outlier.

☒ Apply outlier detection within a specified time window

Start Time for the time window filter. **End Time for the time window filter.**

Start Time (YYYY-MM-DDTHH:MM:SS-0000) End Time (YYYY-MM-DDTHH:MM:SS-0000)

☒ Apply flag 85 if data point is a statistical outlier.

1

☒ Apply outlier detection within a specified time window

Start Time (YYYY-MM-DDTHH:MM:SS-0000)

2022-06-01T00:00:00-0000

End Time (YYYY-MM-DDTHH:MM:SS-0000)

2022-06-01T08:00:00-0000

7.17 Scroll up and click the **Refresh Table Using Settings Below** button to apply **Flag 85**.

7.18 Click the **Flagged (-)** column to sort the table and display 85 flagged entries.

	timestamp(UTC)	longitude(deg)	latitude(deg)	elevation(m)	id(-)	count(-)	pm25_corrected_hourly(ug/m3)	flagged(-)	note(-)
564	2022-06-01T00:01:12-0000	-71.10436	42.38097	6.096	3642	25	-1.63504	85	Dane St & Village St.
1807	2022-06-01T03:00:08-0000	-72.2593	42.93803	241.4016	1224	23	0.95097	85	KSC 17
2337	2022-06-01T03:03:14-0000	-71.10436	42.38097	6.096	3642	26	-1.91622	85	Dane St & Village St.
2499	2022-06-01T04:00:09-0000	-72.2593	42.93803	241.4016	1224	30	0.32927	85	KSC 17
3000	2022-06-01T04:01:15-0000	-71.10436	42.38097	6.096	3642	24	-2.06457	85	Dane St & Village St.
3158	2022-06-01T05:00:10-0000	-72.2593	42.93803	241.4016	1224	30	-0.36902	85	KSC 17
3656	2022-06-01T05:01:15-0000	-71.10436	42.38097	6.096	3642	23	-1.9219	85	Dane St & Village St.
3764	2022-06-01T06:00:10-0000	-72.2593	42.93803	241.4016	1224	30	-0.50745	85	KSC 17
4357	2022-06-01T07:00:11-0000	-72.2593	42.93803	241.4016	1224	30	-0.44966	85	KSC 17
4871	2022-06-01T07:03:16-0000	-71.10436	42.38097	6.096	3642	25	-2.03527	85	Dane St & Village St.

Demo 8: Site Neighbor Plots:

ASNAT has three modes of plotting:

8.1 Single entire dataset

8.2 Compared pair of datasets X, Y - i.e., only points that are neighbors.

8.3 Single selected Dataset X site and its Dataset Y neighbors.

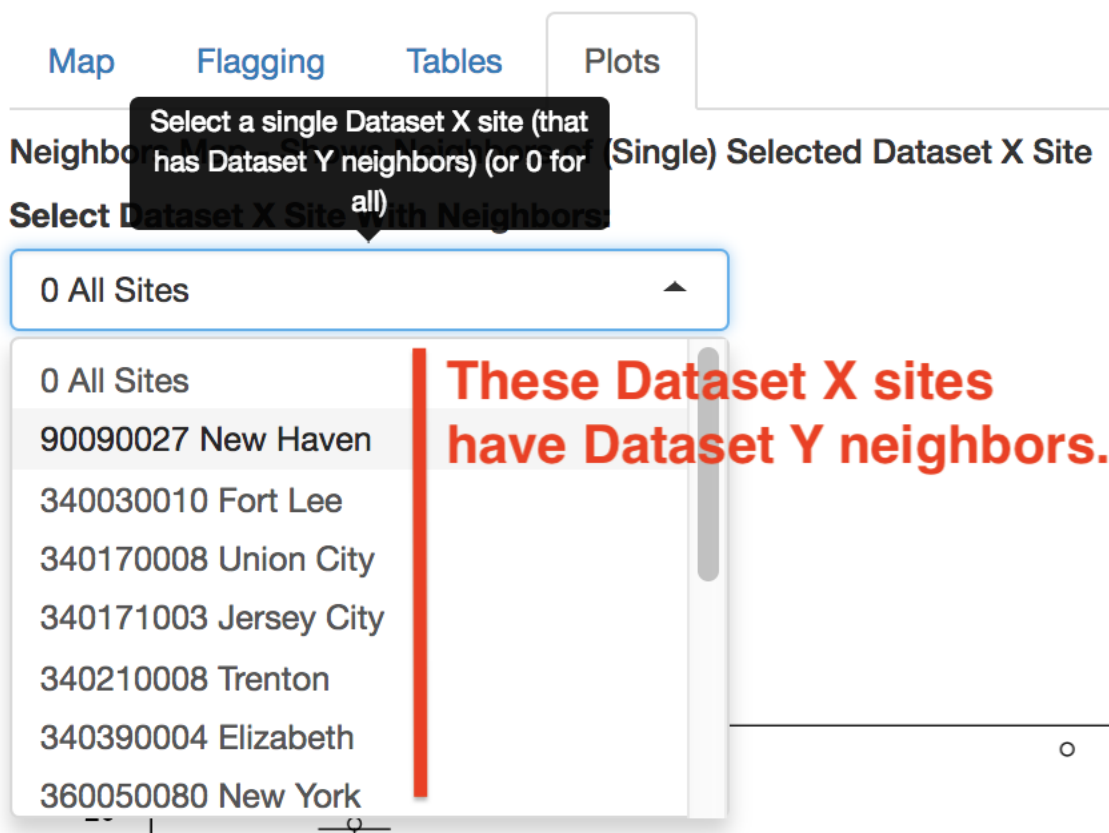
This demo is the last mode.

It will plot a single AQS.pm25 site with its Dataset Y neighbors.

Continuing on from Demo 4 on the Plots tab:

Note the menu 'Select Dataset X Site With Neighbors'.

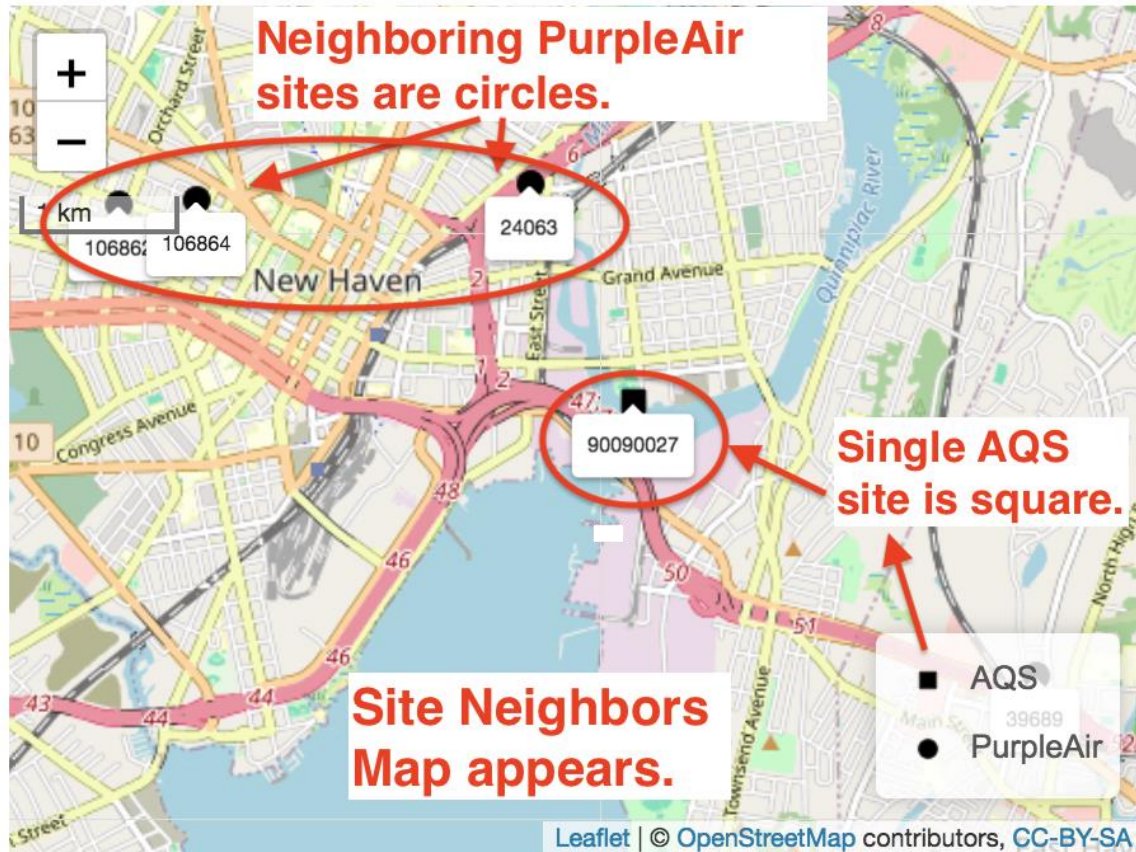
It defaults to '0 - All Sites' which is used in the first two modes.



Click the menu to see a list of Dataset X sites that have Dataset Y neighbors. **Click on '90090027 New Haven'** to select this AQS site (and its PurpleAir neighbors).

[Map](#)[Flagging](#)[Tables](#)[Plots](#)

Neighbors Map - Shows Neighbors of (Single) Selected Dataset X Site

☒ Show site labels**3. Select 'Show site labels'.****Select Dataset X Site With Neighbors:****1. Select New Haven**

90090027 New Haven

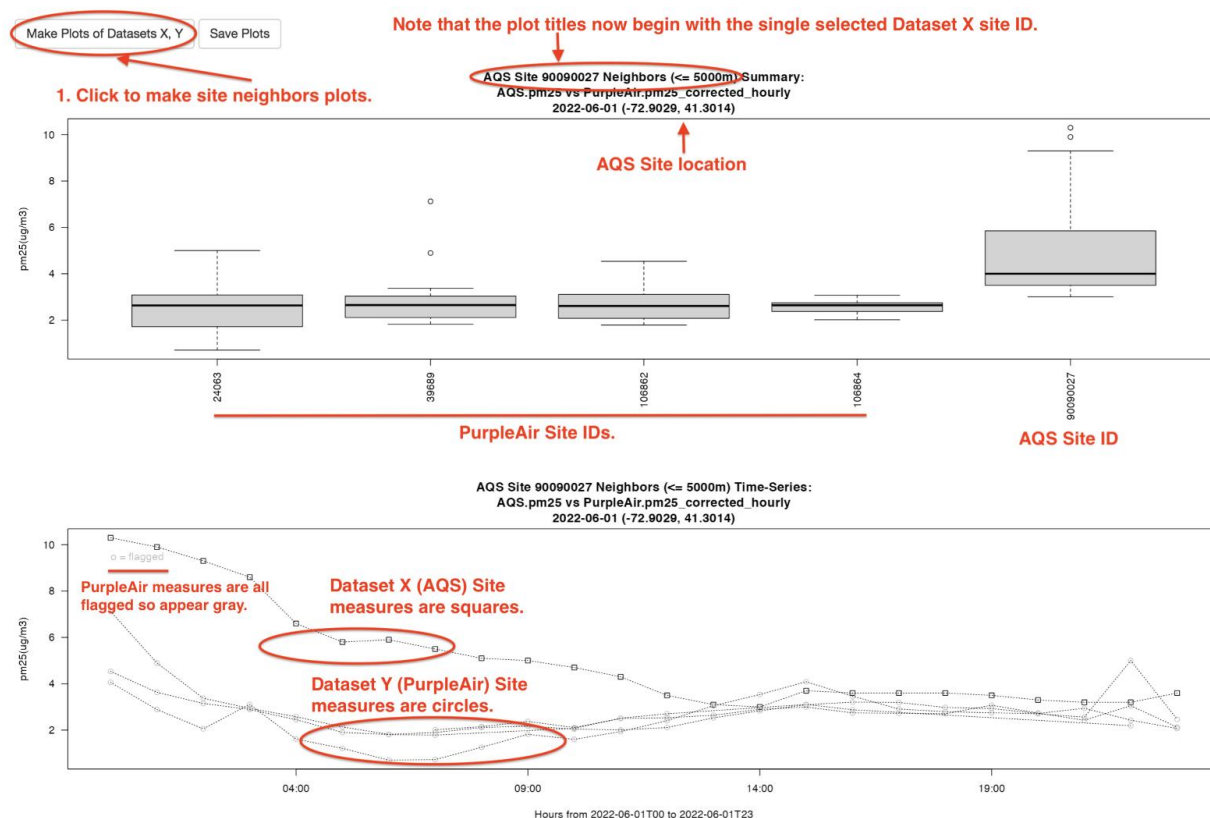
The Neighbors Map shows the location of the single selected Dataset X site and its Dataset Y neighbors. In this example, since the Dataset X is AQS, its site appears as a square glyph on the map. And since Dataset Y is PurpleAir, its sites appear as circles as shown by the map legend.

The 'Show site labels' button adds the site ID to each site displayed.

Clicking on a site pops up a larger label that shows the source, id, and note for that site.

Click the 'Make Plots of Dataset X, Y' button.

This changes the plots from mode 2 to mode 3 - i.e., from Compared Dataset to a single site plus its neighbors.



The Summary boxplot now shows the single selected Dataset X site (AQS 90090027) along with its four Dataset Y (PurpleAir) neighbors - IDs 24063, 39689, 106882, 106864).

The plot titles now begin with the single selected Dataset X site and show its longitude, latitude location, and maximum neighbor distance.

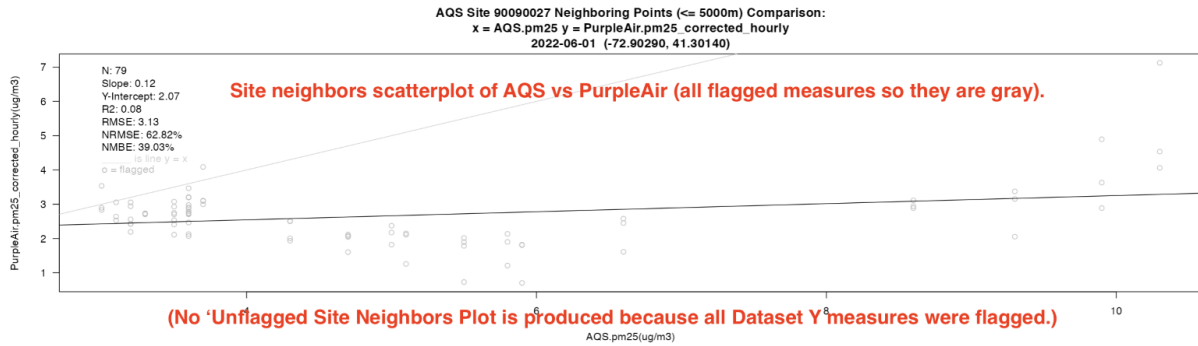
The second plot is the time series of these five sites.

Note that the Dataset X site measures use square points while the Dataset Y measures use circles.

Also, in this example, due to extensive flagging, all Dataset Y measures were flagged so they appear gray instead of black.

Because there are no unflagged measures, there is no 'Unflagged' scatterplot.

(No Dataset Y Summary Boxplot or Time-series plots are produced since we are doing Site Neighbor Plots.)



Demo 9: Moving Average:

ASNAT can correct repetitive/constant values with a moving average.

To use this feature, please refer to Demo 7: Additional Flagging, and ensure that flag 83 is enabled and applied (by clicking the "**Refresh Table Using Settings Below**" button).

Relaunch ASNAT

Load the sample data AQS_pm25_2022-06-01_to_2022-06-02.tsv through "load data from a standard-format file" in \ASNAT\ sample_data\ standard_format_file

Purple Air Sites:

0 All sensors in view ▼

Retrieve Data

Load data from a standard-format file:

Browse...

No file selected

Load data from sensor instrument files

Name	Date modified	Type	Size
Home-Rubber-SD-card-CSV-June2022	10/18/2024 2:04 PM	File folder	
standard_format_file	10/18/2024 2:05 PM	File folder	

Name	Date modified	Type	Size
AQS_pm25_2022-06-01_to_2022-06-02.tsv	10/18/2024 2:03 PM	TSV File	504 KB
PurpleAir_pm25_corrected_hourly_2022-06-01_t...	10/18/2024 2:03 PM	TSV File	2,479 KB

Go to the **“Correction”** tab.
Click the **“Moving Average”** button.

Map Flagging Tables Plots Corrections Network Summary

Regression Correction **Moving Averages**

Moving Average Panel

ⓘ Please ensure you have applied the constant value flag (flag 83) before proceeding with moving averages.
Records with constant values will be displayed below.

Records with Constant Values (Flag 83)

Show 10 ▼ entries

Search:

Records Flagged with Constant Value (83)

Click the “**id(-)**” to sort the repetitive values by ID.
Select all entries from ID 90010010.

Records with Constant Values (Flag 83)

Show entries

Search:

Records Flagged with Constant Value (83)

	timestamp(UTC)	longitude(deg)	latitude(deg)	id(-)	pm25(ug/m3)	flagged(-)	note(-)
827	2022-06-01T06:00:00-0000	-73.1947	41.1708	90010010	6.5	83	Bridgeport
985	2022-06-01T07:00:00-0000	-73.1947	41.1708	90010010	6.5	83	Bridgeport
1142	2022-06-01T08:00:00-0000	-73.1947	41.1708	90010010	6.5	83	Bridgeport
2543	2022-06-01T17:00:00-0000	-73.1947	41.1708	90010010	4.3	83	Bridgeport
2700	2022-06-01T18:00:00-0000	-73.1947	41.1708	90010010	4.3	83	Bridgeport
2859	2022-06-01T19:00:00-0000	-73.1947	41.1708	90010010	4.3	83	Bridgeport
6959	2022-06-02T21:00:00-0000	-72.6799	41.7714	90030025	7.2	83	Hartford
7118	2022-06-02T22:00:00-0000	-72.6799	41.7714	90030025	7.2	83	Hartford
7276	2022-06-02T23:00:00-0000	-72.6799	41.7714	90030025	7.2	83	Hartford
2395	2022-06-01T16:00:00-0000	-72.9029	41.3014	90090027	3.6	83	New Haven

Showing 1 to 10 of 484 entries

Previous 2 3 4 5 ... 49 Next

☒ Select All

☐ Unselect All

Under “**Number of records selected**” you can confirm the number of records and devices selected.

Enter 9 in the Window Size (the window size should be greater than the selected records to achieve the best moving average result).

Click “**Calculate Moving Average**” to generate the result.

☒ Select All
 ☒ Unselect All

Selected Records:

Number of records selected: 6

Selected Devices:

Device: 90010010 (6 records)

Window Size (hours):

9

Calculate Moving Average

Select Device ID:

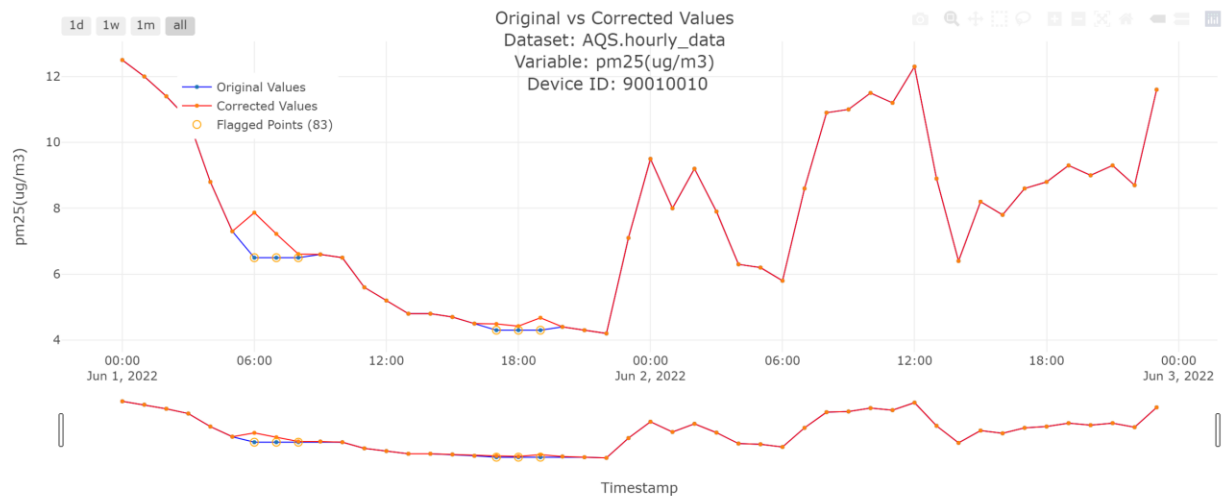
▼

From the drop-down menu, select “Device ID” to switch to the results of different devices (if any records are selected in the prior steps).

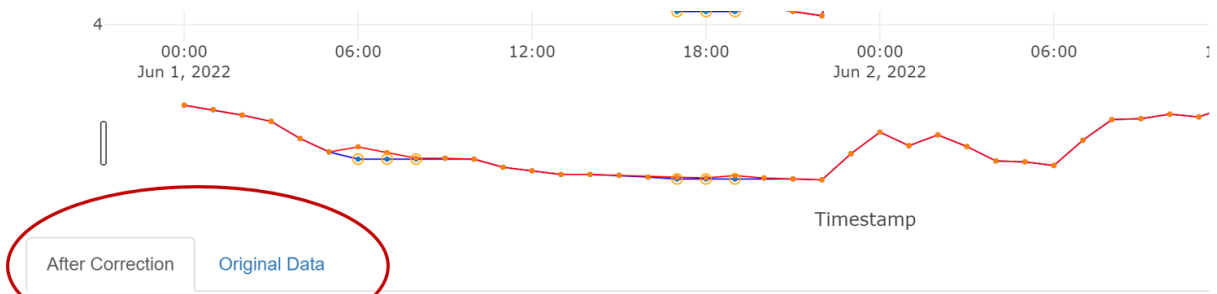
Select Device ID:

90010010 ▼

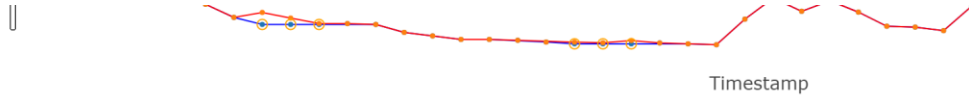
Results



Click “After Correction” and “Original Data” tabs to switch between different table views for corrected and original data.



timestamp(UTC)	longitude(deg)	latitude(deg)	id(-)	pm25(ug/m3)	flagged(-)	note(-)
2022-06-01T00:00:00-0000	-73.19	41.17	90010010	12.50	0	Bridgeport
2022-06-01T01:00:00-0000	-73.19	41.17	90010010	12.00	0	Bridgeport
2022-06-01T02:00:00-0000	-73.19	41.17	90010010	11.40	0	Bridgeport
2022-06-01T03:00:00-0000	-73.19	41.17	90010010	10.70	0	Bridgeport
2022-06-01T04:00:00-0000	-73.19	41.17	90010010	8.80	0	Bridgeport
2022-06-01T05:00:00-0000	-73.19	41.17	90010010	7.30	0	Bridgeport
2022-06-01T06:00:00-0000	-73.19	41.17	90010010	7.87	83	Bridgeport
2022-06-01T07:00:00-0000	-73.19	41.17	90010010	7.22	83	Bridgeport
2022-06-01T08:00:00-0000	-73.19	41.17	90010010	6.61	83	Bridgeport
2022-06-01T09:00:00-0000	-73.19	41.17	90010010	6.60	0	Bridgeport
2022-06-01T10:00:00-0000	-73.19	41.17	90010010	6.50	0	Bridgeport
2022-06-01T11:00:00-0000	-73.19	41.17	90010010	5.60	0	Bridgeport
2022-06-01T12:00:00-0000	-73.19	41.17	90010010	5.20	0	Bridgeport
2022-06-01T13:00:00-0000	-73.19	41.17	90010010	4.80	0	Bridgeport
2022-06-01T14:00:00-0000	-73.19	41.17	90010010	4.80	0	Bridgeport
2022-06-01T15:00:00-0000	-73.19	41.17	90010010	4.70	0	Bridgeport
2022-06-01T16:00:00-0000	-73.19	41.17	90010010	4.50	0	Bridgeport
2022-06-01T17:00:00-0000	-73.19	41.17	90010010	4.49	83	Bridgeport
2022-06-01T18:00:00-0000	-73.19	41.17	90010010	4.42	83	Bridgeport
2022-06-01T19:00:00-0000	-73.19	41.17	90010010	4.68	83	Bridgeport
2022-06-01T20:00:00-0000	-73.19	41.17	90010010	4.40	0	Bridgeport
2022-06-01T21:00:00-0000	-73.19	41.17	90010010	4.30	0	Bridgeport



After Correction

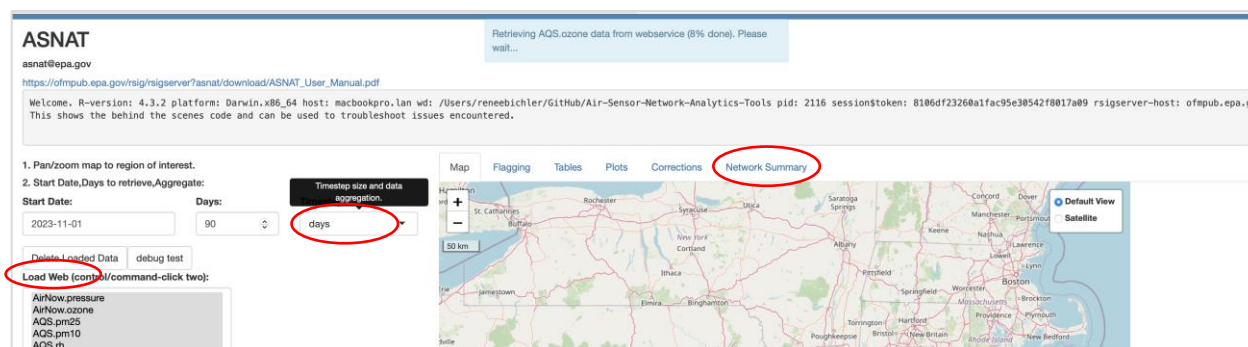
Original Data

timestamp(UTC)	longitude(deg)	latitude(deg)	id(-)	pm25(ug/m3)	flagged(-)	note(-)
2022-06-01T00:00:00-0000	-73.19	41.17	90010010	12.50	0	Bridgeport
2022-06-01T01:00:00-0000	-73.19	41.17	90010010	12.00	0	Bridgeport
2022-06-01T02:00:00-0000	-73.19	41.17	90010010	11.40	0	Bridgeport
2022-06-01T03:00:00-0000	-73.19	41.17	90010010	10.70	0	Bridgeport
2022-06-01T04:00:00-0000	-73.19	41.17	90010010	8.80	0	Bridgeport
2022-06-01T05:00:00-0000	-73.19	41.17	90010010	7.30	0	Bridgeport
2022-06-01T06:00:00-0000	-73.19	41.17	90010010	6.50	83	Bridgeport
2022-06-01T07:00:00-0000	-73.19	41.17	90010010	6.50	83	Bridgeport
2022-06-01T08:00:00-0000	-73.19	41.17	90010010	6.50	83	Bridgeport
2022-06-01T09:00:00-0000	-73.19	41.17	90010010	6.60	0	Bridgeport
2022-06-01T10:00:00-0000	-73.19	41.17	90010010	6.50	0	Bridgeport
2022-06-01T11:00:00-0000	-73.19	41.17	90010010	5.60	0	Bridgeport
2022-06-01T12:00:00-0000	-73.19	41.17	90010010	5.20	0	Bridgeport
2022-06-01T13:00:00-0000	-73.19	41.17	90010010	4.80	0	Bridgeport
2022-06-01T14:00:00-0000	-73.19	41.17	90010010	4.80	0	Bridgeport
2022-06-01T15:00:00-0000	-73.19	41.17	90010010	4.70	0	Bridgeport
2022-06-01T16:00:00-0000	-73.19	41.17	90010010	4.50	0	Bridgeport
2022-06-01T17:00:00-0000	-73.19	41.17	90010010	4.30	83	Bridgeport
2022-06-01T18:00:00-0000	-73.19	41.17	90010010	4.30	83	Bridgeport
2022-06-01T19:00:00-0000	-73.19	41.17	90010010	4.30	83	Bridgeport
2022-06-01T20:00:00-0000	-73.19	41.17	90010010	4.40	0	Bridgeport
2022-06-01T21:00:00-0000	-73.19	41.17	90010010	4.30	0	Bridgeport

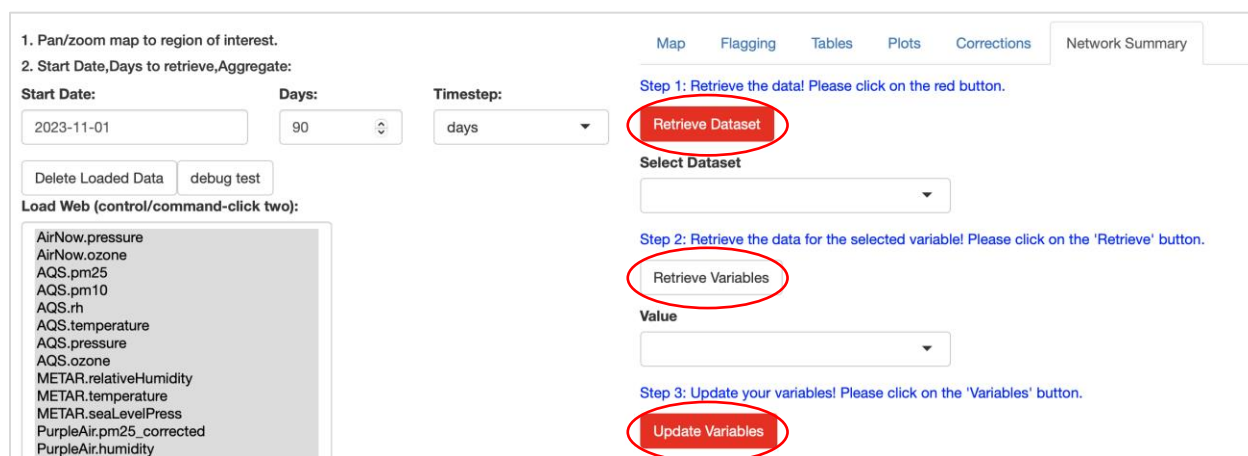
Demo 10: Network Summary:

The Network Summary Panel will give insights about basic statistics regarding selected data from the “**Load Web**” panel. Some plots depend on the selected timestep size which means hourly or daily data.

After loading the data frames via “**Load Web**” move over to the “**Network Summary**” panel.



After clicking on “**Retrieve Dataset**”, select a dataset from the dropdown menu for example AQS.pm25. The dropdown menu will offer different selection options depending on the loaded data from “**Load Web**”.



Map Flagging Tables Plots Corrections Network Summary

Step 1: Retrieve the data! Please click on the red button.

Retrieve Dataset

Select Dataset

AQS.pm25

AirNow.pressure

AirNow.ozone

AQS.pm25

AQS.pm10

AQS.rh

AQS.temperature

AQS.pressure

AQS.ozone

Update Variables

In the next step click on the **“Retrieve Variables”** button. This triggers the retrieval of the column names of the selected data frame. If own datasets were uploaded the selection will show different options.

Map Flagging Tables Plots Corrections Network Summary

Step 1: Retrieve the data! Please click on the red button.

Retrieve Dataset

Select Dataset

AQS.pm25

Step 2: Retrieve the data for the selected variable! Please click on the 'Retrieve' button.

Retrieve Variables

Value

pm25_daily_average(ug/m3)

pm25_daily_average(ug/m3)

Update Variables

With **“Update Variables”** the selected input variables will be used to process the data frame.

Map Flagging Tables Plots Corrections Network Summary

Step 1: Retrieve the data! Please click on the red button.

Retrieve Dataset

Select Dataset

AQS.pm25

Step 2: Retrieve the data for the selected variable! Please click on the 'Retrieve' button.

Retrieve Variables

Value

pm25_daily_average(ug/m3)

Step 3: Update your variables! Please click on the 'Variables' button.

Update Variables

AQS.pm25

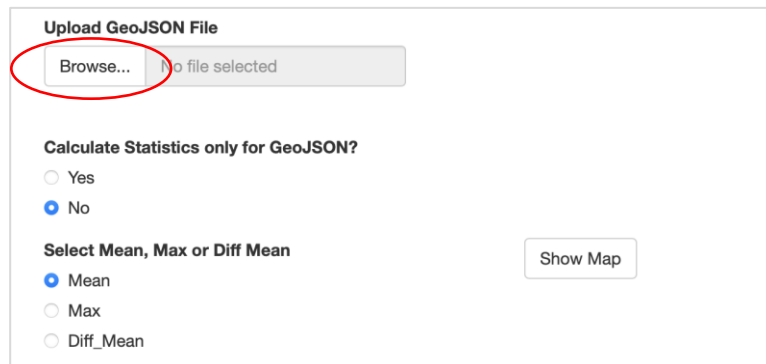
pm25_daily_average(ug/m3)

It is also possible to upload a GeoJSON file to either visually check if there are measurement stations in the polygon or to process the network summary statistics only for observations within the GeoJSON file. **This is an optional feature and can be skipped if not needed.**

To create a GeoJSON it is possible to use ArcGIS, QGIS, or simply visit www.geojson.io to create and download a polygon file.

Afterward, select if the statistics should be carried out with measurement sites inside the polygon or not. The default setting is set to “No” and can always be changed later on.

The selection below will show the mean or maximum value over the selected period for each measurement site. It is also possible to visualize how big the difference of the mean value for each site differs from the overall mean of all selected sites or only for sites within the polygon.



Upload GeoJSON File

No file selected

Calculate Statistics only for GeoJSON?

☐ Yes

☒ No

Select Mean, Max or Diff Mean

☒ Mean

☐ Max

☐ Diff_Mean

After checking the selected dataset and variables click on “**Show Map**”.

Map Flagging Tables Plots Corrections Network Summary

Step 1: Retrieve the data! Please click on the red button.

Retrieve Dataset

Select Dataset

AQS.pm25

Step 2: Retrieve the data for the selected variable! Please click on the 'Retrieve' button.

Retrieve Variables

Value

pm25_daily_average(ug/m3)

Step 3: Update your variables! Please click on the 'Variables' button.

Update Variables

AQS.pm25 pm25_daily_average(ug/m3)

Optional: Upload your own GeoJSON. An easy way to create one is under the following link: geojson.io

Upload GeoJSON File

Browse... map-2.geojson

Upload complete

Calculate Statistics only for GeoJSON?

☐ Yes

☒ No

Select Mean, Max or Diff Mean

☒ Mean

☐ Max

☐ Diff_Mean

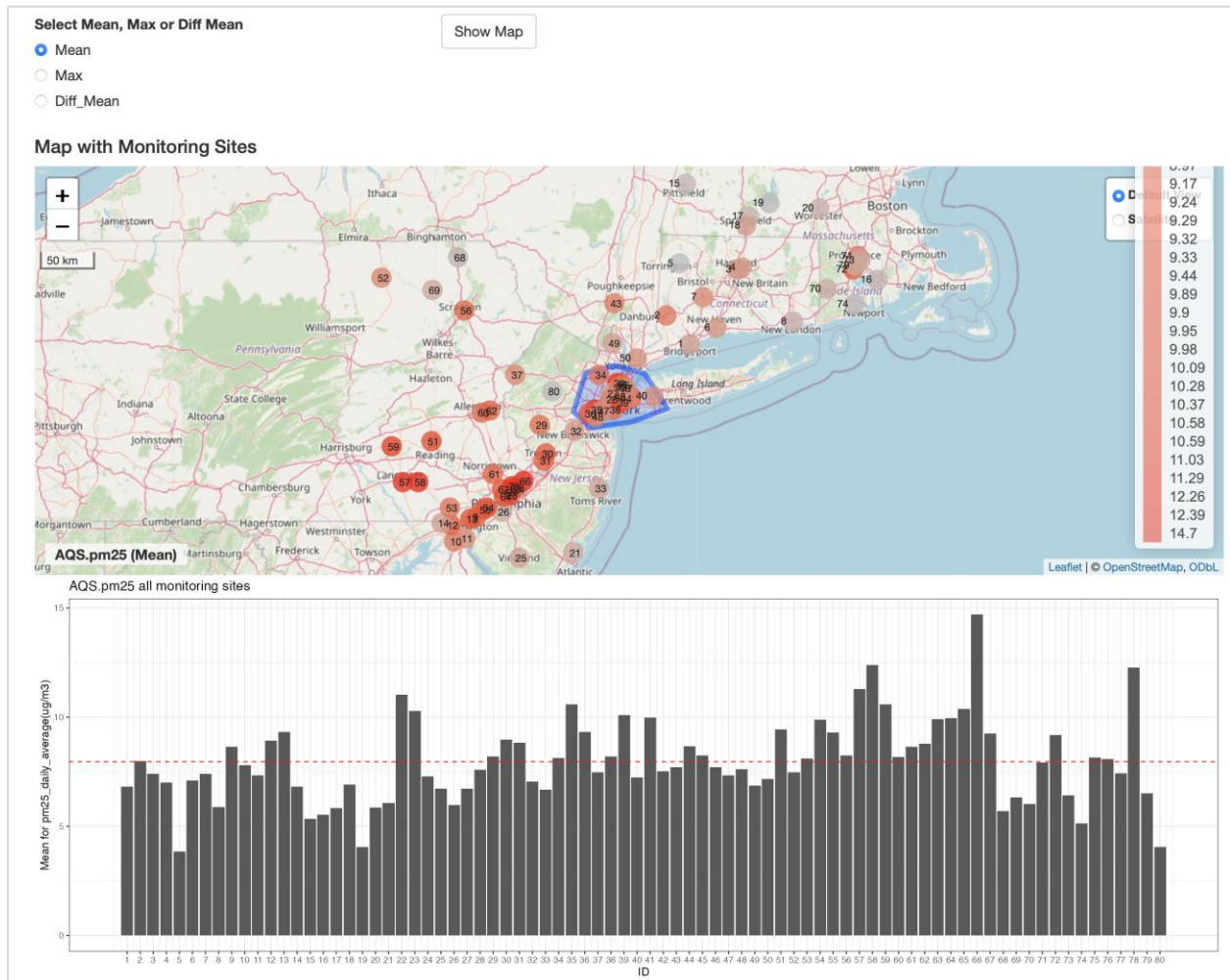
Show Map

A leaflet map with the uploaded GeoJSON should be displayed. The map title will appear in the bottom left corner. The values in the circle show the new ID number that is connected to the bar chart below.

Note: The new ID is not the same as the original measurement site ID.

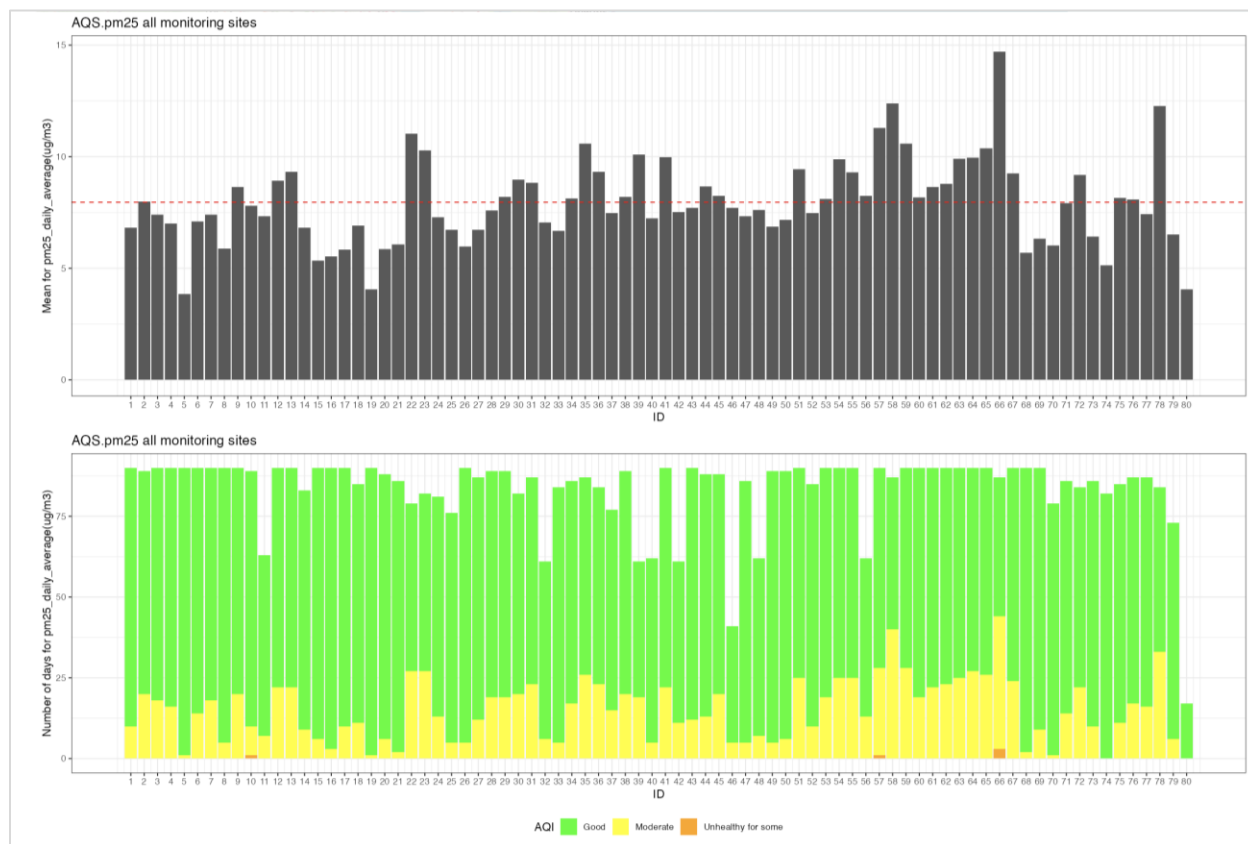
When clicking on one of the circle markers a popup feature will appear.

The red dotted line in the bar chart below presents the overall mean value for all monitoring sites displayed in the leaflet map. If the maximum values or the difference values for the mean value shall be displayed, simply select the option and click on “**Show Map**” again.

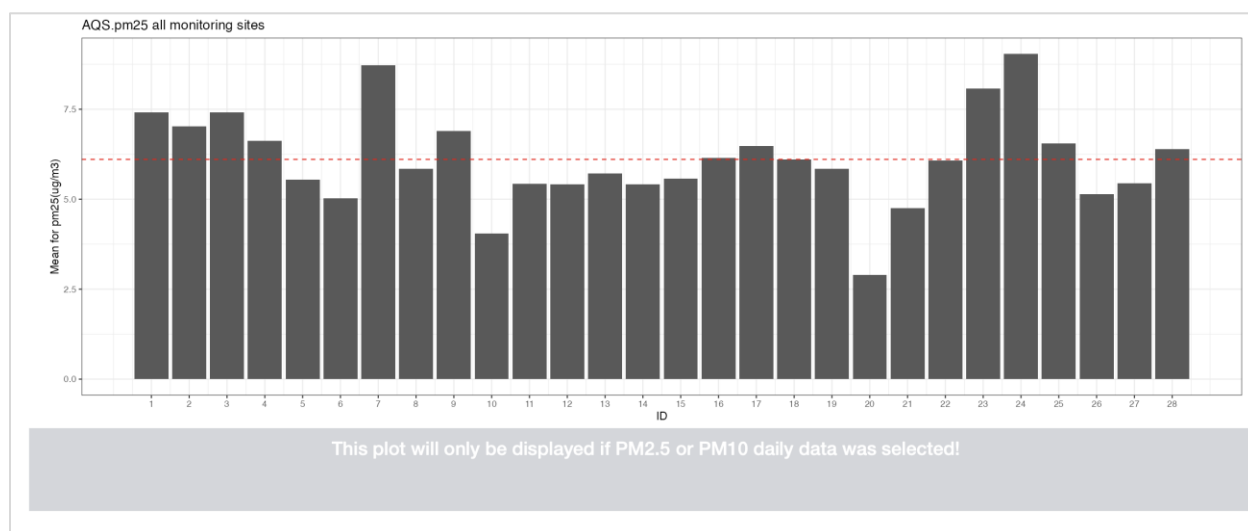


In this example, AQS.pm25 daily data was loaded. As a result, a second bar chart will be displayed that counts the number of days with a certain air quality during the selected time range.

Note: This plot only appears if PM2.5 or PM10 daily data was selected! Therefore, the plot will not appear if PM2.5 hourly data is loaded.



When hourly PM2.5 dataset was selected for example, the plot which represents the number of days with a certain air quality will not appear in the dashboard.



The network summary offers the possibility to produce a time series plot, a calendar series plot as well as a normal calendar plot. On top of that, the calendar plot offers to option to show the mean values of a day in the calendar or the number of days.

Plot daily mean Statistics

Create Time Series Create Calendar Series Create Calendar Plot

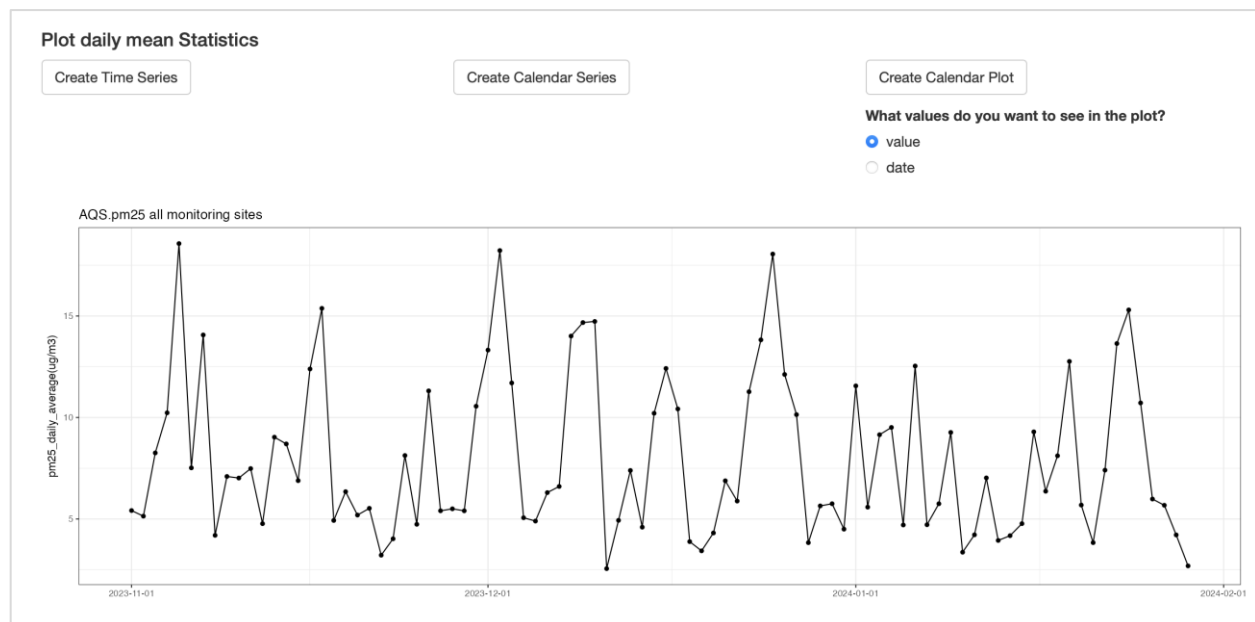
What values do you want to see in the plot?

☒ value

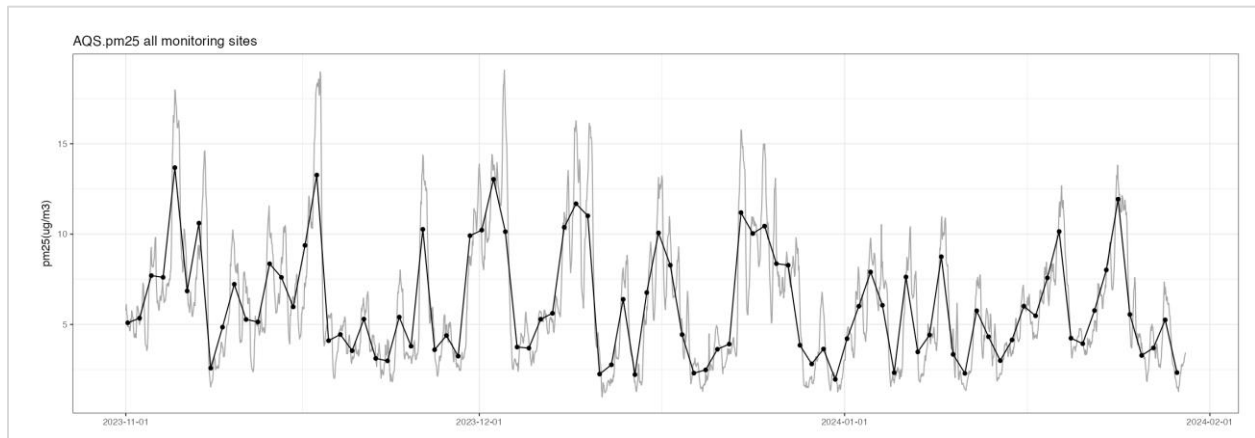
☐ date

When clicking on “**Create Time Series**” a time series of the daily mean of the selected data frame is processed.

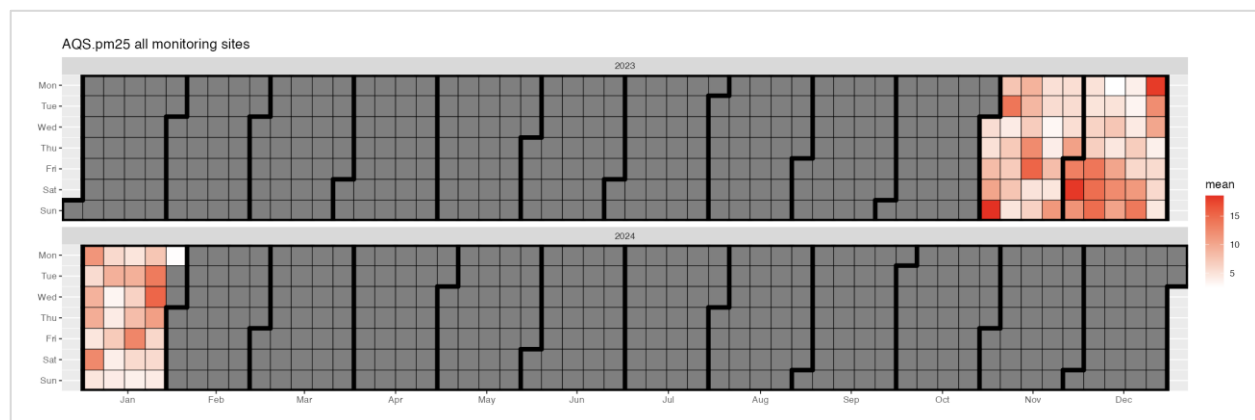
Note: When a GeoJSON was uploaded and only measurements of monitoring sites within the polygon are analyzed, the title of the plots will change. This indicates what area of interest was used, either “**all monitoring sites**” or “**within GeoJSON**”.



When a data frame with hourly data is loaded the time series plot will look like the following plot. The gray line represents the hourly data and the black line represents the daily mean based on the hourly input data.

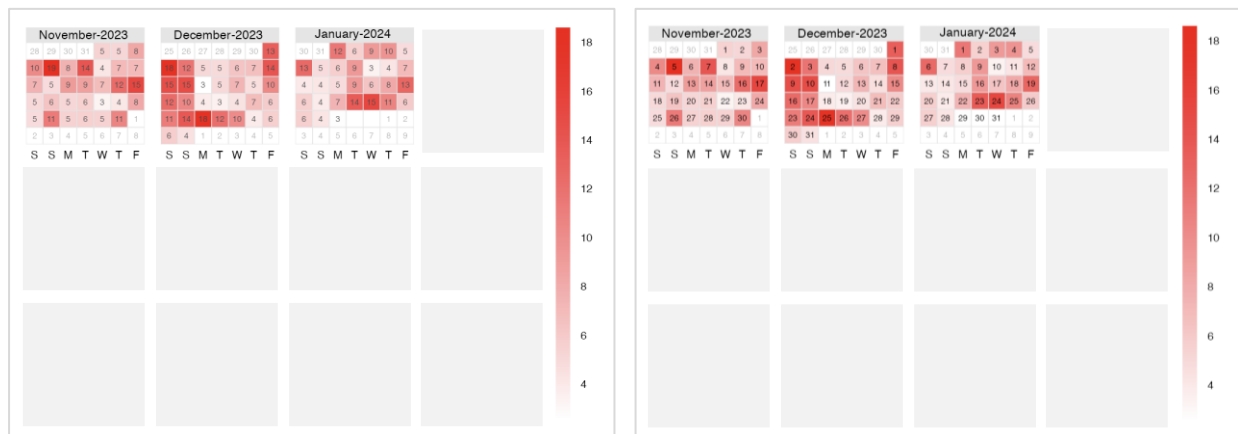


The **“Create Calendar Series”** displays the daily mean of a time series. The y-axis shows the name of the day from Monday to Sunday, and the x-axis represents the month. Darker red colors imply higher mean values, whereas lighter red colors represent lower mean values.



The **“Create Calendar Plot”** presents the daily mean of a time series and comes in a 3x4 months matrix. Thereby, it is no problem to show a time range that starts in 2023 and ends in 2024 for example. However, this plot cannot plot more than 12 months in total.

The darker red color means higher values, and light red color presents lower values. The values in the calendar plot can either display the daily mean value or represent the day of a month.



The “**Network Summary**” panel also offers the possibility to compare two datasets with each other. To do so select the two data frames and retrieve the variables/column names that should be used.

Compare two selected Datasets with each other.

For the Time Variation Plot it makes sense to compare dataset with the same unit!

DF1

AirNow.pressure

Retrieve DF1 Variables

Variable DF1

Update Variables

Create Time Variation Plot

DF2

AirNow.pressure

Retrieve DF2 Variables

Variable DF2

Create Scatter Plot

Compare two selected Datasets with each other.

For the Time Variation Plot it makes sense to compare dataset with the same unit!

DF1

AQS.pm25

Retrieve DF1 Variables

Variable DF1

pm25_daily_average(ug/m3)

Update Variables

Create Time Variation Plot

DF2

PurpleAir.pm25_corrected_daily

Retrieve DF2 Variables

Variable DF2

pm25_corrected_daily(ug/m3)

elevation(m)

count(-)

pm25_corrected_daily(ug/m3)

Create Scatter Plot

Do not forget to click on “**Update Variables**” to confirm the choices.

Compare two selected Datasets with each other.

For the Time Variation Plot it makes sense to compare dataset with the same unit!

DF1

Variable DF1

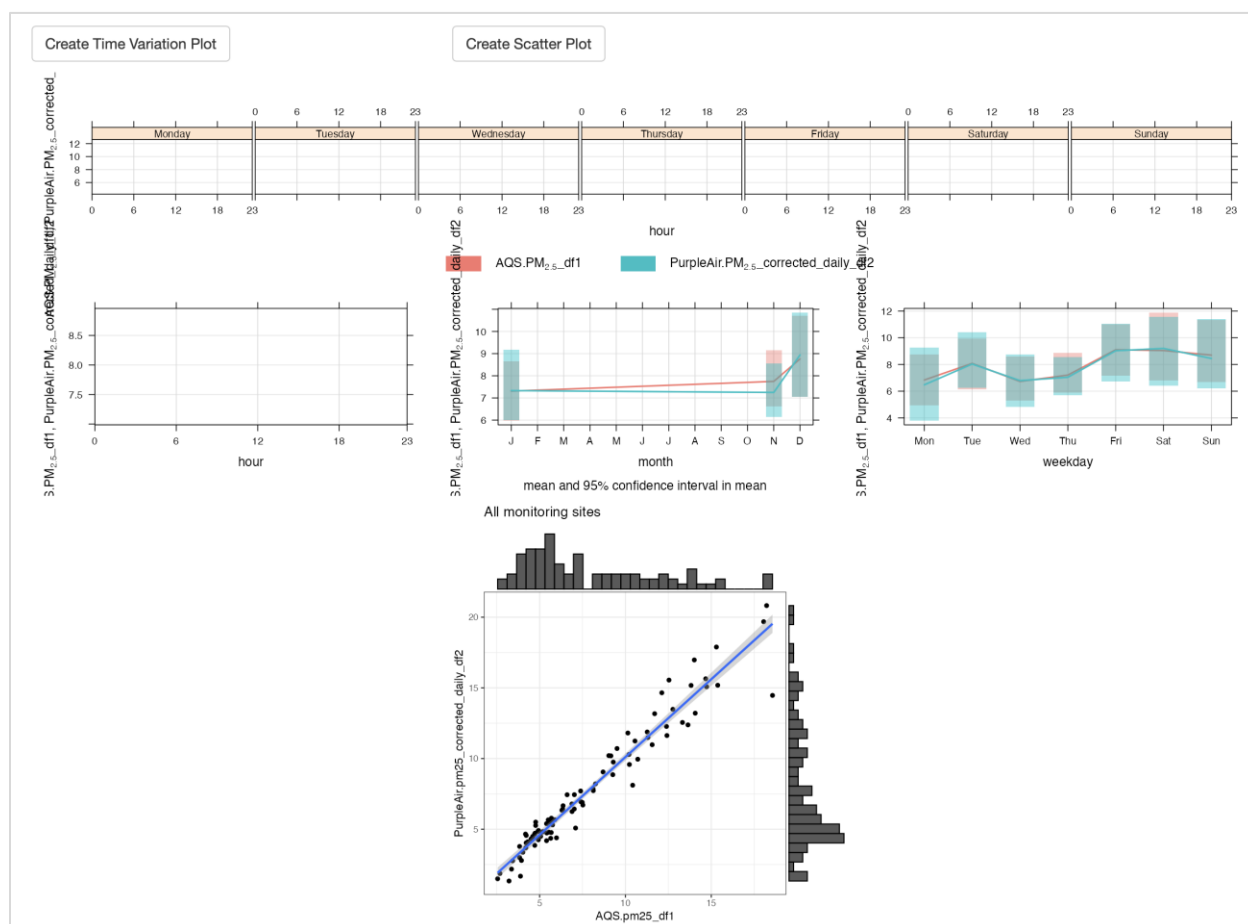
AQS.pm25
PurpleAir.pm25_corrected_daily

DF2

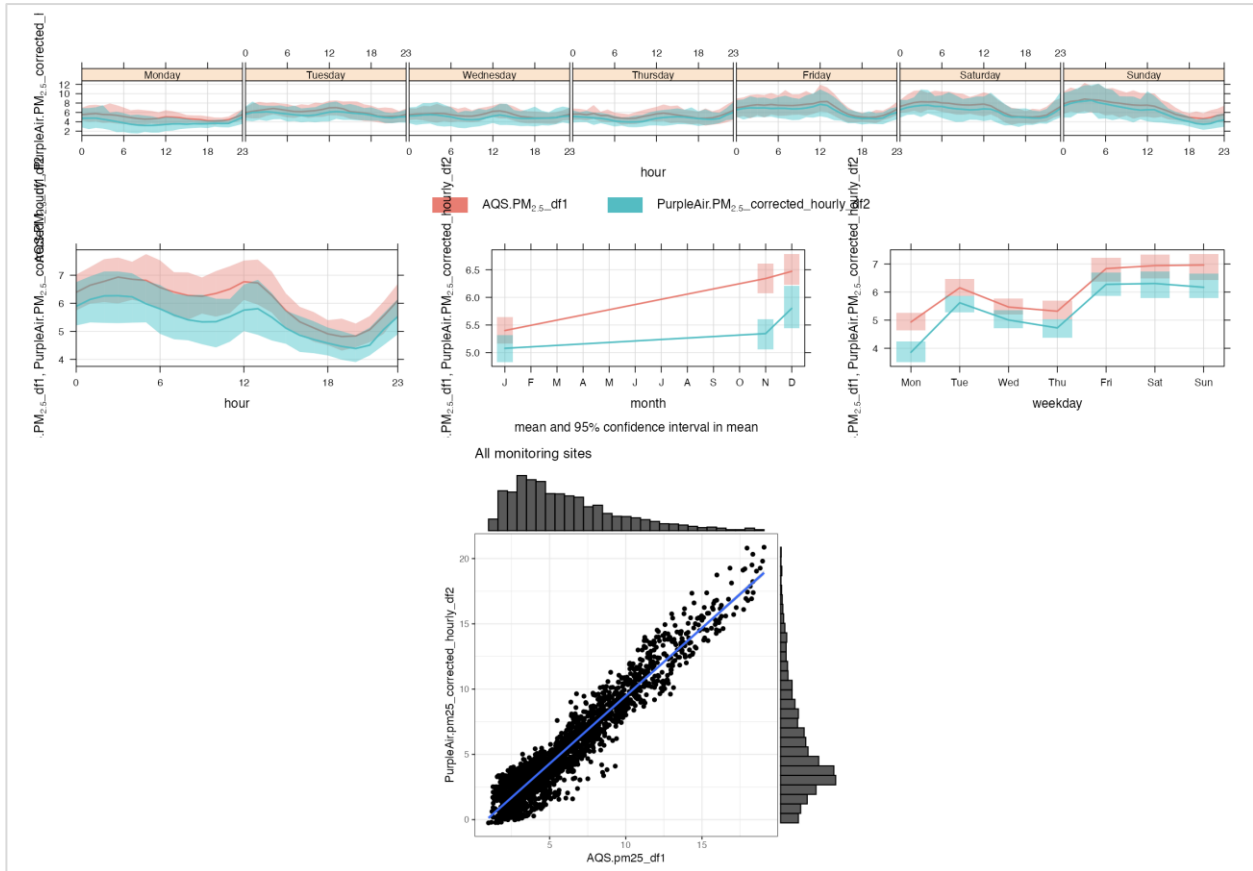
Variable DF2

pm25_daily_average(ug/m3)
pm25_corrected_daily(ug/m3)

The data frames used to create those plots are daily observations. In that case, some parts of the time variation plot cannot be displayed since hourly observations are needed.



Here is an example of how the plot looks like with hourly PM2.5 observations.



Finally, download the plots and the original data frames that were used to create them in the first place.

ASNAT “**Network Summary**” plots and original data frames will be downloaded as zip files and include only visible plots from the “**Network Summary**” panel. If the process of the time series plot is skipped, for example, the download will not include the time series plot.

Note: The leaflet map will be downloaded as an HTML file.

Download all current plots!

Download all current data frames!

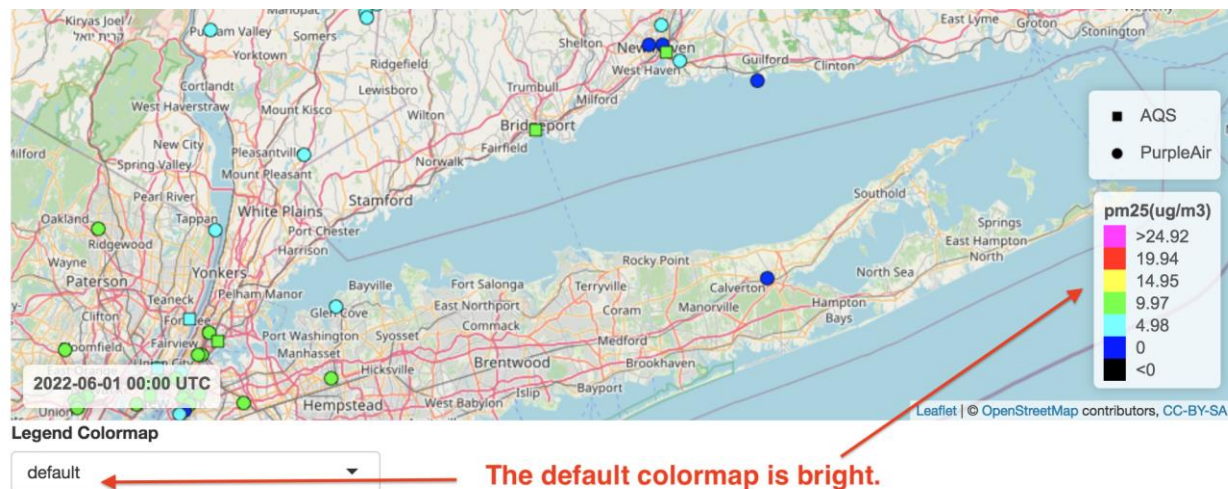
Downloads		
Name	Größe	Art
ASNAT_Network_Summary_DF_2024-12-11 13_40_41.185025	--	Ordner
df2.csv	14,9 MB	CSV-Dokument
df1.csv	16,2 MB	CSV-Dokument
df.csv	16,2 MB	CSV-Dokument
ASNAT_Network_Summary_Plots_2024-12-11 13_40_41.183653	--	Ordner
scatterplot.png	109 KB	PNG-Bild
calendar_series.png	123 KB	PNG-Bild
time_series.png	442 KB	PNG-Bild
barchart_aqi.png	119 KB	PNG-Bild
barchart.png	106 KB	PNG-Bild
leaflet.html	510 KB	HTML-Dokument
ASNAT_Network_Summary_DF_2024-12-11 13_40_41.185025.zip	4,8 MB	ZIP-Archiv
ASNAT_Network_Summary_Plots_2024-12-11 13_40_41.183653.zip	922 KB	ZIP-Archiv

Colormaps/legends:

ASNAT has a few colormaps/legends.

Each has a half-dozen hues since human vision cannot distinguish many subtly differing hues.

The default colormap contains bright most easily discernable hues.



The AQI colormap is familiar to the general public but the standard is not defined for all measures available in ASNAT.

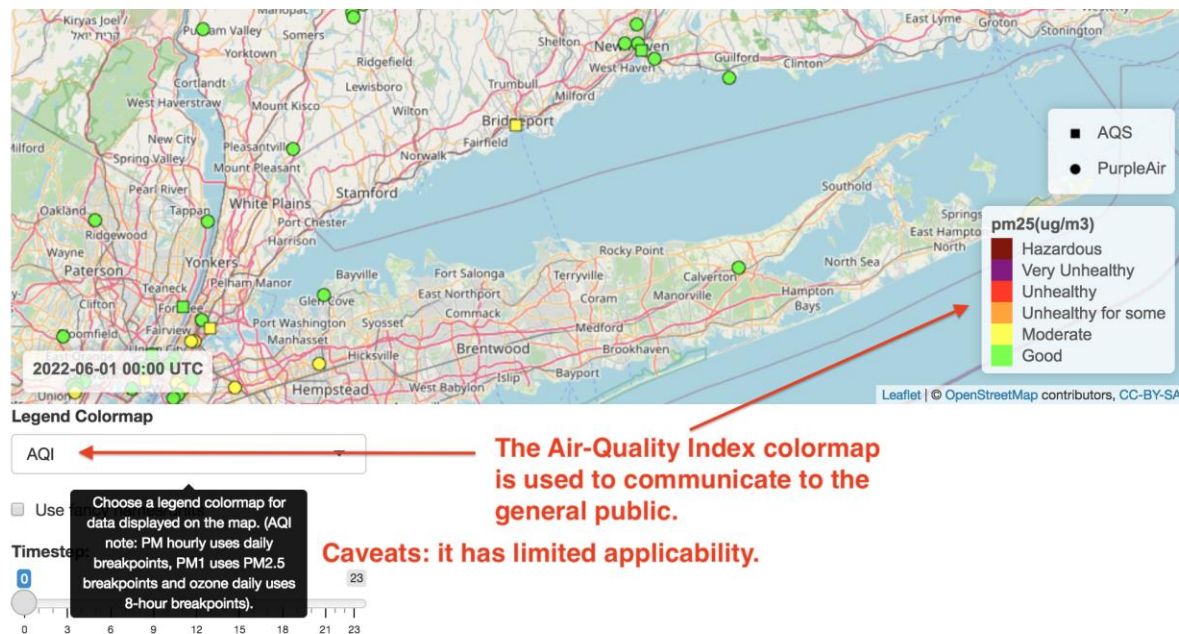
https://aqs.epa.gov/aqsweb/documents/codetables/aqi_breakpoints.html

For hourly PM2.5 and PM10, ASNAT uses daily breakpoints.

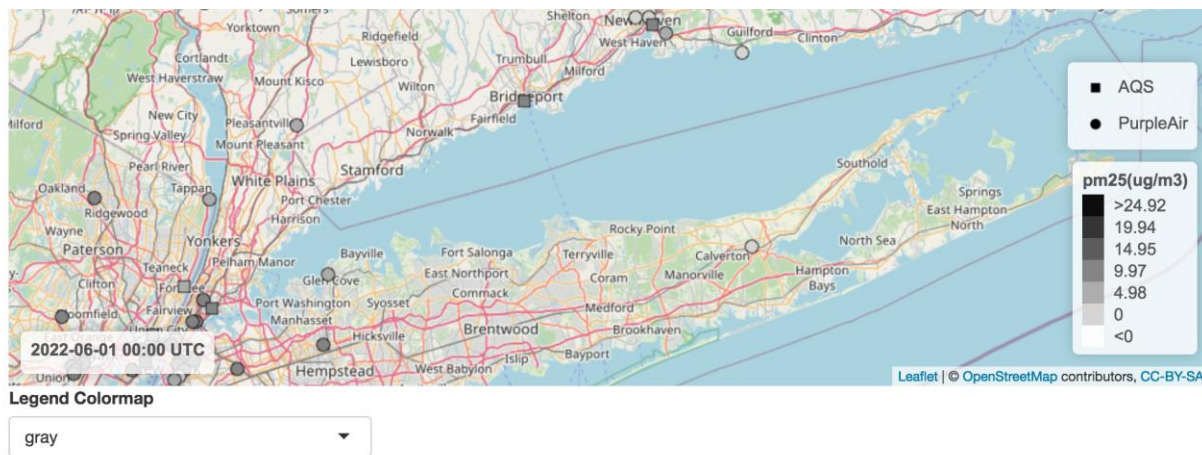
For PM1 ASNAT uses PM2.5 breakpoints.

For daily ozone, ASNAT uses 8-hour average breakpoints.

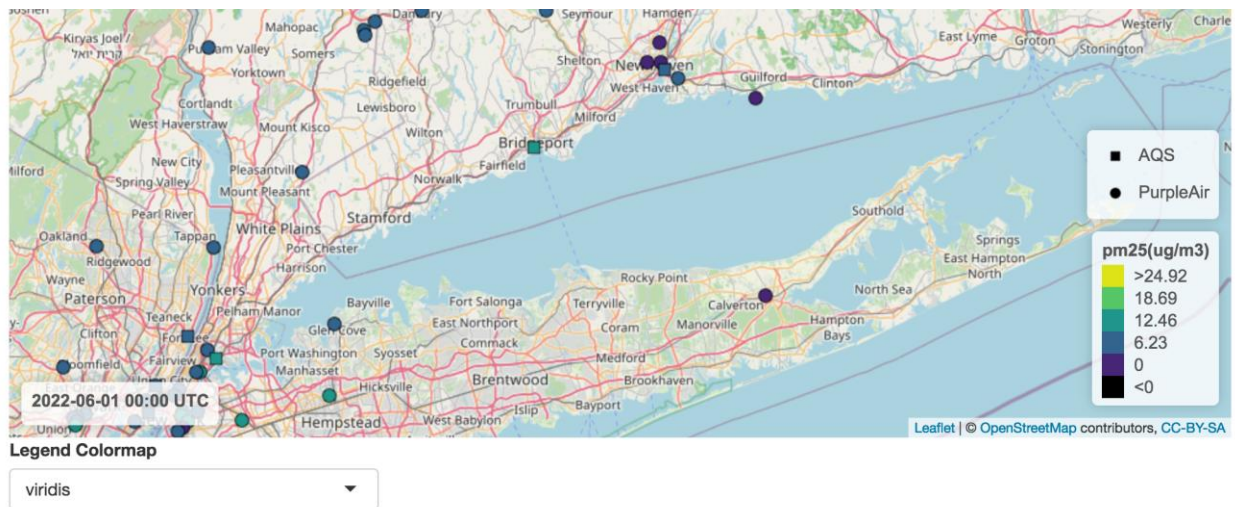
If a variable is selected that is not in the AQI standard, then the default colormap will be used instead.



The grayscale colormap can work well for saved map images for publication and color-impaired viewers.



The last colormap, viridis (Latin for green) is supposedly also suitable for color-impaired viewers.



Demo 11: PM2.5 Parameter Codes, Neighbor Statistics and AQI Plots

This demo will show retrieval by PM2.5 parameter codes, Neighbor Statistics and AQI plots.

11.1 To start, **Quit** ASNAT and **restart** it and make the selections shown in the following figures.

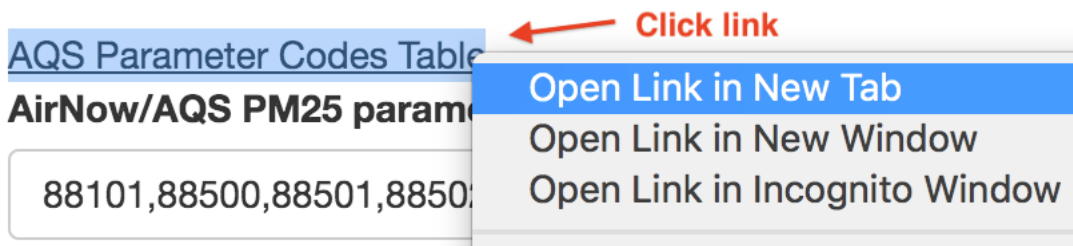
The screenshot shows the ASNAT interface with the following elements and instructions:

- Start Date:** 2022-06-01 (Instruction: 1. Select this date.)
- Days:** 1 (Instruction: 2. Delete Data.)
- Timestep:** hours (Instruction: 2. Delete Data.)
- Delete Loaded Data** button (Instruction: 2. Delete Data.)
- Load Web (control/command-click two):** A list of variables with three selected: AQS.pm25, AQS.pm10, and AQS.rh (Instruction: 3. Select these 3 variables.)
- PurpleAir Key:** your_key (Instruction: 4. Enter your PurpleAir key.)

<https://www2.purpleair.com/pages/attribution>

11.2 For AQS.pm25, we will specify only data designated with a particular parameter code will be retrieved.

Click on the AQS Parameter Codes Table hyperlink.



This will display the web page below. Enter 8850 partial code and the table will show PM2.5 parameters that start with those digits:

← → ↻ <https://aqs.epa.gov/aqsweb/documents/codetables/parameters.html> 🔍 R ⚙️ 🗖️ 👤

EPA United States Environmental Protection Agency 🔍

◦ Contact Us

You are here: EPA Home » Air Quality System » Code Tables

Parameters

February 12, 2025

AQS Reference Table

JavaScript must be enabled to view this page.

To sort on a column, click the column header. That column will be (subtly) highlighted. Particularly useful is the search box at the top of the table on the right. Separate multiple search terms with a space. Search looks across all columns in the table.

Download a csv version: [parameters.csv](#)

Show 25 entries Search: 8850

Parameter Code	Parameter	Parameter Abbreviation	Parameter Alternate Name	CAS Number	Standard Units	Still Valid	Round or Truncate
88500	PM2.5 Total Atmospheric				Micrograms/cubic meter (LC)	YES	T
88501	PM2.5 Raw Data				Micrograms/cubic meter (LC)	YES	T
88502	Acceptable PM2.5 AQI & Speciation Mass	PM2.5			Micrograms/cubic meter (LC)	YES	T
88503	PM2.5 Volatile Channel				Micrograms/cubic meter (LC)	YES	T

Showing 1 to 4 of 4 entries (filtered from 1,479 total entries)

Previous 1 Next

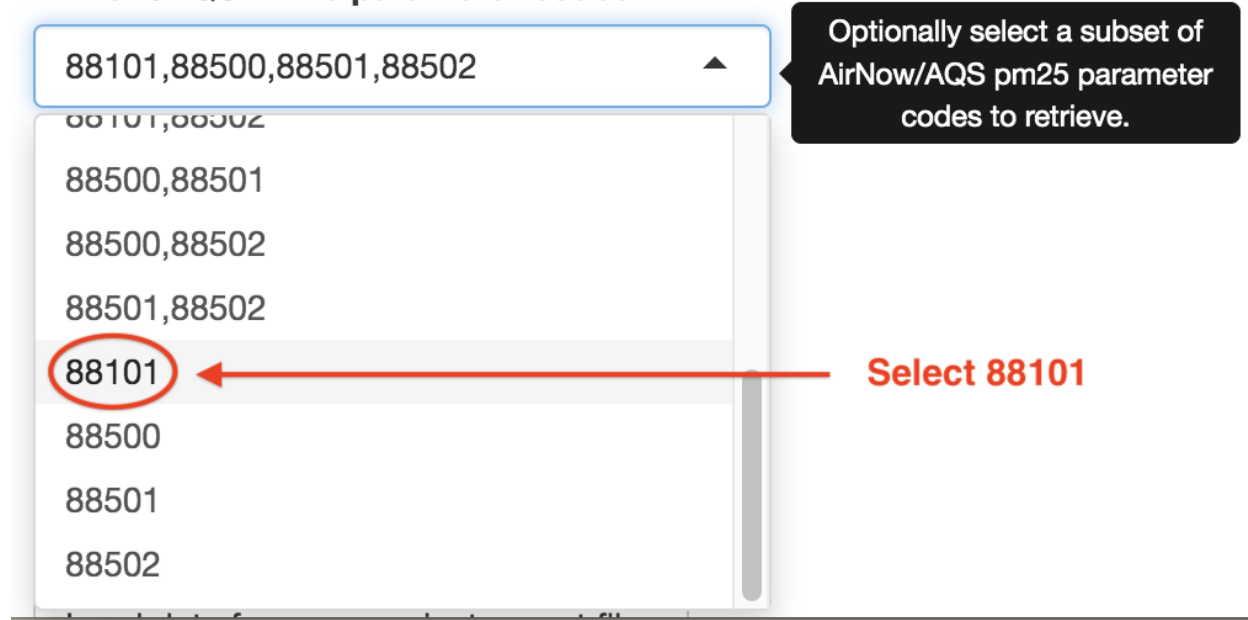
Enter partial parameter code

11.3 In ASNAT, Select 88101 in the AirNow/AQS PM25 parameter codes menu.

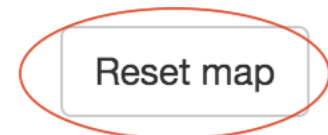
This specifies that AirNow/AQS PM25 data retrieved will only match code 88101. Note the menu lists all possible subsets of the four PM25 parameter codes.

AQS Parameter Codes Table

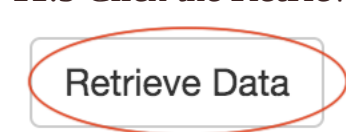
AirNow/AQS PM25 parameter codes:



11.4 Click the Reset map button (below the map image).



11.5 Click the Retrieve Data button.



11.6 Select the following Datasets X, Y and Z and Neighbor distance shown below. Dataset Z specifies an optional humidity variable to be used to color scatterplot points. *The humidity dataset must match either Dataset X or Dataset Y*

- in this case, either AQS.rh or PurpleAir.humidity. We will choose the latter since it contains more data points.

Select Dataset X: **Make these menu selections.**

AQS.pm25 ▼

Select Variable Of Dataset X:

pm25(ug/m3) ▼

Select Comparison Dataset Y (optional):

PurpleAir.pm25_corrected_hourly ▼

Select Variable Of Dataset Y:

pm25_corrected_hourly(ug/m3) ▼

Select Coloring Dataset Z (optional):

PurpleAir.humidity_hourly ▼

Select Variable Of Dataset Z:

humidity_hourly(%) **Humidity will be used to color scatterplots.** ▼

Neighbor distance(m): ⓘ

5000

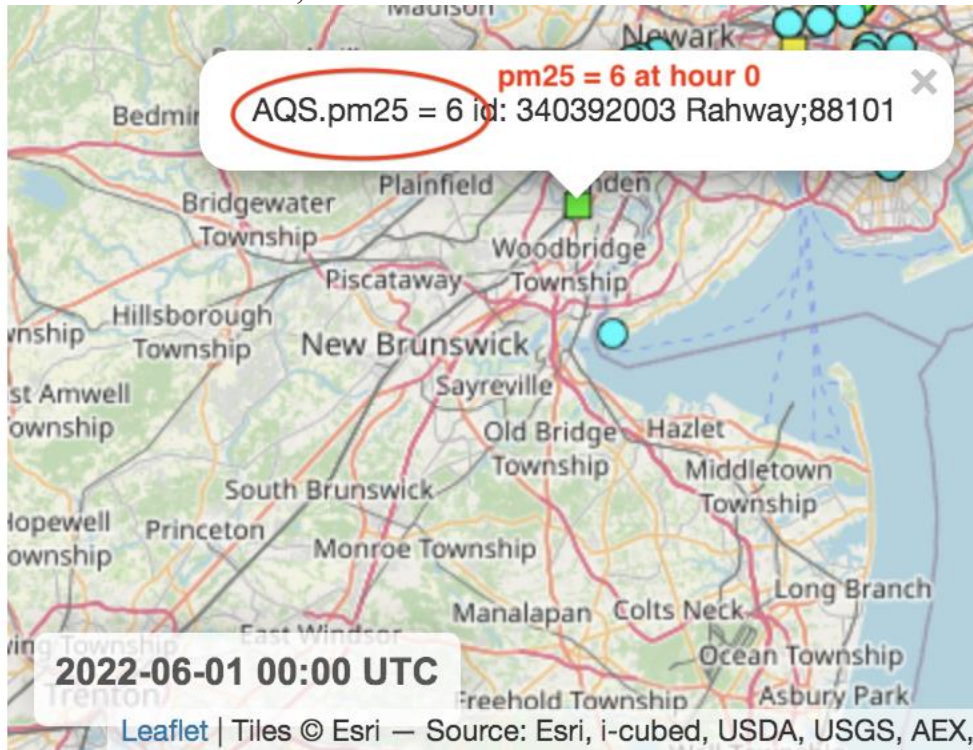
11.7 Select AQI in the Legend Colormap menu.

Legend Colormap

AQI

← **Select AQI Legend Colormap** ▼

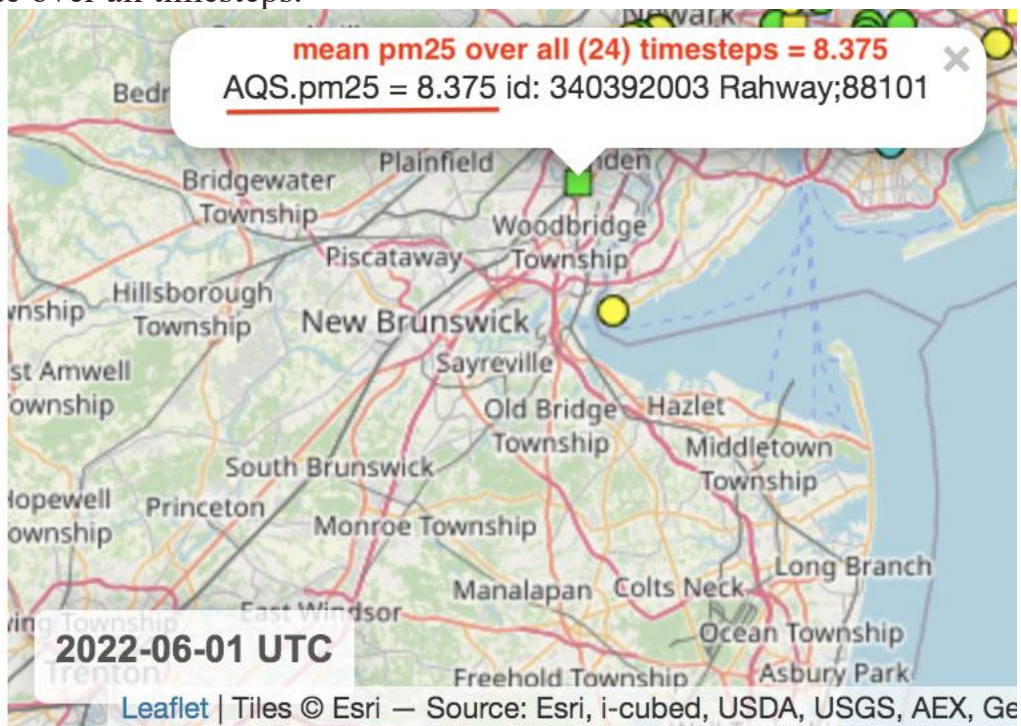
11.8 Look at the map and **find and click on the AQS site shown below**. The pop-up shows the AQS site #340392003 labeled *Rahway;88101*. Note that site numbers are unique but site labels are not. The label shows the parameter code of that measure - in this case, 88101. This is the measure at hour 0.



11.8 ASNAT can quickly display the mean of each site's measures - over all loaded timesteps -by simply clicking the **Show mean values button**.

☒ Show mean values over all timesteps

11.9 Now **click on the AQS site again** to see the mean value (8.375 ug/m3) of that site over all timesteps.



11.10 Select the Flagging tab.



11.11 Select the following flags to be applied to Dataset Y (PurpleAir.pm25_corrected_hourly).

☒ Apply flag 80 if difference >	Enable each of these.	5
☒ Apply flag 81 if percent difference >		70
☒ Apply flag 82 if R-squared <		0.5

11.12 Select **Tables** tab and click the **Summarize** button.



Dataset Summary: AQS.pm25 vs PurpleAir.pm25_corrected_hourly

AQS.pm25: 2022-06-01 to 2022-06-01 (-74.2762, -72.9029) (40.6039, 41.3992)

Site_Id(-)	Timestamp_First_Measurement(UTC)	Timestamp_Last_Measurement
90011123	2022-06-01T15:00:00-0000	2022-06-01T23:00:00-0000
90090027	2022-06-01T00:00:00-0000	2022-06-01T23:00:00-0000

11.13 Scroll to the bottom of the page to see the **AQI Category Statistics** summary that compares each AQS measure with its *unflagged* PurpleAir neighbor measures. Looking at the first row, we see there are 84 AQS paired measures not exceeding 9 ug/m³ which is *Good* category and PurpleAir neighboring measures differences are summarized by the statistics: Normalized Mean Bias Error, etc.

Similarly for the 46 AQS paired measures in the Moderate category.

Unflagged Neighboring Hourly Point Measures (<= 5000m) AQI Category
Statistics: AQS.pm25 vs PurpleAir.pm25_corrected_hourly

2022-06-01 (-74.2762, -72.9029) - (40.6039, 41.3992)

1. Scroll to bottom.

AQI_Category(-)	Upper_Limit(ug/m3)	Count(-)	NMBE(%)	RMSE(ug/m3)	NRMSE(%)
Good	9.0000	84	13.5458	1.8907	31.3630
Moderate	35.4000	46	22.7443	3.1372	27.9078
Unhealthy for some	55.4000	0	0.0000	0.0000	0.0000
Unhealthy	125.4000	0	0.0000	0.0000	0.0000
Very Unhealthy	225.4000	0	0.0000	0.0000	0.0000
Hazardous	1000000.0000	0	0.0000	0.0000	0.0000

AQI category statistics table.

11.14 Select the Plots tab and Make Plots button.

Map Flagging Tables **Plots** Corrections Network Summary

Neighbors Map - Shows Neighbors of (Single) Selected Dataset X Site

Select Dataset X Site With Neighbors:

0 All Sites ▼

☐ Use interactive plots

Make Plots of Datasets X, Y Save Plots

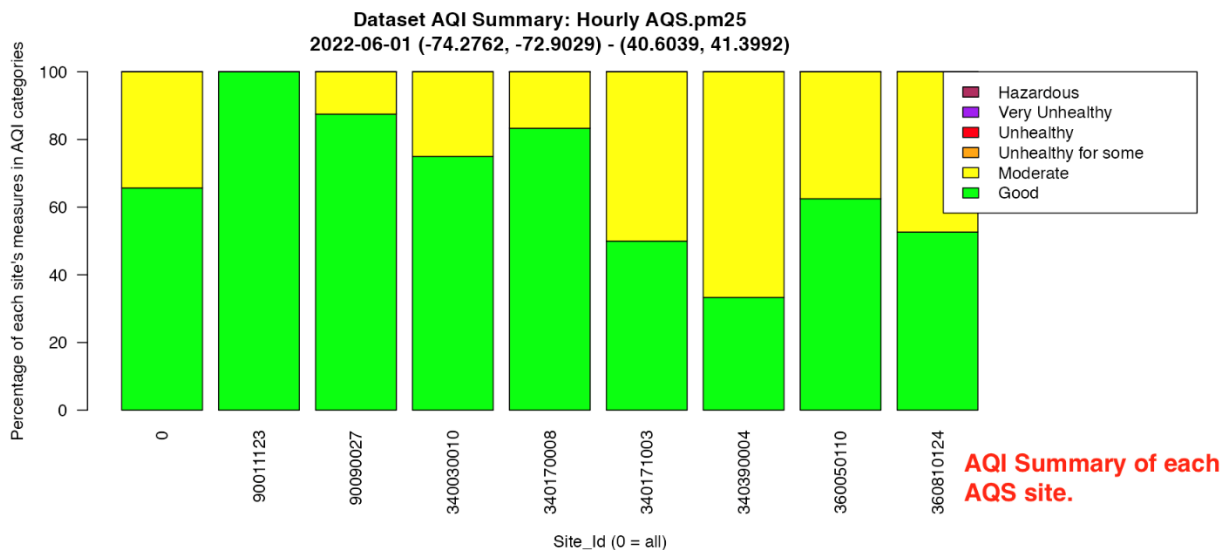
1. Click Plots tab.

2. Click Make Plots.

11.15 Scroll to the 2nd plot to see the AQI Summary Plot of AQS PM25.

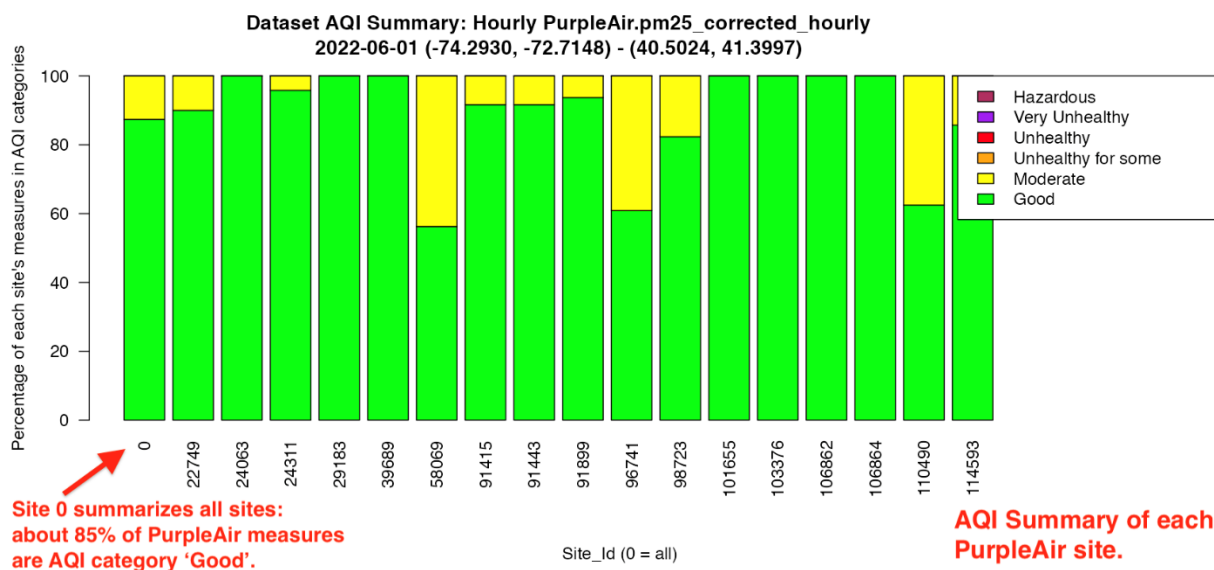
This shows the percentage each site's measures in eachAQI category.

Site 0 is a summary of all sites. We see that site # 90011123 measures are all in the Good category, while overall (labeled site 0) about 65% of all AQS measures are in the Good category with the remainder in the Moderate category.

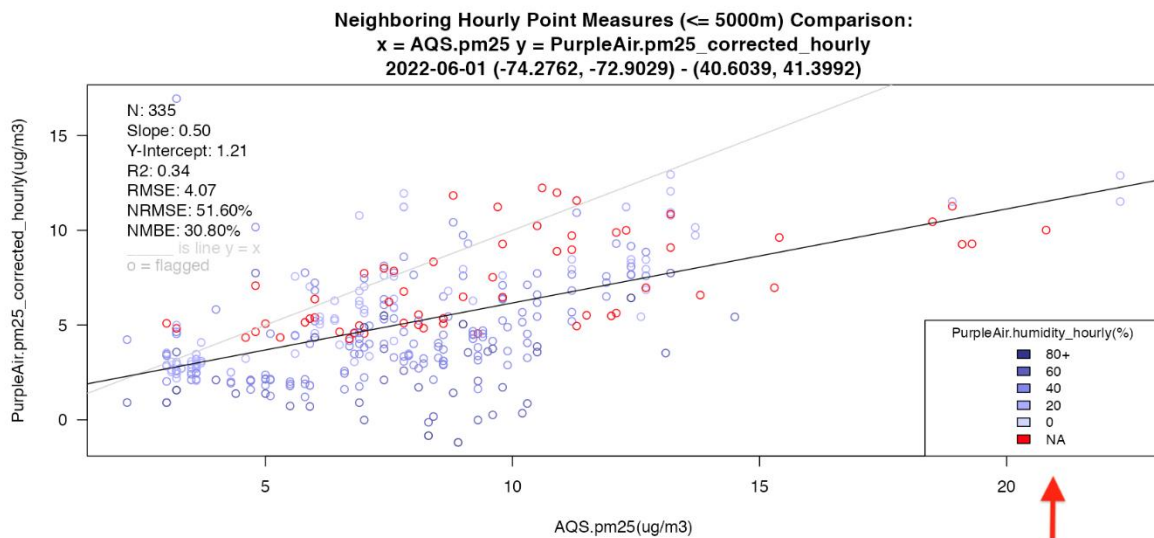


11.16 Scroll to the 5th plot to see the AQI Summary Plot of PurpleAir PM25.

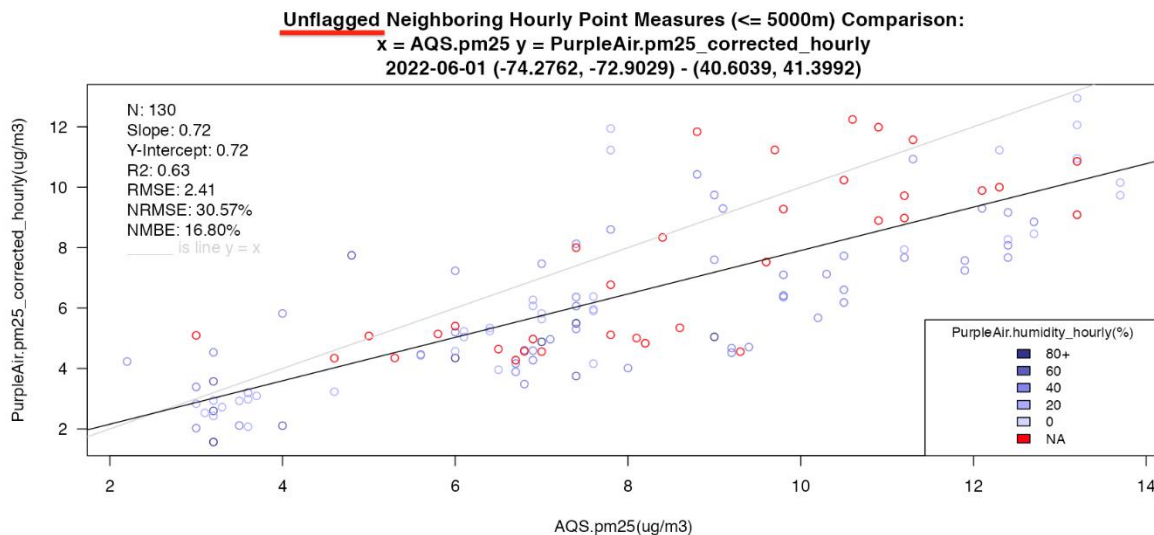
This shows that, overall, about 85% of PurpleAir unflagged measures are in the Good category with the remainder in the Moderate category.



11.17 Scroll down to the scatterplots. The top one shows all neighbor pair measures (AQS vs PurpleAir) and the bottom one shows only unflagged neighbors. These points are *colored by PurpleAir humidity* when there is a coincident PurpleAir humidity measure. Not all PurpleAir points had a humidity measure and in those cases the point is red to indicate NA (not available).



Scatterplot points colored by PurpleAir humidity.



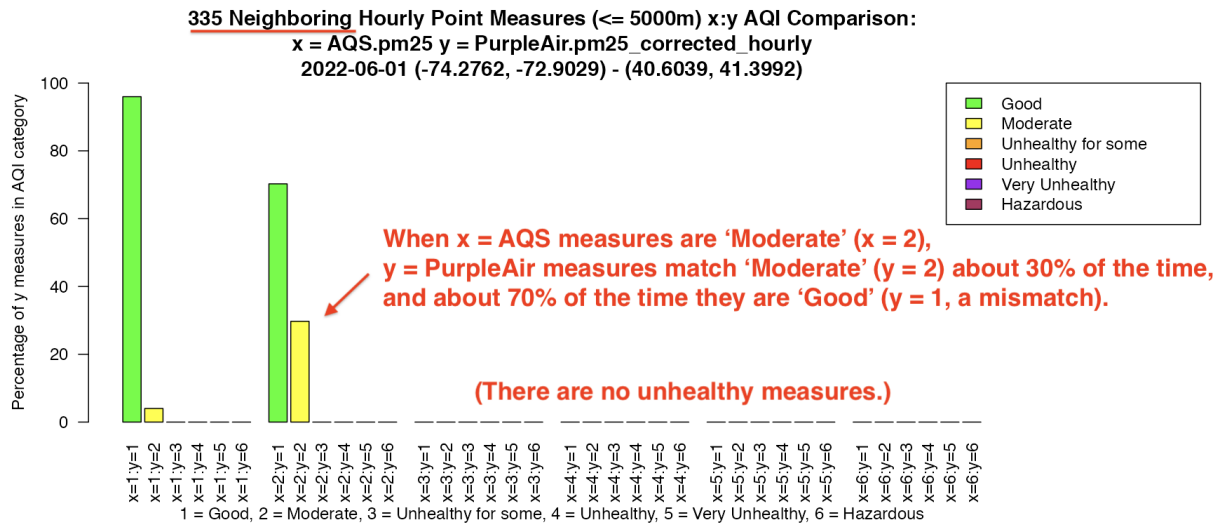
11.18 The next pair of plots compare the AQI categories of neighboring measurement pairs - in this case, $x = \text{AQS PM}_{2.5}$ vs $y = \text{PurpleAir PM}_{2.5}$ - with the bottom plot showing unflagged PurpleAir measures.

There are 6 AQI categories numbered from 1=Good, 2=Moderate ... 6=Hazardous.

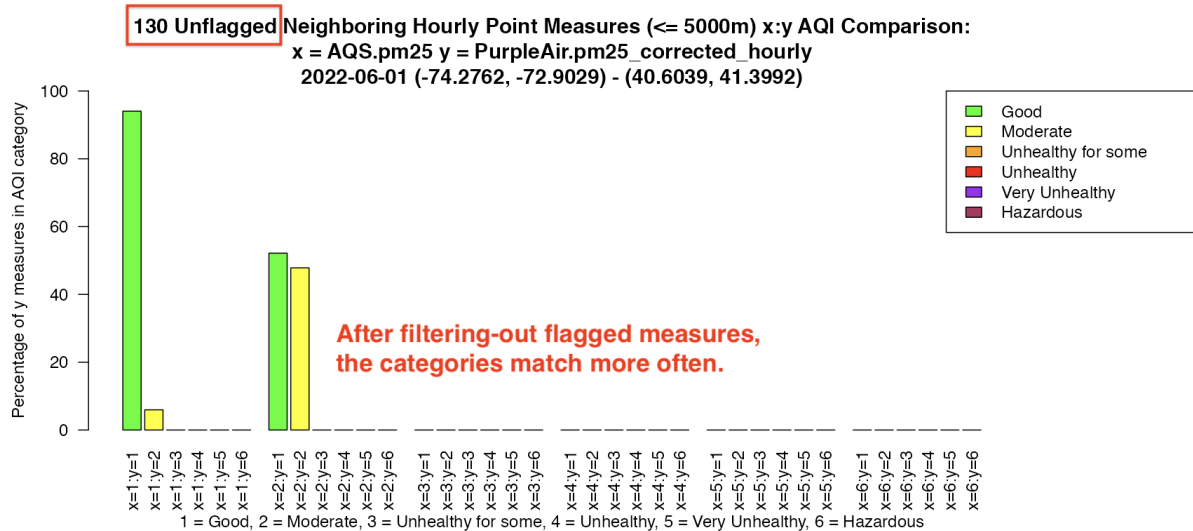
The left-most green bar labelled $x=1:y=1$ indicates that when AQS measures are 1=Good, 95% of PurpleAir measures are 1=Good (matched) and about 5% of PurpleAir measures are 2=Moderate (2nd yellow bar labelled $x=1:y=2$).

Next, $x=2:y=1$ means when AQS is 2=Moderate, about 70% of PurpleAir measures are 1=Good (unmatched) with the remaining 30% 2=Moderate (matched). There are no measures in the unhealthy categories.

The bottom plot shows the *unflagged* neighboring measure comparisons. Of course, after filtering-out various mismatches, the remaining paired measures match categories better. In this example, there are no instances of mismatches differing by more than one category.



Neighboring measures AQI Comparison



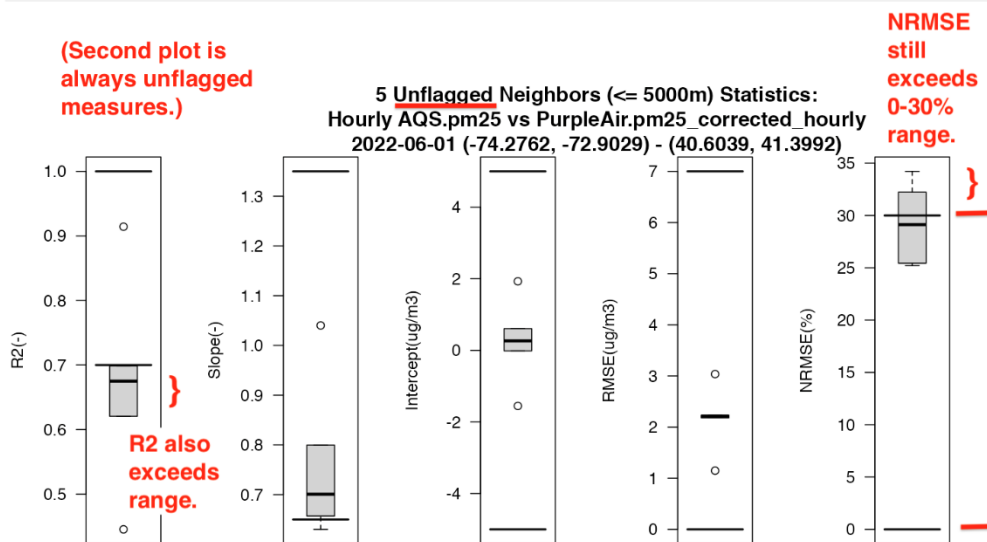
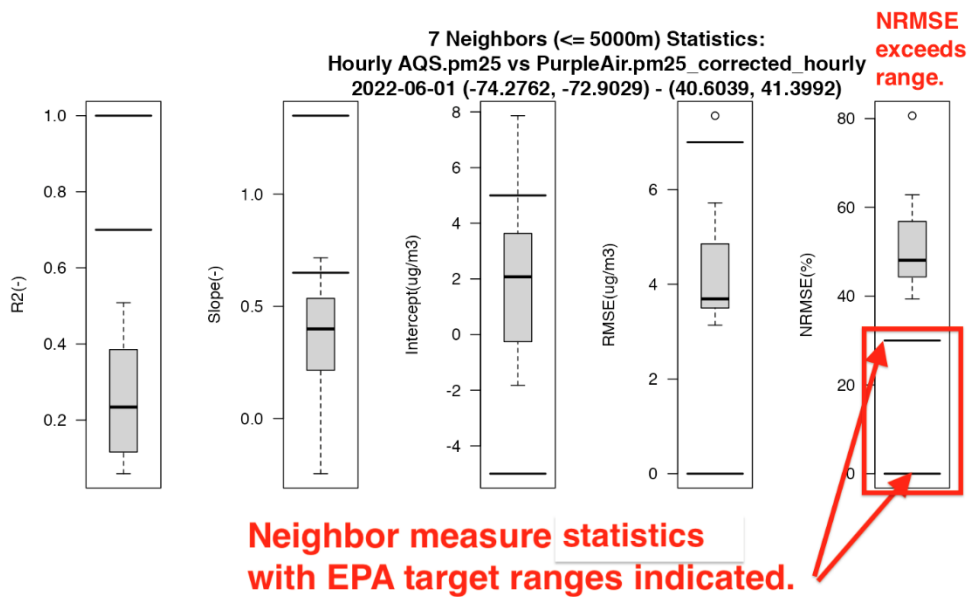
11.19 Scroll to the bottom of the page to see the Neighbor Statistics Plots.

These plots show several statistics for each set of neighbor measures.

As before, the bottom plot shows only unflagged neighbor measures.

If the variable has an EPA target range (currently PM1, PM2.5, PM10, O3, NO2, SO2, CO) for each statistic then that range is indicated with the bold lines.

For example, The Normalized Root Mean Square Error target range for PM2.5 is [0, 30%] and both plots show exceedance. They both show exceedance of the R-squared statistic target range [0.7, 1].

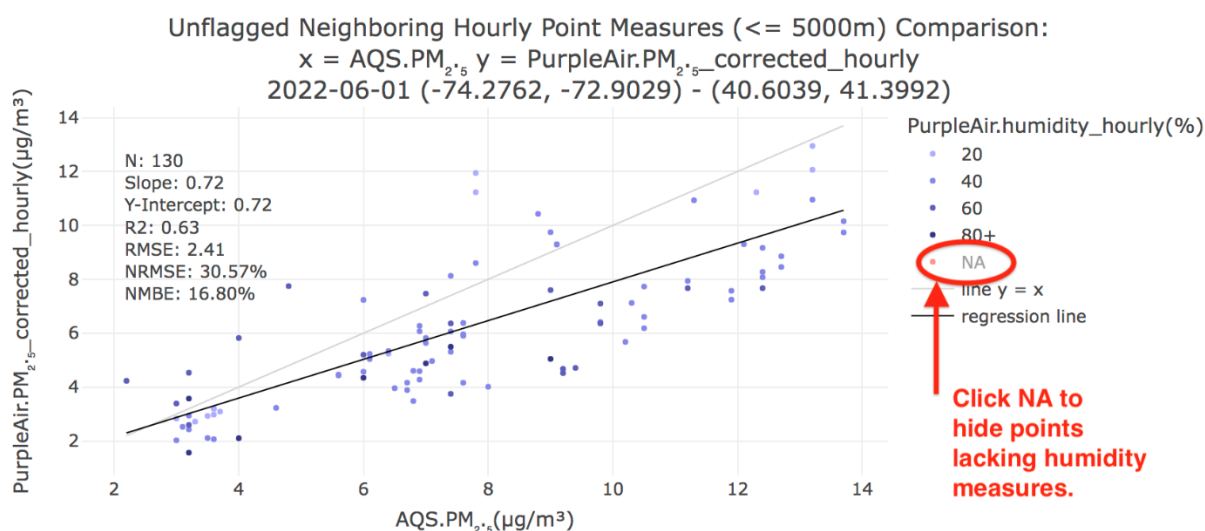


11.20 The Interactive Plots provide more detail.

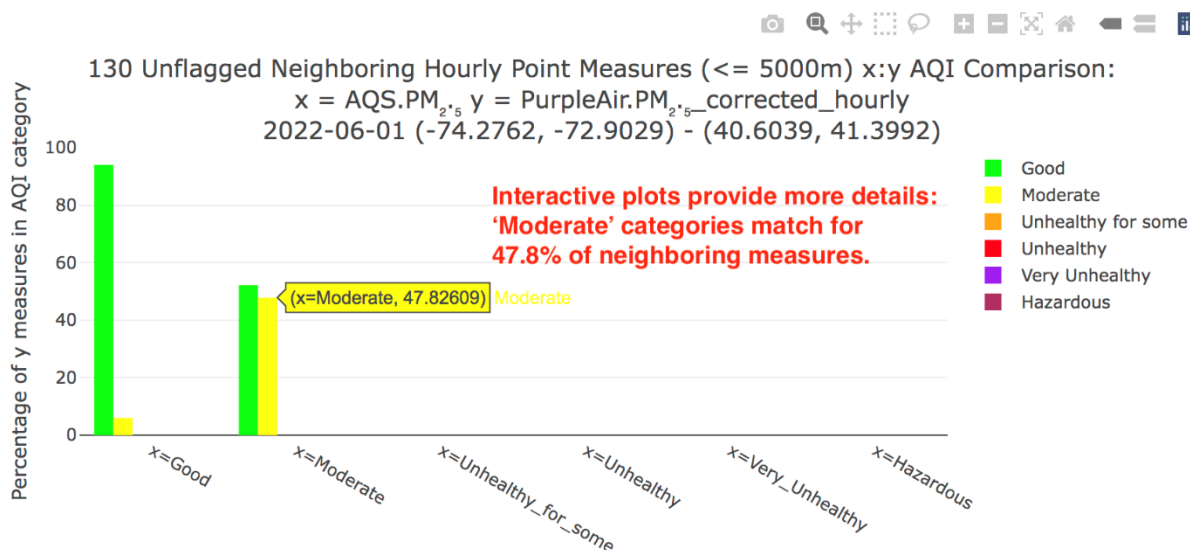
Click the Use interactive plots button and then click the Make Plots button.



11.21 Scroll down to the 2nd scatterplot. Then click on NA in the legend to hide the red points that indicate missing humidity. Click again to toggle it or click other values such as 80+ to hide points with humidity greater than 80%. Moving the mouse over points will display their measures as (AQS, PurpleAir), humidity.

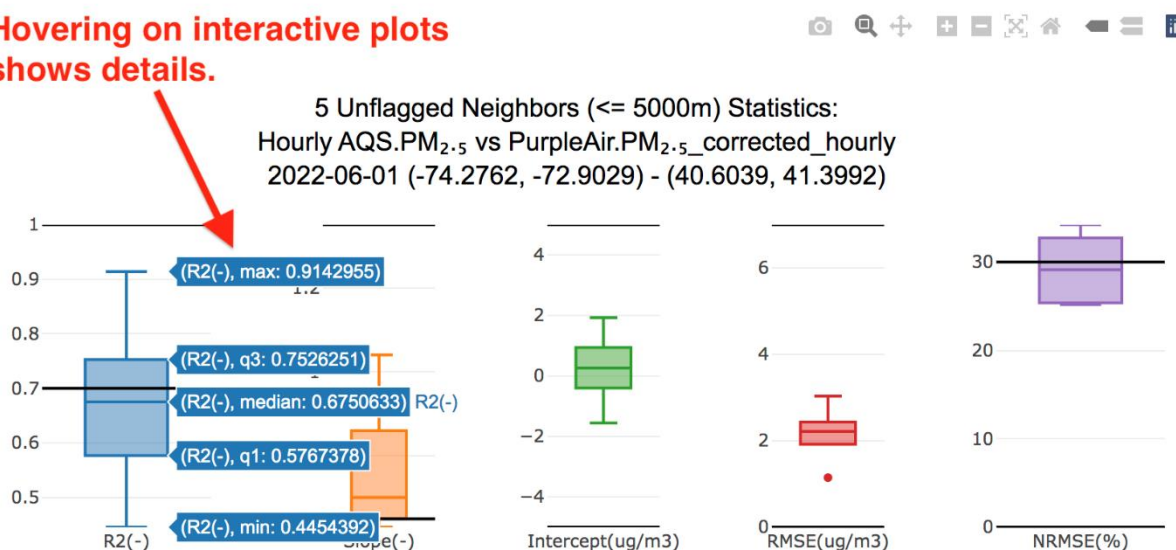


11.22 Below the scatterplots are the **Interactive AQI Comparison** plots. These are labelled more clearly and moving the mouse pointer over the bars shows numeric values. For example, in the bottom plot of *unflagged* neighboring measures, when AQS is *Moderate*, 47.8% of PurpleAir measures are *Moderate* (matched) and 52.2% are *Good* (mismatched by one category).



11.23 Scroll to the bottom of the page to see the **Interactive Neighbor Statistics Plots**. Moving the mouse pointer over a boxplot will show the numeric values of the quartiles.

Hovering on interactive plots shows details.



11.24 Scroll to the top of the page and click **Save Plots** if desired. Note these will be the basic plots.

Map

Flagging

Tables

Plots

Neighbors Map - Shows Neighbors of (Single) Site

Select Dataset X Site With Neighbors:

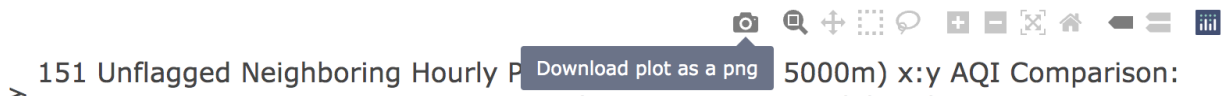
0 All Sites

☒ Use interactive plots

Make Plots of Datasets X, Y

Save Plots

To save the interactive plots **click the camera icon** in each plot's menu.



11.25 Click the **Quit button** to exit ASNAT.

Output File Format: tsv

Save Data

Quit

Demo 12: Nowcast

EPA's Nowcast algorithm computes an hourly ozone, PM10 or PM2.5 concentration estimate based on a number of preceding hourly measures (window).

Mostly used to forecast a daily AQI based on yesterday's hourly measures, it can be used for any sequence of hourly values for a dataset loaded into ASNAT, provided the selected variable is one of: ozone(ppb), pm10(ug/m3), pm25(ug/m3). For ozone the window is 8 hours with no minimum weight factor. For PM the window is 12 hours with a minimum weight factor of 0.5. Note that the Nowcast hourly concentrations cannot be computed until the window of preceding measures are available therefore the value NA will be used for the first window hours.

Detailed descriptions of the Nowcast algorithm are available from these websites:
[https://en.wikipedia.org/wiki/NowCast_\(air_quality_index\)](https://en.wikipedia.org/wiki/NowCast_(air_quality_index))

<https://www.epa.gov/sites/default/files/2018-01/documents/nowcastfactsheet.pdf>

There are different implementations of Nowcast code. ASNAT's implementation is based on code from jonathan.s.callahan@gmail.com downloaded on 2025-02-07 from:

https://rdr.io/cran/AirMonitor/src/R/monitor_nowcast.R

To demo Nowcast:

12.1 **Quit/restart** ASNAT and load the same datasets as in Demo 1:

Reset map

Start Date: 2022-06-01

Days: 1

Timestep: hours

Delete Loaded Data

Load Web:

AQS.pm25

PurpleAir.pm25_corrected

Purple Air Key:

your_key

Retrieve Data

Select Comparison Dataset Y:

PurpleAir.pm25_corrected_hourly

Neighbor Distance(m): 5000

ASNAT should appear as the image below.

1. Pan/zoom map to region of interest.
2. Start Date, Days to retrieve, Aggregate:

Start Date: 2022-06-01
Days: 1

Timestamp:
hours

Delete Loaded Data

Load Web (control/command-click two):

AirNow.pm25
AirNow.pm10
AirNow.rh
AirNow.temperature
AirNow.pressure
AirNow.ozone
AirNow.no2
AirNow.co
AirNow.so2
AQS.pm25
AQS.pm10
AQS.rh
AQS.temperature
AQS.pressure
AQS.ozone
AQS.no2
AQS.co
AQS.so2
METAR.relativeHumidity
METAR.temperature
METAR.seaLevelPress
PurpleAir.pm25_corrected
PurpleAir.humidity
PurpleAir.temperature

<https://www2.purpleair.com/pages/attribution>

PurpleAir Key:

your_key

Validate Purple Air Key

Purple Air Sites:

0 All sensors in view

[AQS Parameter Codes Table](#)

AirNow/AQS PM25 parameter codes:

88101,88500,88501,88502

Retrieve Data

Load data from standard-format file:

Browse... No file selected

Load data from sensor instrument files

Select Dataset X:

AQS.pm25

Select Variable Of Dataset X:

pm25(ug/m3)

Select Comparison Dataset Y (optional):

PurpleAir.pm25_corrected_hourly

Select Variable Of Dataset Y:

pm25_corrected_hourly(ug/m3)

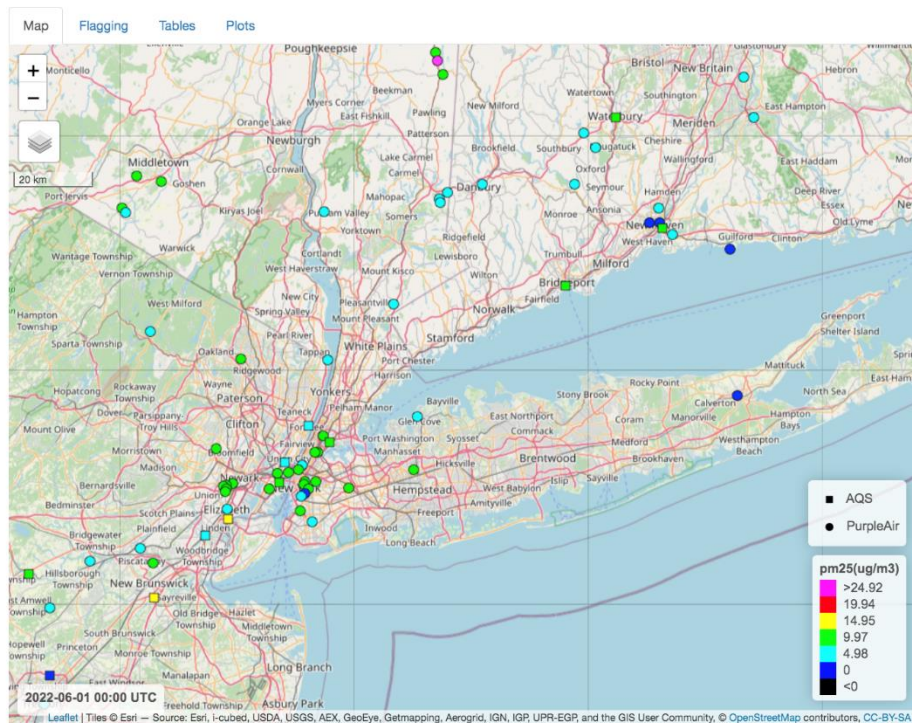
Select Coloring Dataset Z (optional):

none

Select Variable Of Dataset Z:

Compute Hourly Nowcast NowCast Info

Neighbor distance(m): 5000



Reset map

Legend Colormap

default

☐ Use fancy names/units

☐ Show mean values over all timesteps

Timestamp:

0 23

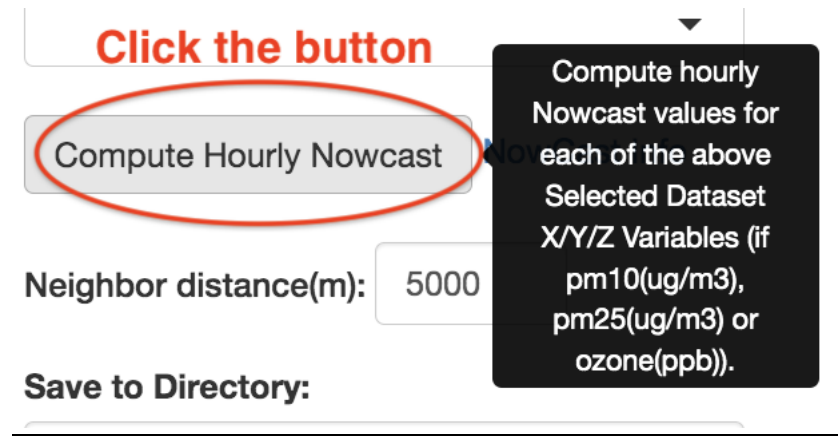
Zoom To Data

Save Map Image

Save Map Movie

12.2 Click the Compute Hourly Nowcast button.

Hovering over the button will pop-up a tooltip explaining that hourly Nowcast values will be computed for each of the selected Dataset variables - provided they are compatible with the Nowcast algorithm (including units).



12.3 The message area will report that the new **_nowcast** variables have been added to their datasets:

```
Added variable AQS.pm25_nowcast(ug/m3)
Added variable PurpleAir.pm25_corrected_hourly_nowcast(ug/m3)
```

12.4 Select these **_nowcast** variables from the menus:

Select Dataset X:

AQS.pm25 ▼

Select Variable Of Dataset X:

pm25_nowcast(ug/m3) ▼

Select Comparison Dataset Y (optional):

PurpleAir.pm25_corrected_hourly ▼

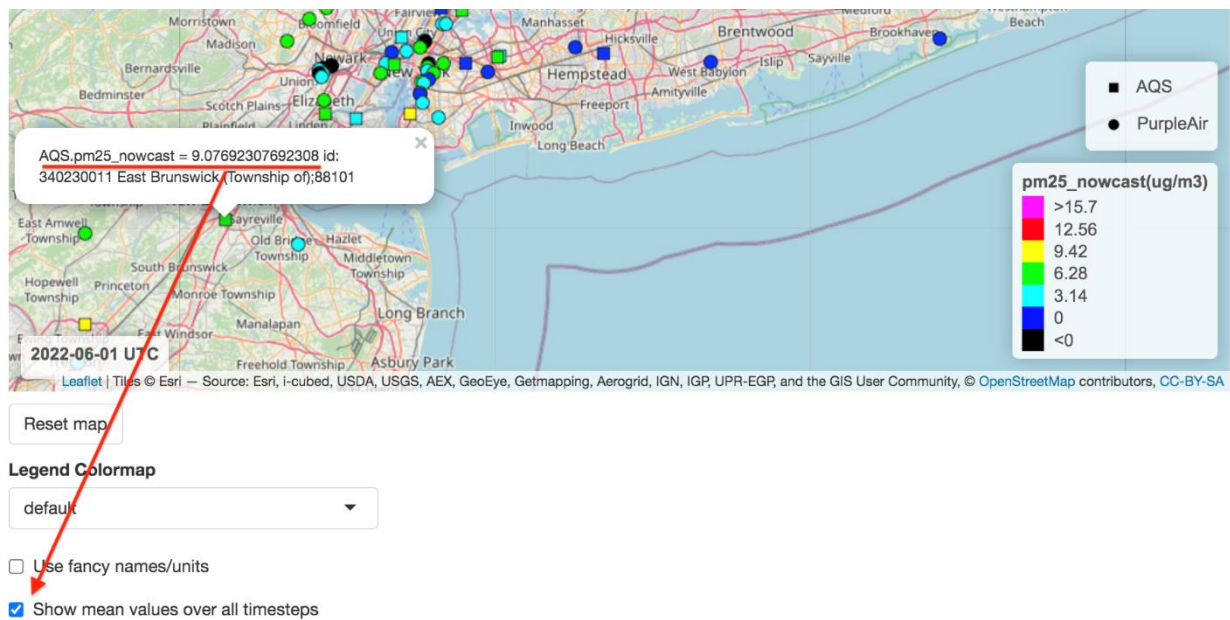
Select Variable Of Dataset Y:

pm25_corrected_hourly_nowcast(ug/m3) ▼

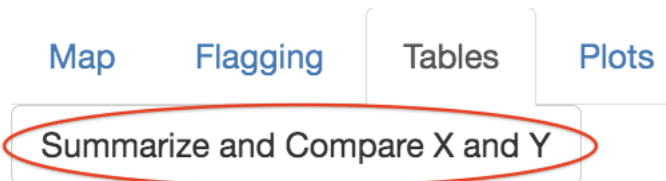
12.5 Note that the map shows the nowcast measures as **black indicating NA** since the Nowcast hourly value is not available for the first window = 12 hours.



12.6 You may **click the checkbox *Show mean values over all timesteps*** to see the means instead.



12.7 Click the **Summarize and Compare X and Y** button to compute tables.

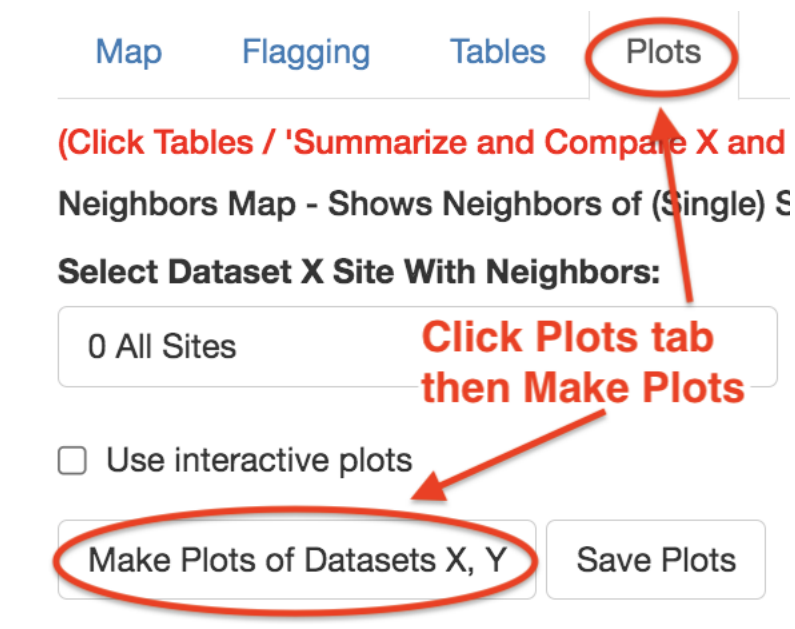


12.8 Note that in the Comparison Table, Nowcast values start at hour 1, the end of the 12-hour window for PM.

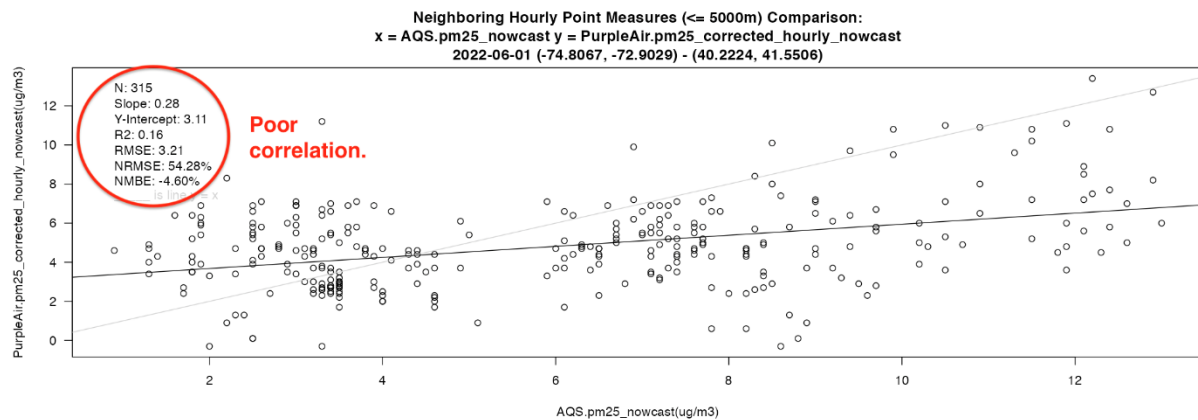
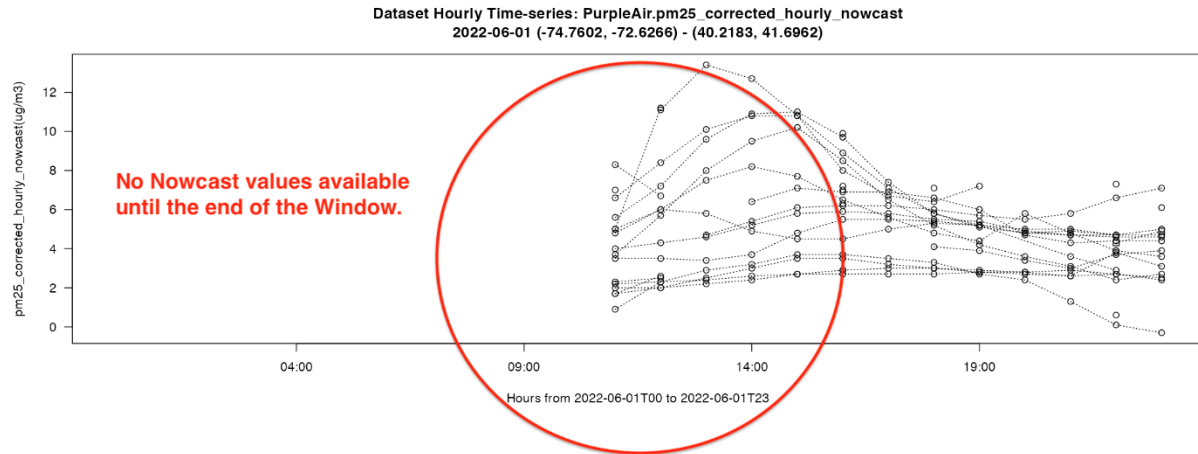
Neighboring Hourly Point Measures (<= 5000m) Comparison: AQS.pm25_nowcast vs PurpleAir.pm25_corrected_hourly_nowcast
2022-06-01 (-74.8067, -72.9029) - (40.2224, 41.5506)

timestamp(UTC)	AQS.id(-)	AQS.pm25_nowcast(ug/m3)	PurpleAir.id(-)	PurpleAir.pm25_corrected_hourly_nowcast(ug/m3)	PurpleAir.flagged(-)
2022-06-01T11:00:00-0000	90090027	4.60	106864	2.20	0
2022-06-01T11:00:00-0000	90090027	Nowcast values start at the end of the 12-window.			0
2022-06-01T11:00:00-0000	90090027	4.60	24063	1.70	0
2022-06-01T11:00:00-0000	90090027	4.60	106862	2.30	0
2022-06-01T11:00:00-0000	340170008	3.50	98723	1.70	0

12.9 Click **Plots** tab then click **Make Plots** button:



12.10: Scroll down to the **PurpleAir time-series**. Note that there are no Nowcast values available until hour 11 (the end of the window). Also, note the poor correlation.



12.11: Click the **Save Data** button to save the datasets with the new `_nowcast` variables and also save the tables. Then click the **Quit** button.

Save to Directory:

/Users/plessel/demo

Output File Format:

tsv ▼

Save Data

Quit

Specification of Standard Format Sensor Files:

Airnow/AQS Parameter codes:

<https://aqs.epa.gov/aqsweb/documents/codetables/parameters.html>

Here are the parameter codes used by the rsigserver web service (used by ASNAT):

pm25: 88101, 88500, 88501, 88502

pm10: 81102

rh: 62201, 62202, 68110

temperature: 62101

pressure: 64101

ozone: 44201

AirNow data are available near real-time.

AQS data are available on a 90-day delay because states have 90 days to report data.

Some data are often reported sooner than 90 days.

AQS data is quality-reviewed (and revised) more than AirNow data so prefer AQS data and, only if not available, try AirNow.

https://aqs.epa.gov/aqsweb/documents/about_aqs_data.html#timeliness-of-data

AirNow data are also QC-flagged and rsigserver webservice (used by ASNAT) filters-out data with a QC flag greater than 4.

https://www.airnowtech.org/resources/AIRNow-I_AQCSV-Final.pdf

Table 7, Page 28.

0 = Valid

1 = Adjusted

2 = Averaged

3 = interpolated

4 = Estimated

5 = Suspect

6 = Suspect (audit failure)

7 = Insufficient data

8 = Missing

9 = Invalid

Flagging Codes:

Neighbor-based Flags

- Flag 80 - Flags data points where the absolute difference between Dataset X and Y exceeds a specified threshold
- Flag 81 - Flags data points where the percent difference between Dataset X and Y exceeds a specified threshold
- Flag 82 - Flags data points where the R-squared value between Dataset X and Y is below a specified threshold

Additional Flags

- Flag 83 - Flags data points where values remain constant for more than a specified number of consecutive measurements
- Flag 84 - Flags data points where data is missing for more than a specified number of consecutive timesteps
- Flag 85 - Flags statistical outliers based on standard deviation from the mean
- Flag 86 - Flags outliers identified by the Hampel filter method (https://mazamascience.github.io/AirSensor/articles/outlier_detection.html)

Invalid Values Flags

- Flag 60 - Flags invalid negative values in pollutant measurements (O3, NO2, CO, etc.)

Cross-field Validation Flags

- Flag 65 - Flags temporal inconsistencies (e.g., end timestamps before start timestamps)

Suspicious Changes Flags

- Flag 70 - Flags sudden spikes in pollutant levels
- Flag 71 - Flags sudden drops in pollutant levels
- Flag 72 - Flags ozone measurements that deviate from typical daily patterns
- Flag 73 - Flags PM measurements that deviate from typical daily patterns

Redundancy check

- Flag 90 - Flags duplicate measurements at the same location and timestamp

Invalid formatting and units

- Flag 95 - Flags timestamps that don't conform to ISO 8601 format (YYYY-MM-DDTHH:MM:SS-ZZZZ)